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Kim et al.

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(54) **FAST RESPONSE ACTIVE CLEARANCE SYSTEMS WITH PIEZOELECTRIC ACTUATOR IN AXIAL, AXIAL/RADIAL COMBINED, AND CIRCUMFERENTIAL DIRECTIONS**

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F03G 7/06 (2006.01)
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(2021.08); **F03G 7/0665** (2021.08);
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H10N 30/206; H10N 30/50; H10N
30/802

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,203,673 A 4/1993 Evans
5,545,007 A 8/1996 Martin
(Continued)

FOREIGN PATENT DOCUMENTS

WO 2010083805 A1 7/2010

OTHER PUBLICATIONS

United States Patent and Trademark Office, "Non-Final Office
Action," issued in connection with U.S. Appl. No. 17/739,766,
dated Apr. 20, 2023, 8 pages.

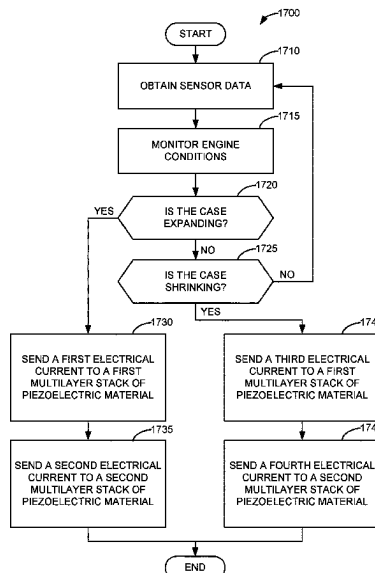
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Zimmerman, LLC

(57) **ABSTRACT**

Certain examples disclose and describe apparatus and meth-
ods to provide fast response active clearance system with
piezoelectric actuator in axial, axial/radial combined, and
circumferential directions. In some examples, an apparatus
includes an actuator to control clearance between a blade
and at least one of a shroud or a hanger, the actuator
including a multilayer stack of material, and wherein the
actuator is outside a case. The apparatus further includes a
rod coupled to the actuator and the at least one of the shroud
or the hanger through an opening in the case, the rod to move
the at least one of the shroud or the hanger in a radial
direction based on axial movement of the multilayer stack of
material.

20 Claims, 18 Drawing Sheets



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H10N 30/80 (2023.01)
- (52) **U.S. Cl.**
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(2023.02); *H10N 30/802* (2023.02)

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,780,957	A *	7/1998	Oliver	B60J 1/17 310/328
6,142,477	A	11/2000	Meinzer	
6,927,528	B2	8/2005	Barillot et al.	
7,125,223	B2	10/2006	Turnquist et al.	
7,448,849	B1	11/2008	Webster et al.	
8,113,768	B2	2/2012	Somanath et al.	
8,939,709	B2	1/2015	Nanukuttan et al.	
8,961,115	B2	2/2015	Rhoden et al.	
9,845,700	B2	12/2017	Dierksmeier	
11,867,068	B2	1/2024	Kim et al.	
2008/0267770	A1	10/2008	Webster et al.	
2011/0215674	A1 *	9/2011	Rinner	H10N 30/073 29/25.35
2012/0156007	A1	6/2012	Bacic	
2022/0254981	A1 *	8/2022	Kuroda	H10N 30/871

* cited by examiner

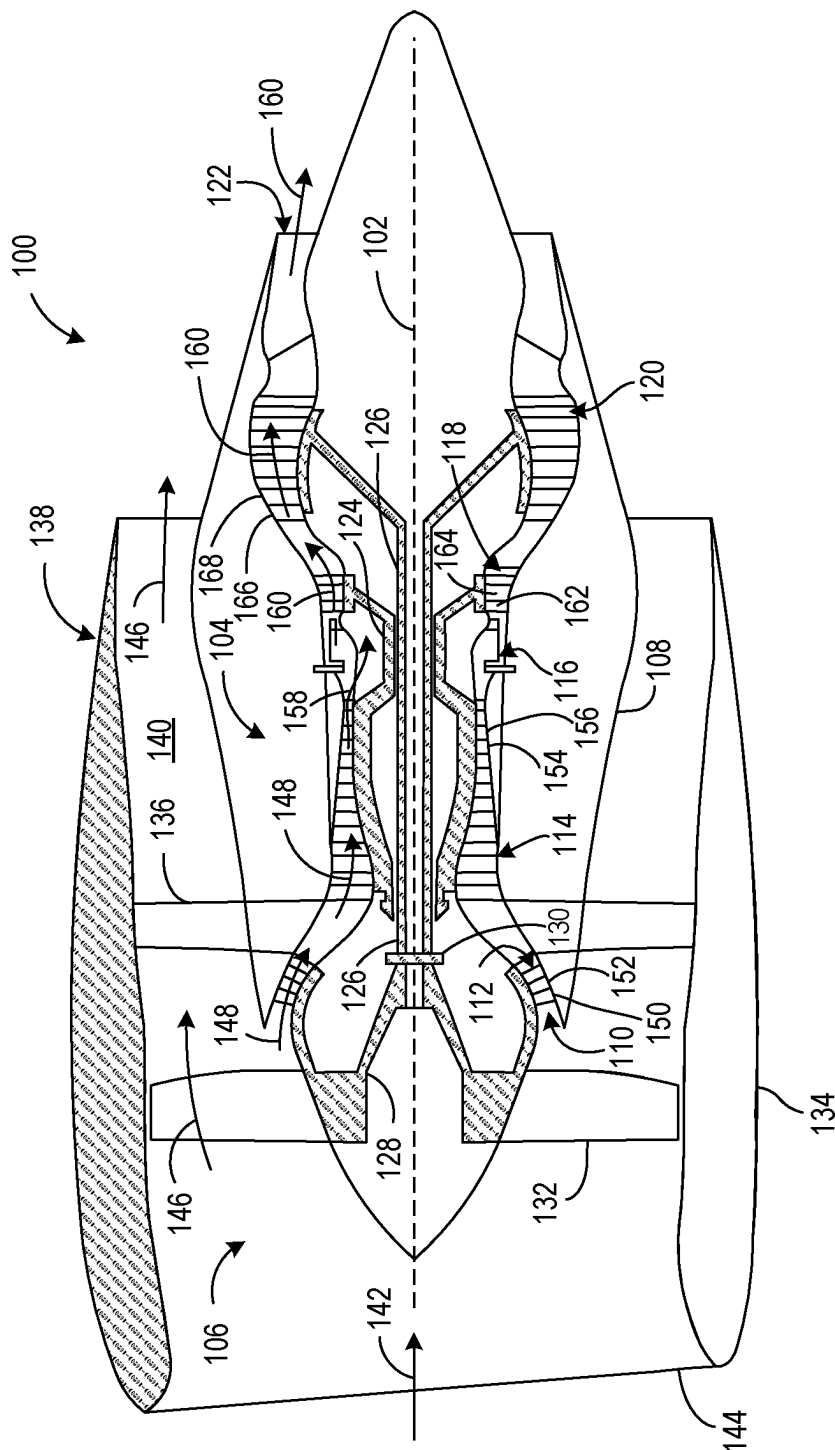


FIG. 1

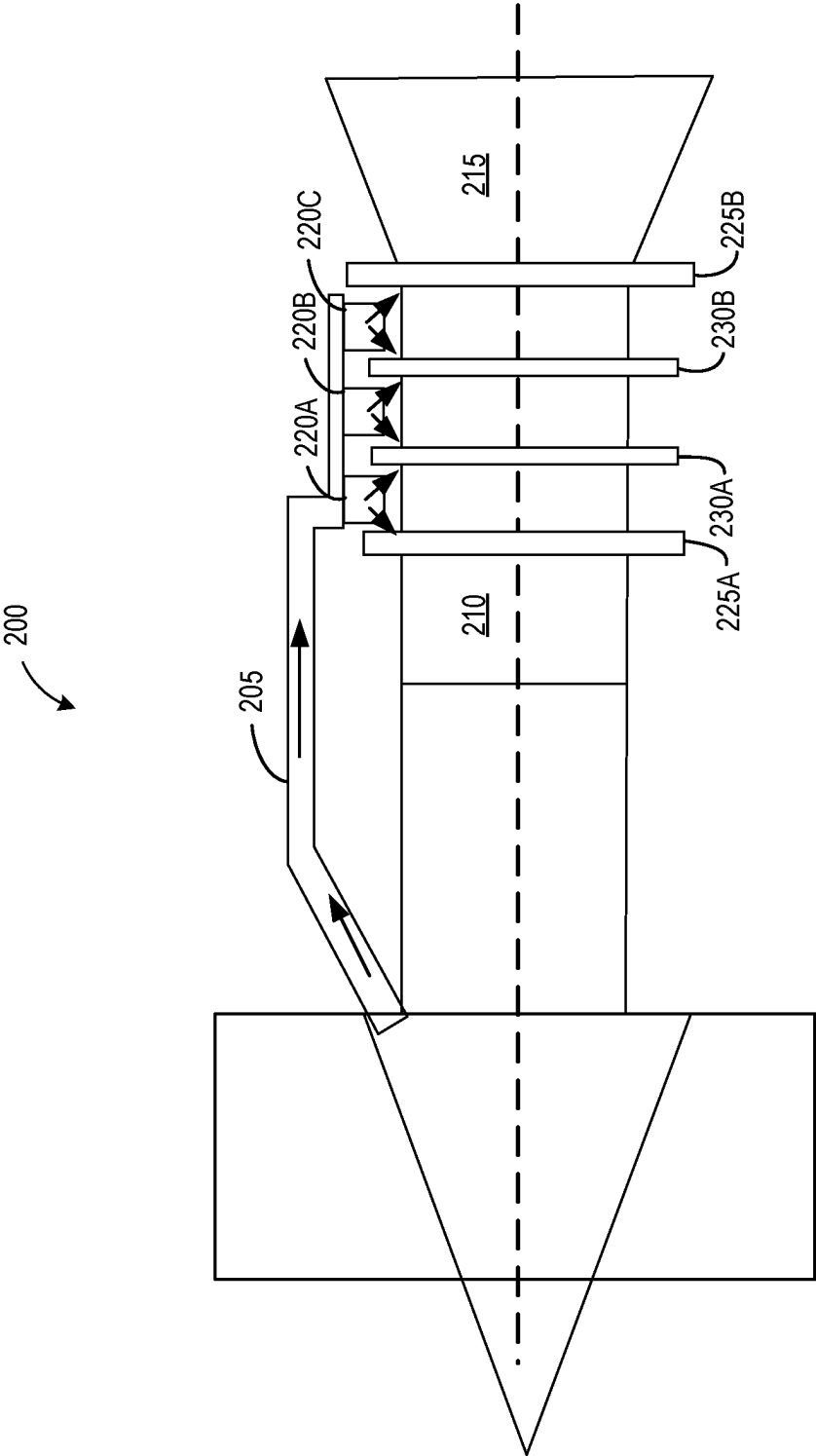


FIG. 2

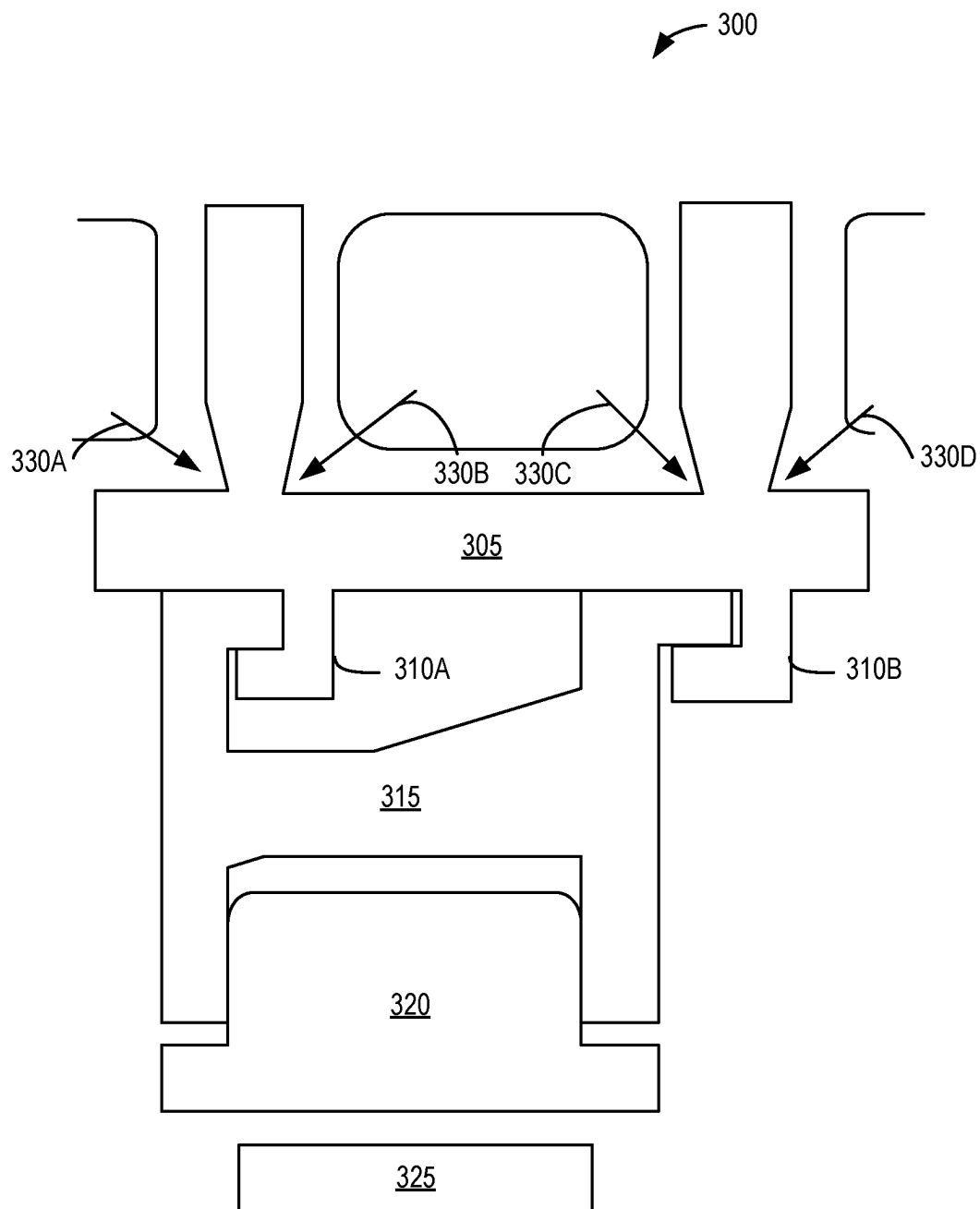


FIG. 3
(PRIOR ART)

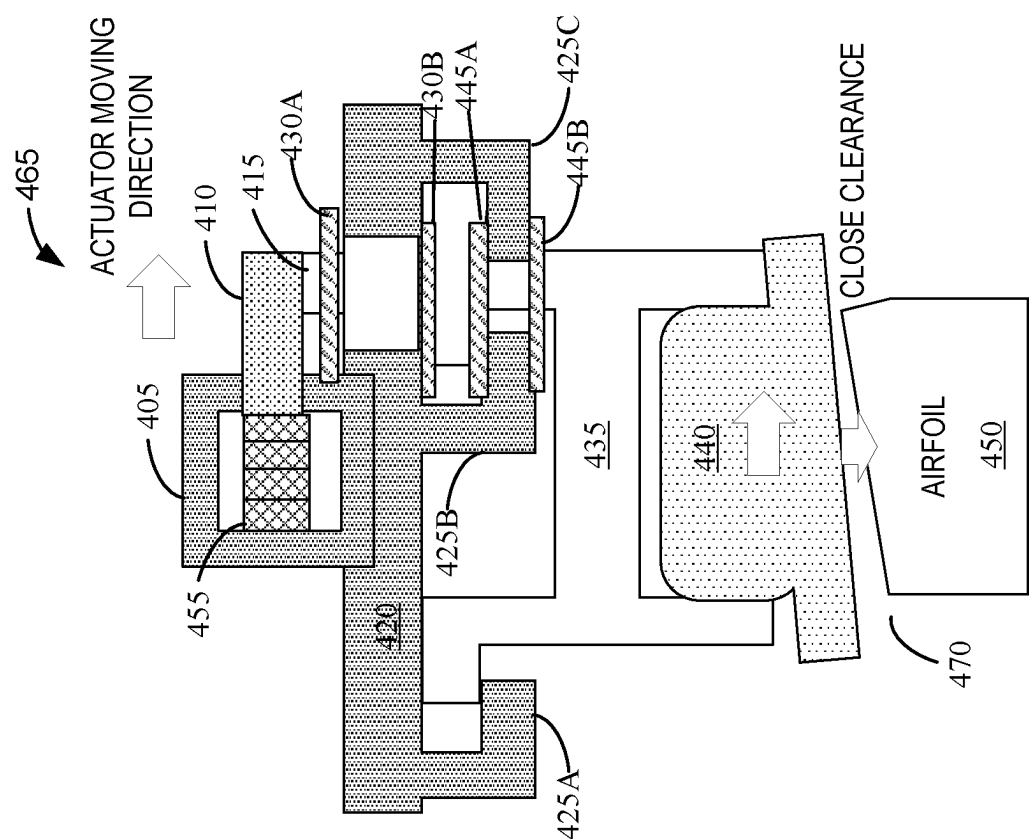


FIG. 4B

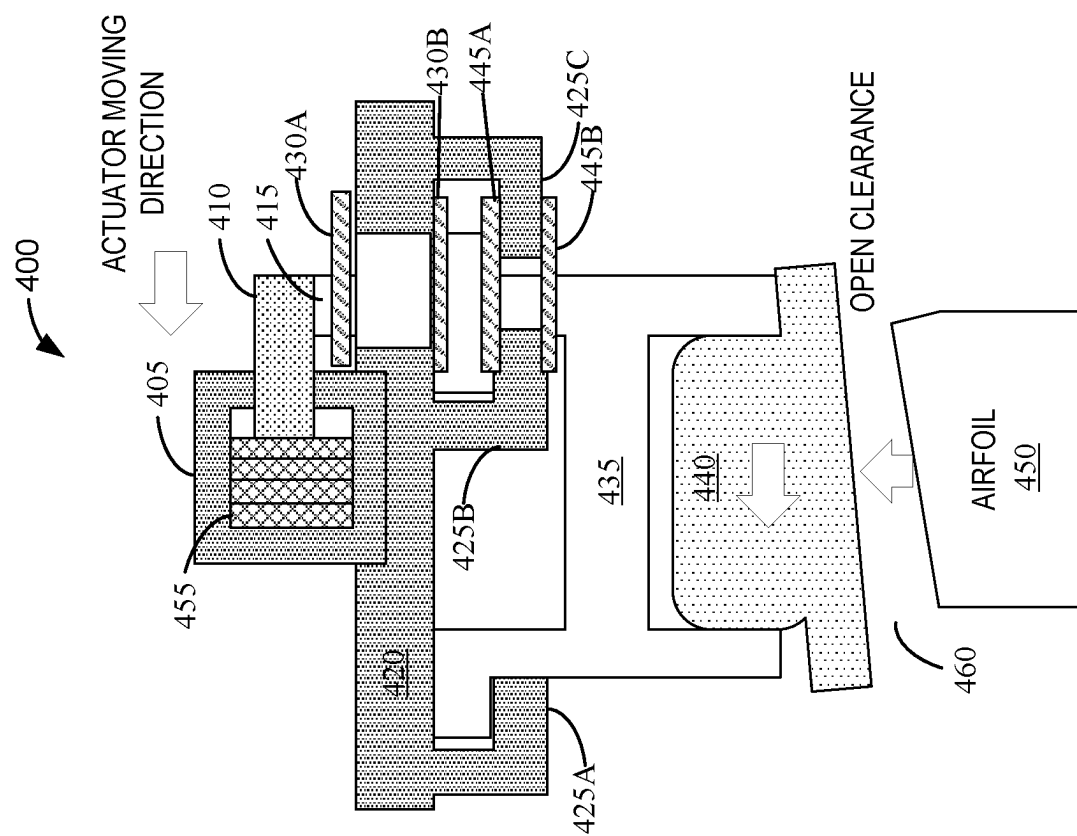


FIG. 4A

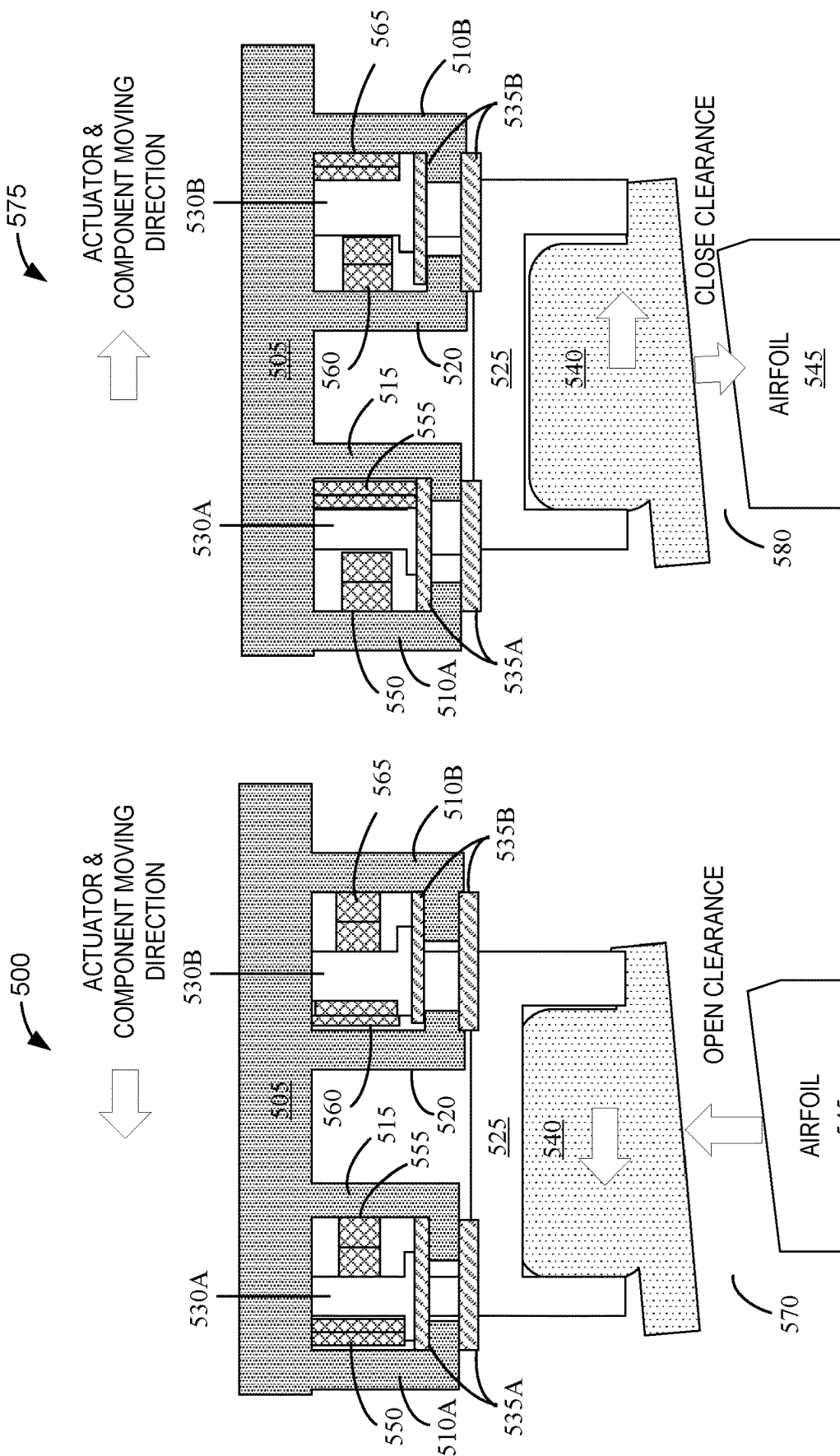


FIG. 5B

FIG. 5A

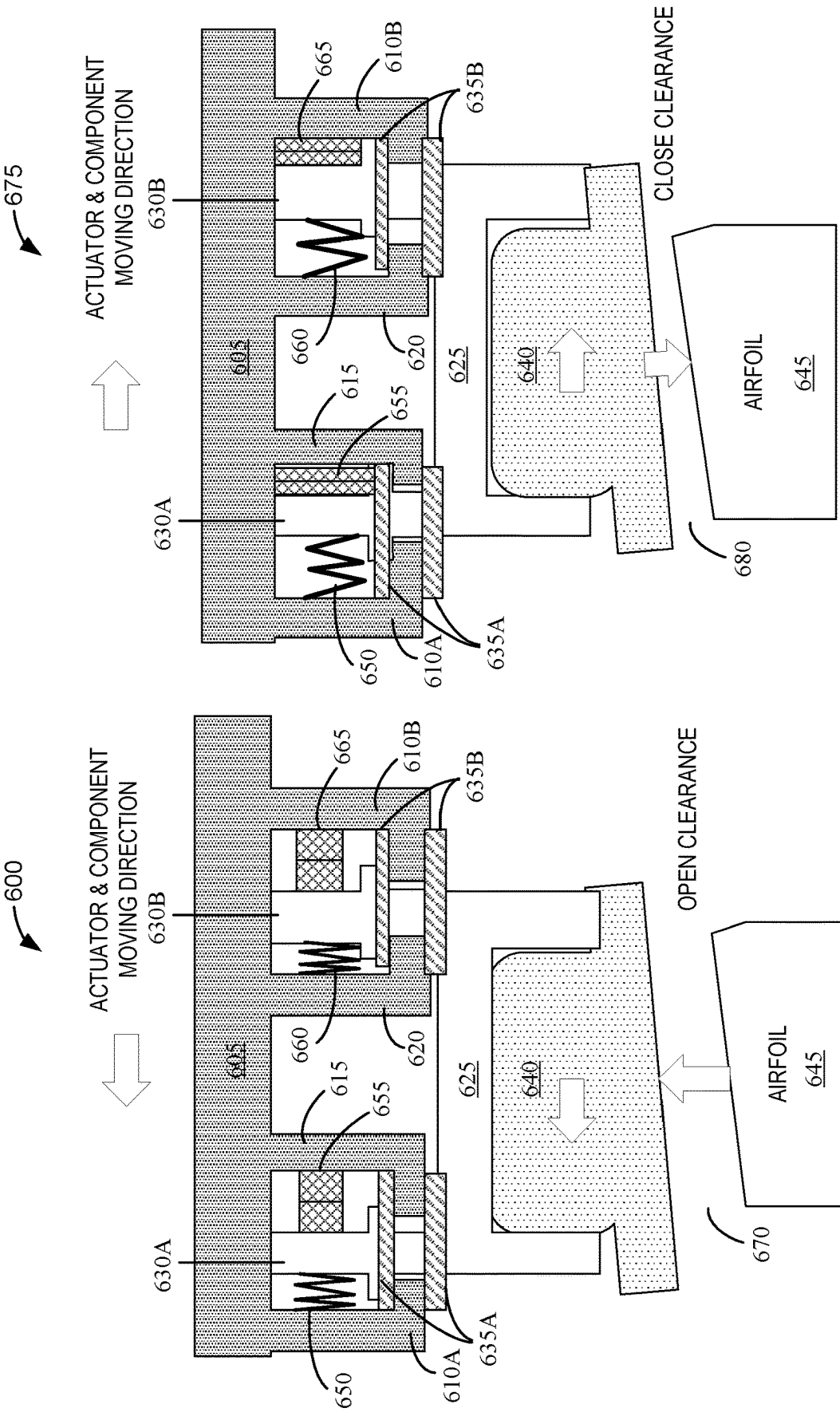


FIG. 6B

FIG. 6A

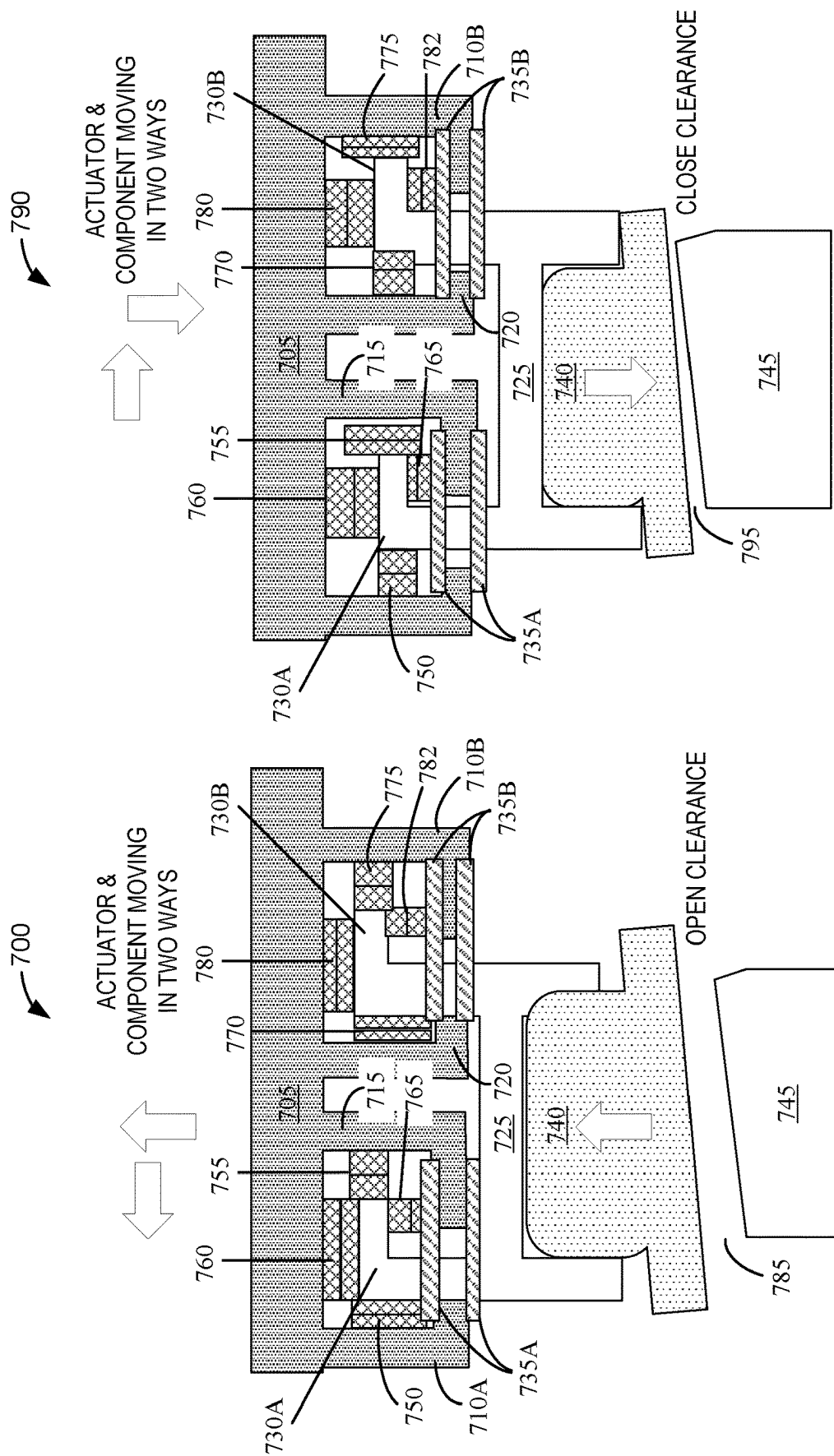


FIG. 7B

FIG. 7A

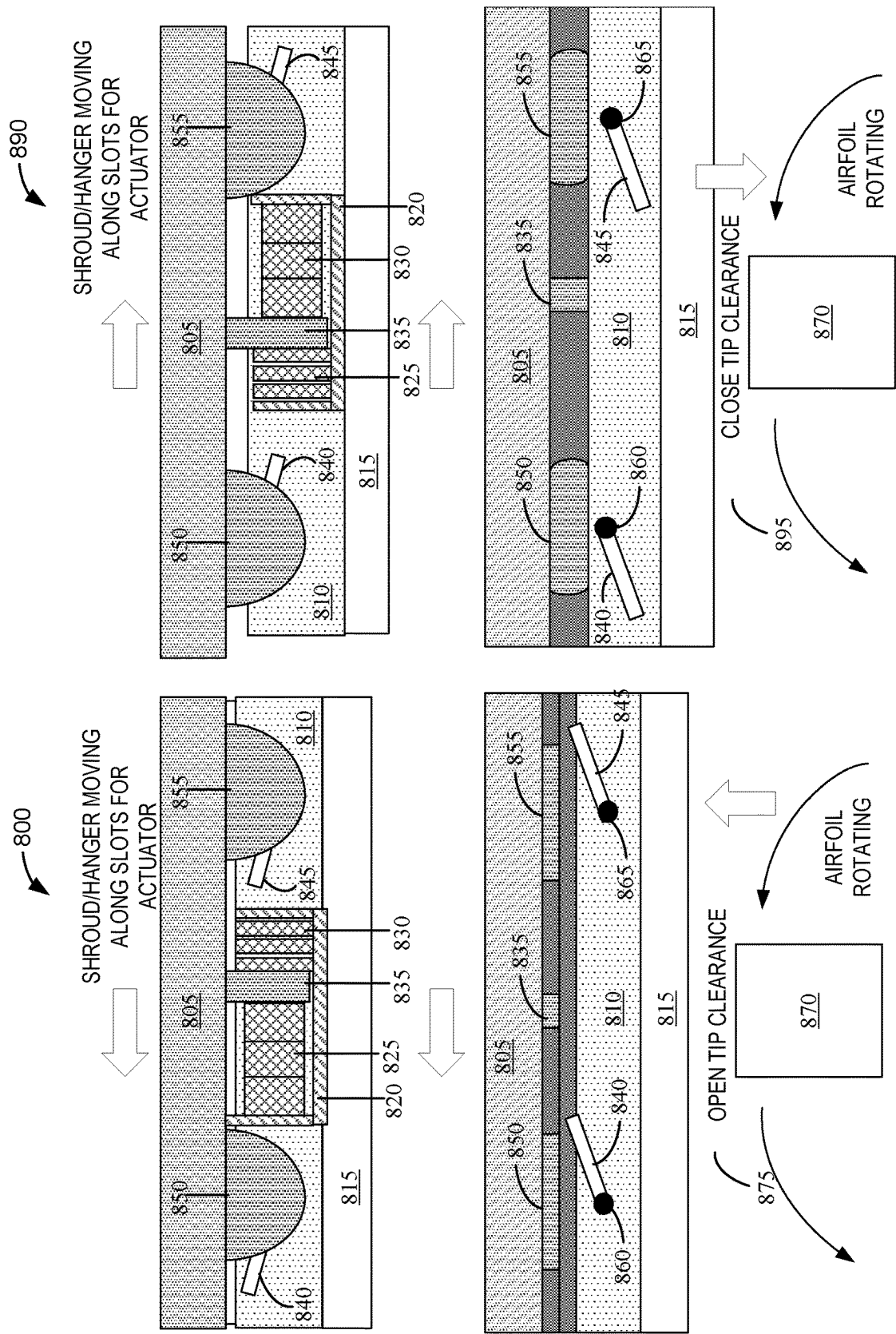
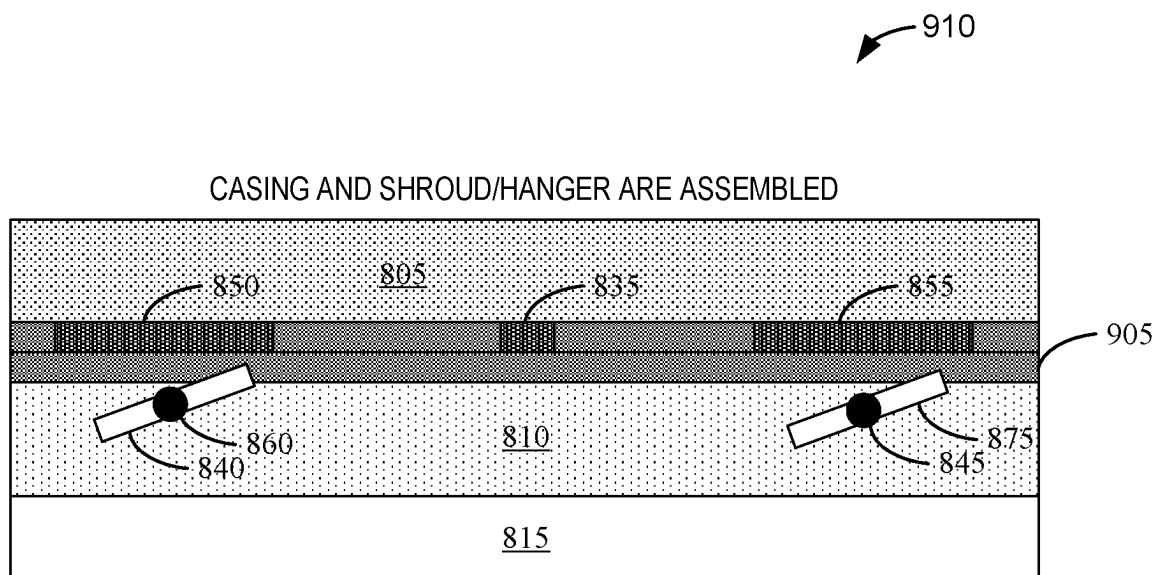
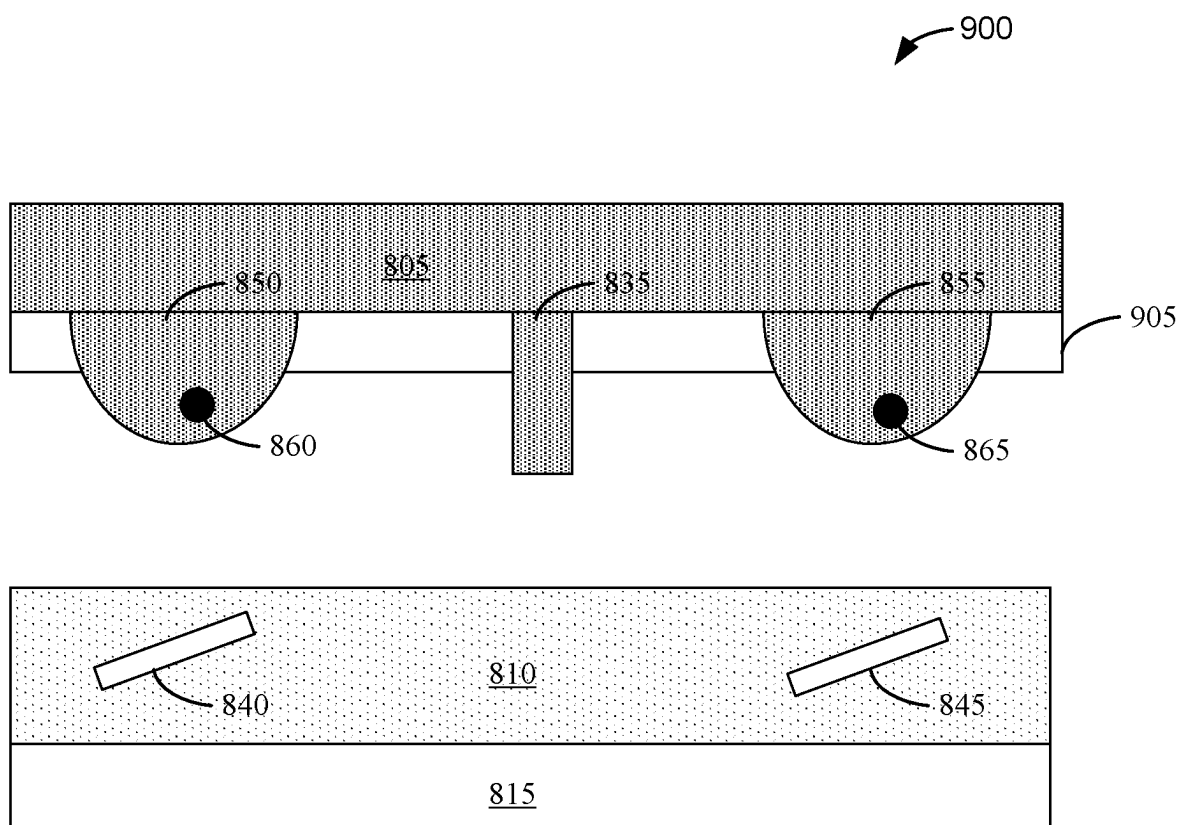


FIG. 8B

FIG. 8A



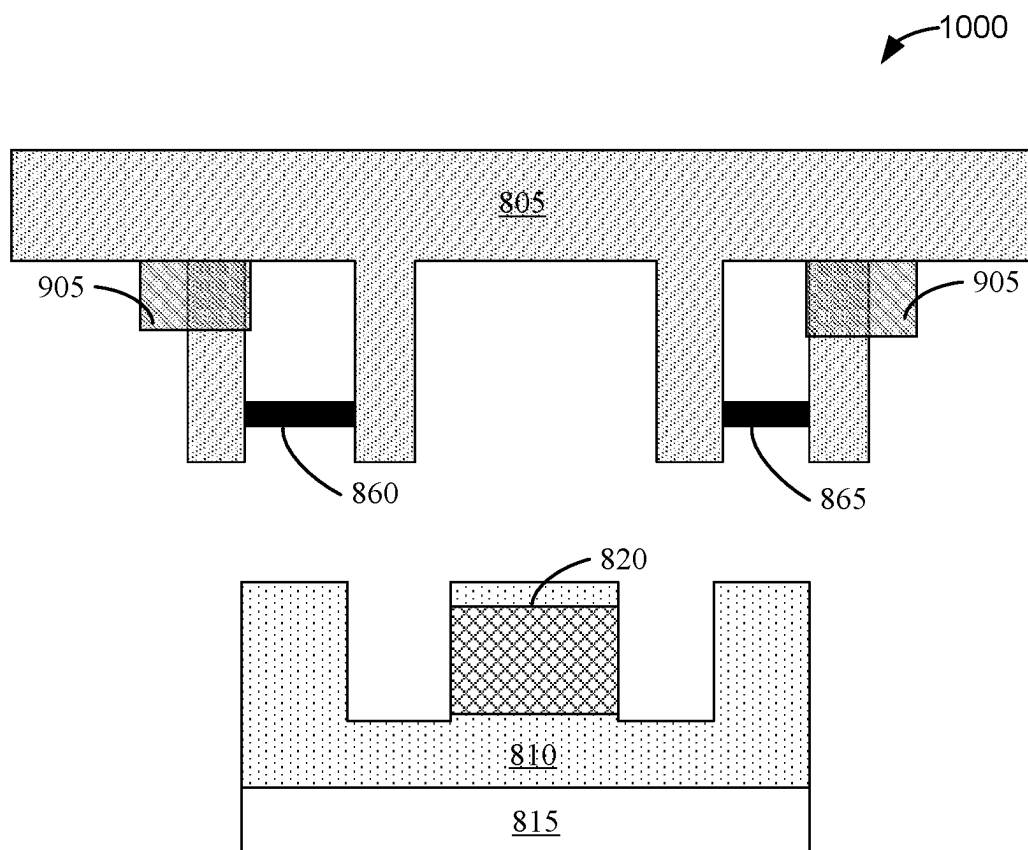


FIG. 10A

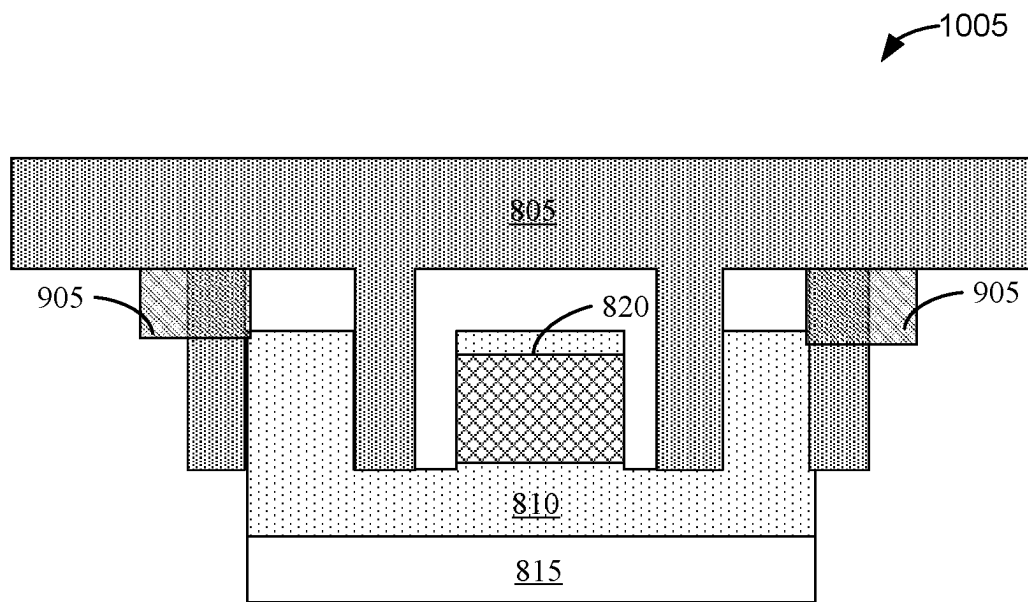


FIG. 10B

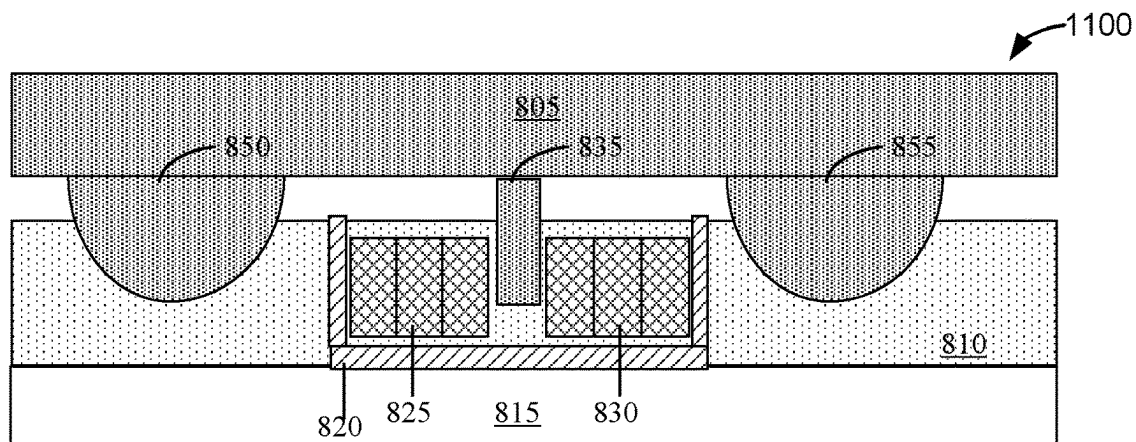


FIG. 11A

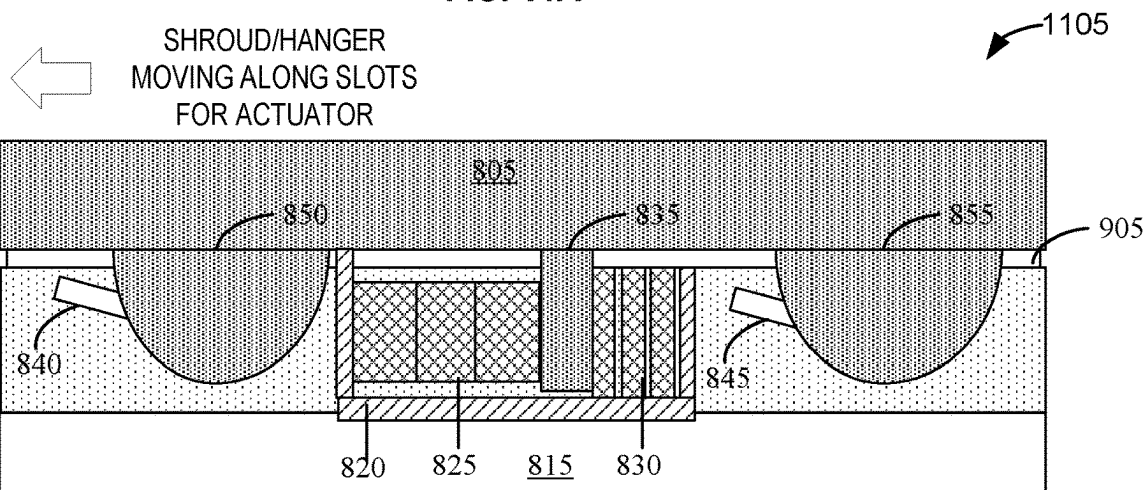


FIG. 11B

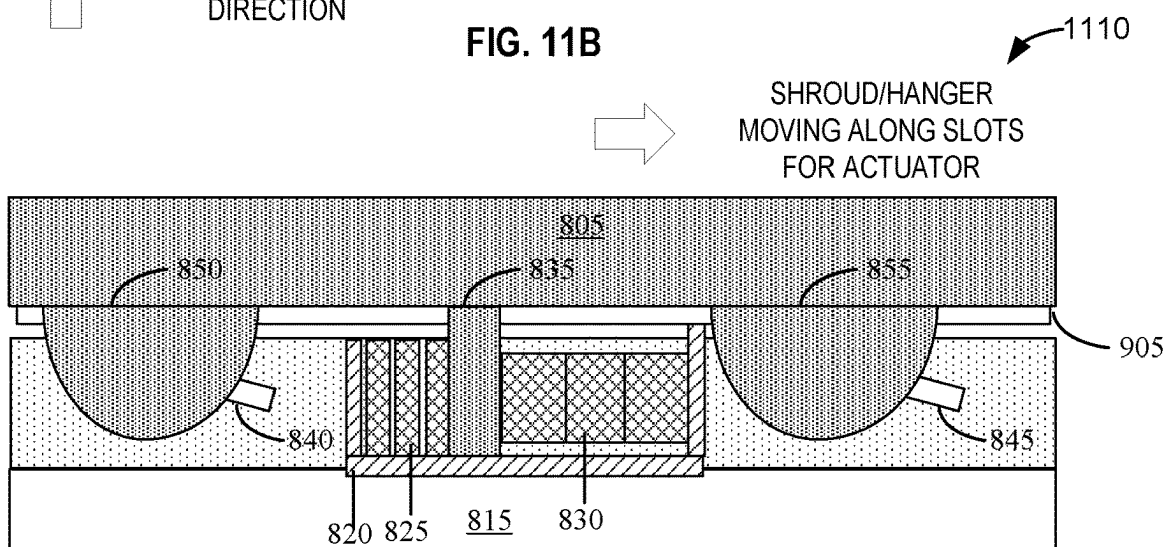
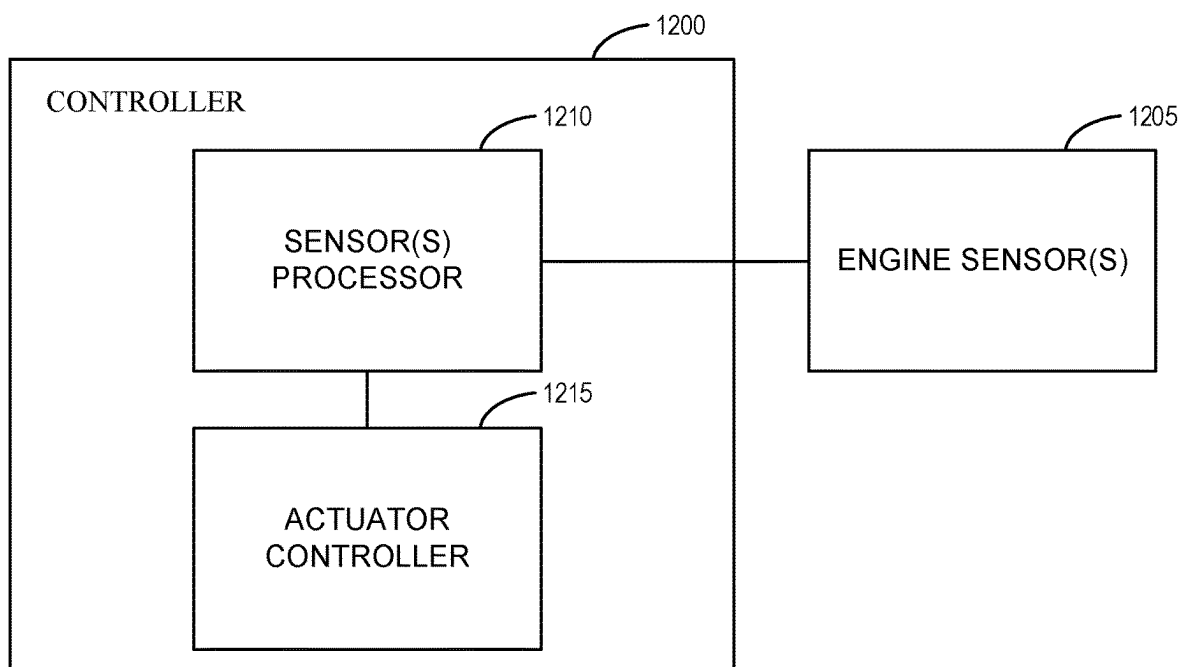


FIG. 11C

**FIG. 12**

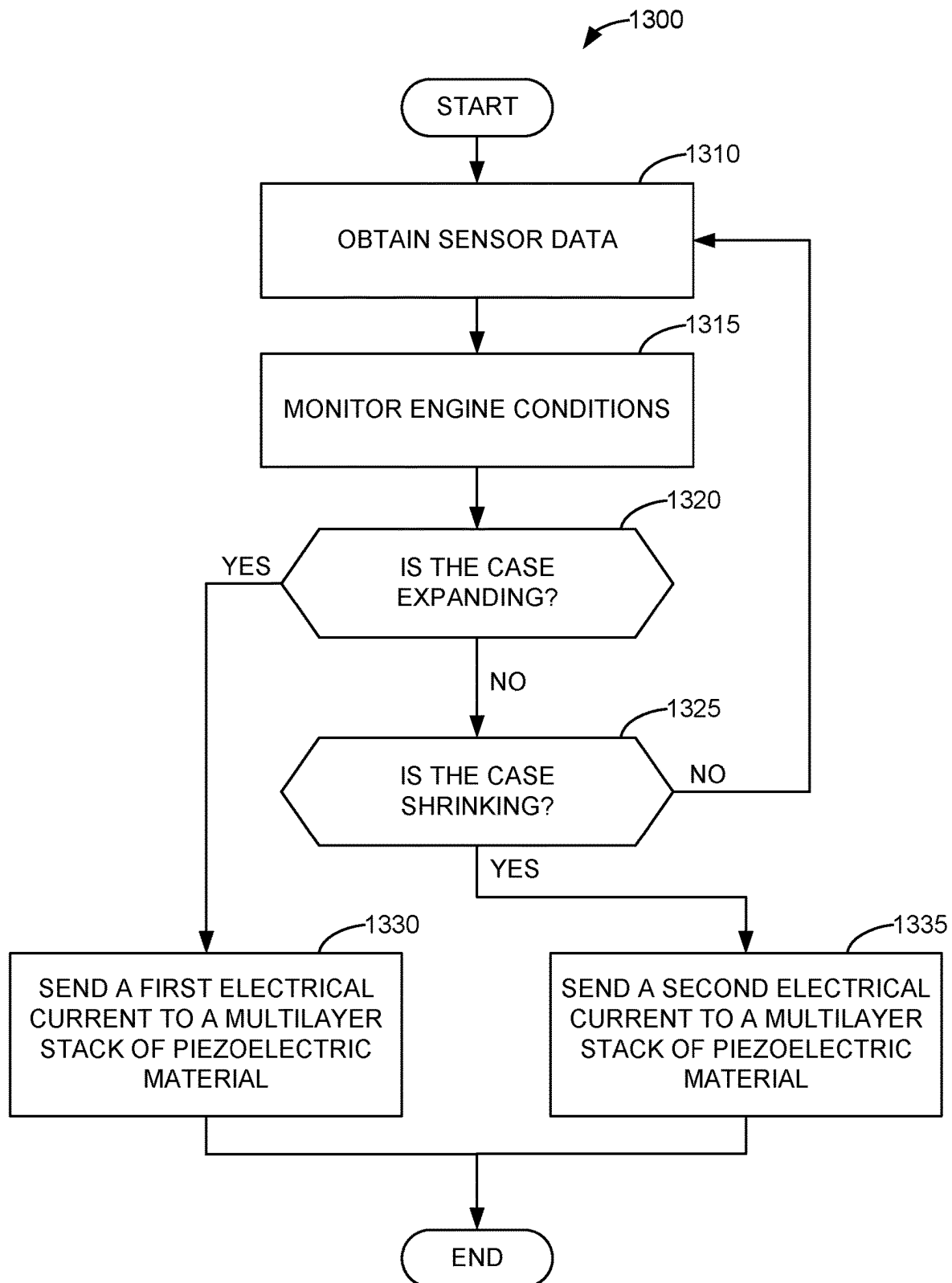


FIG. 13

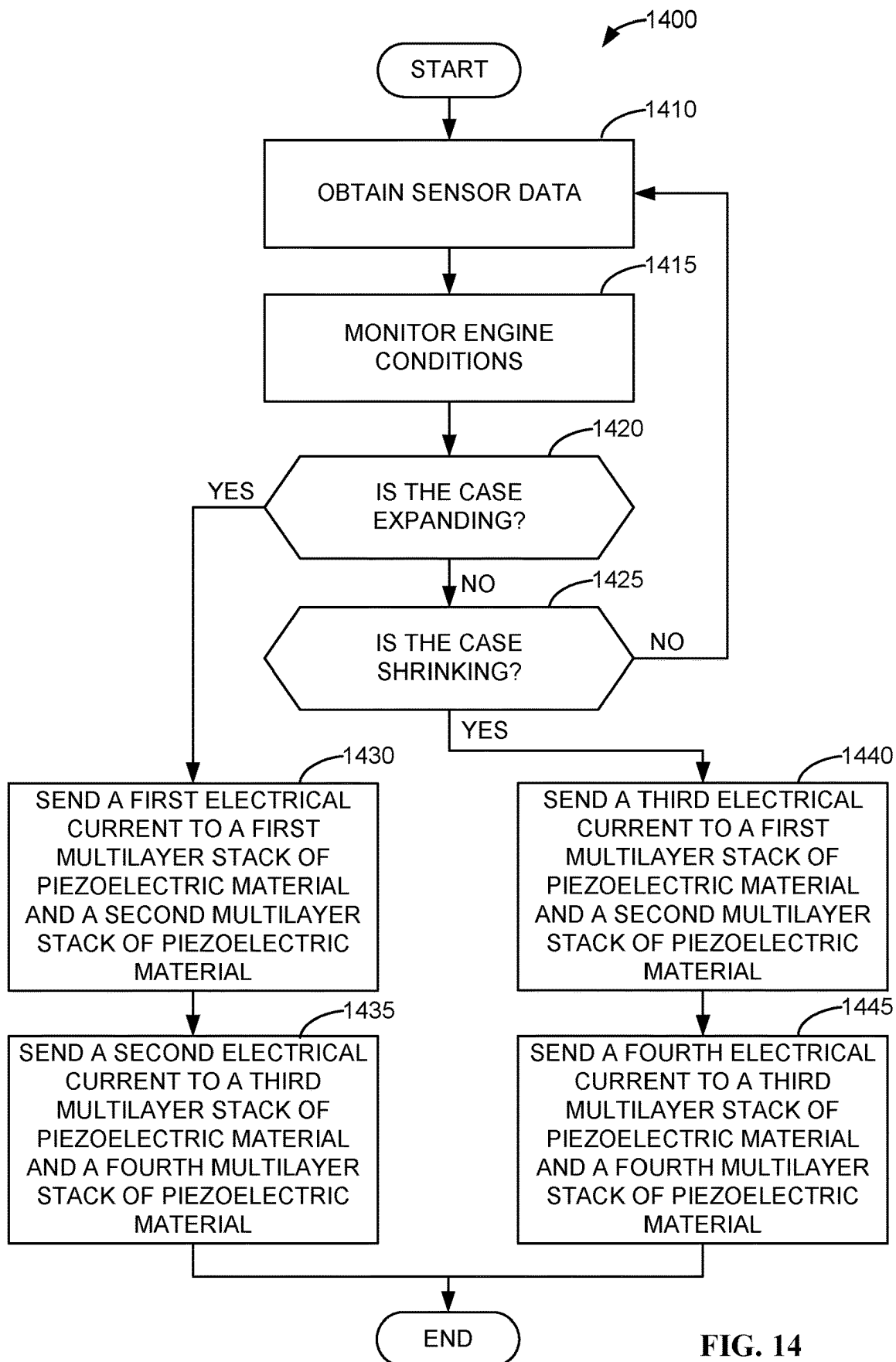


FIG. 14

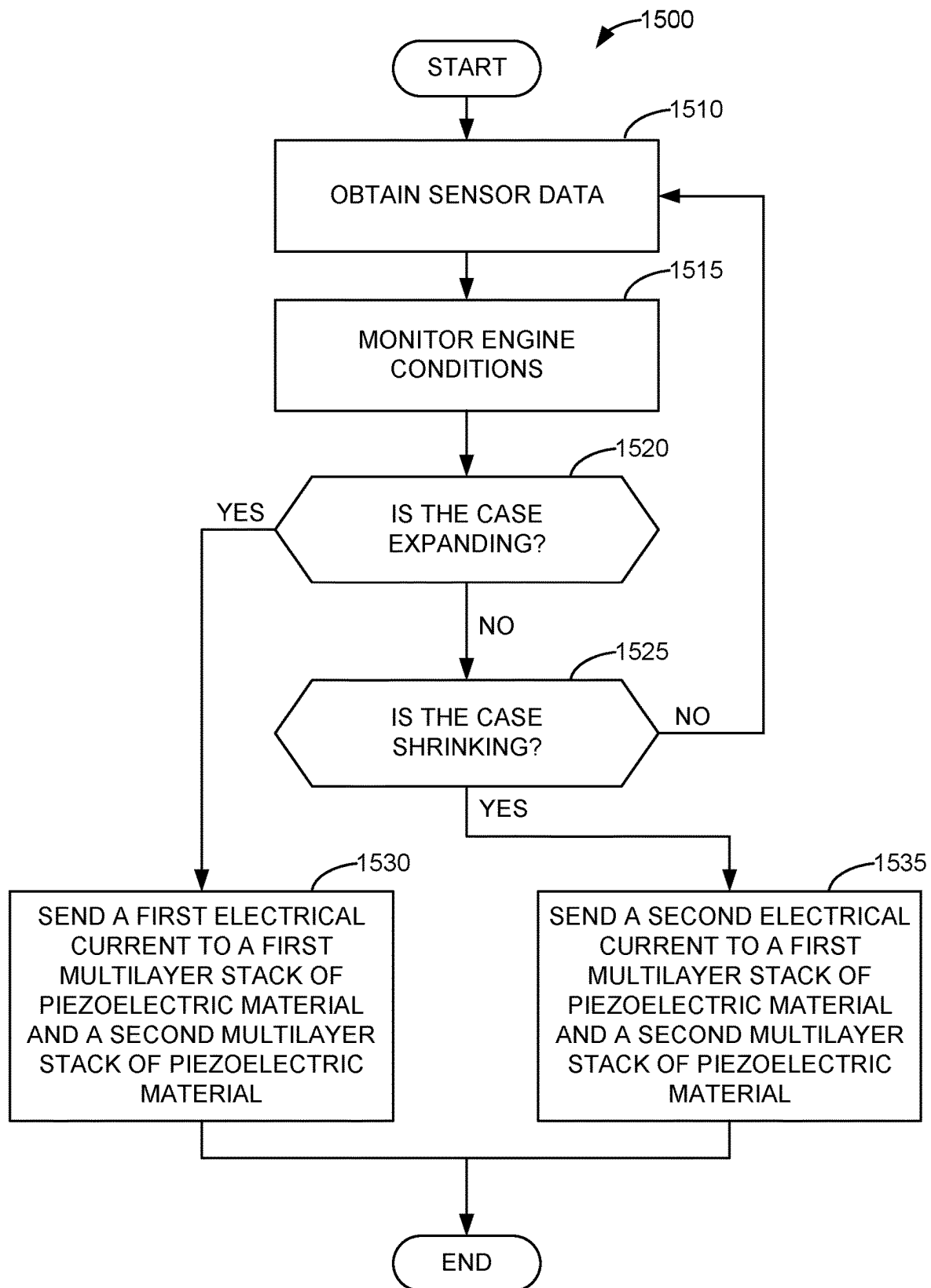


FIG. 15

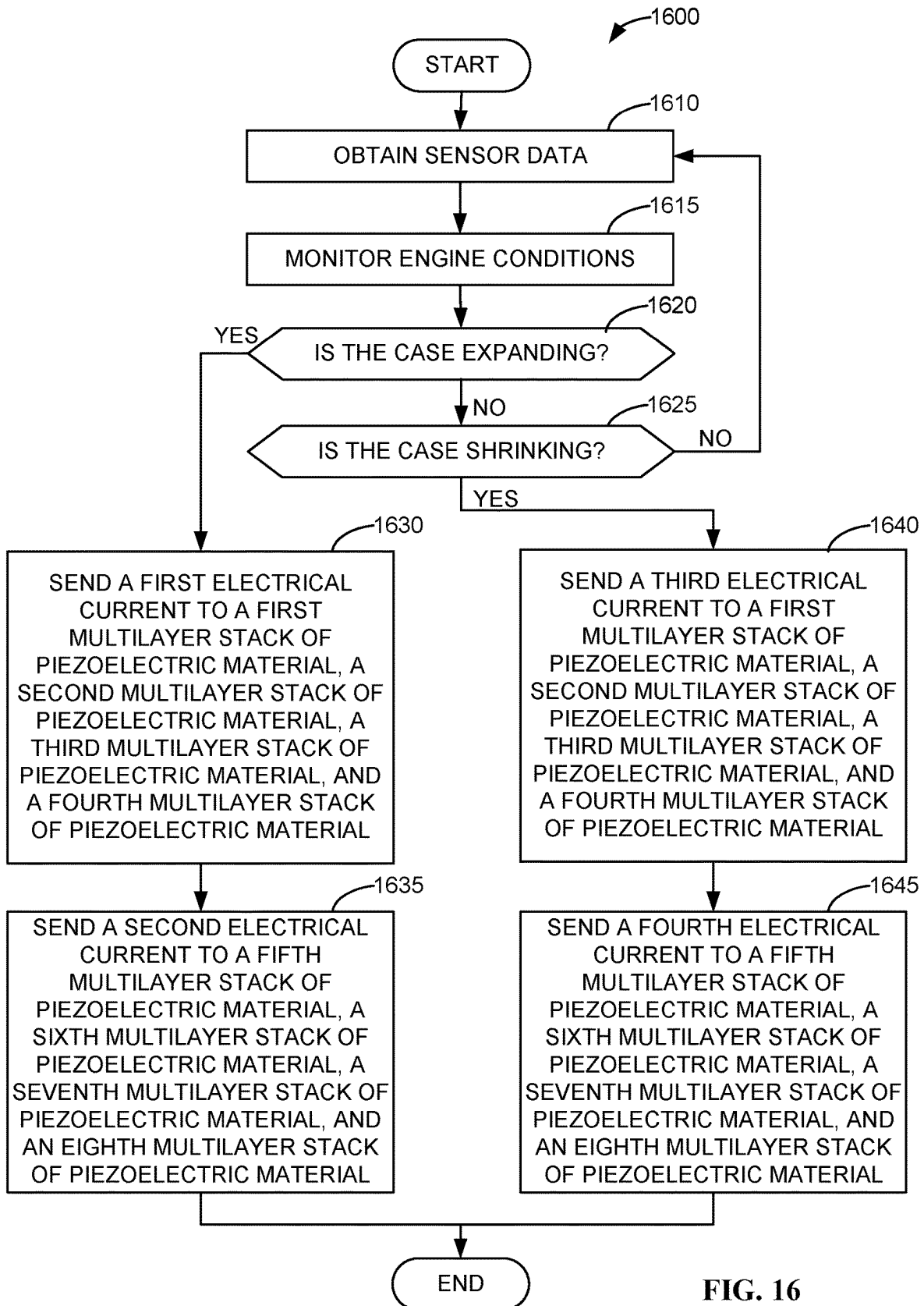


FIG. 16

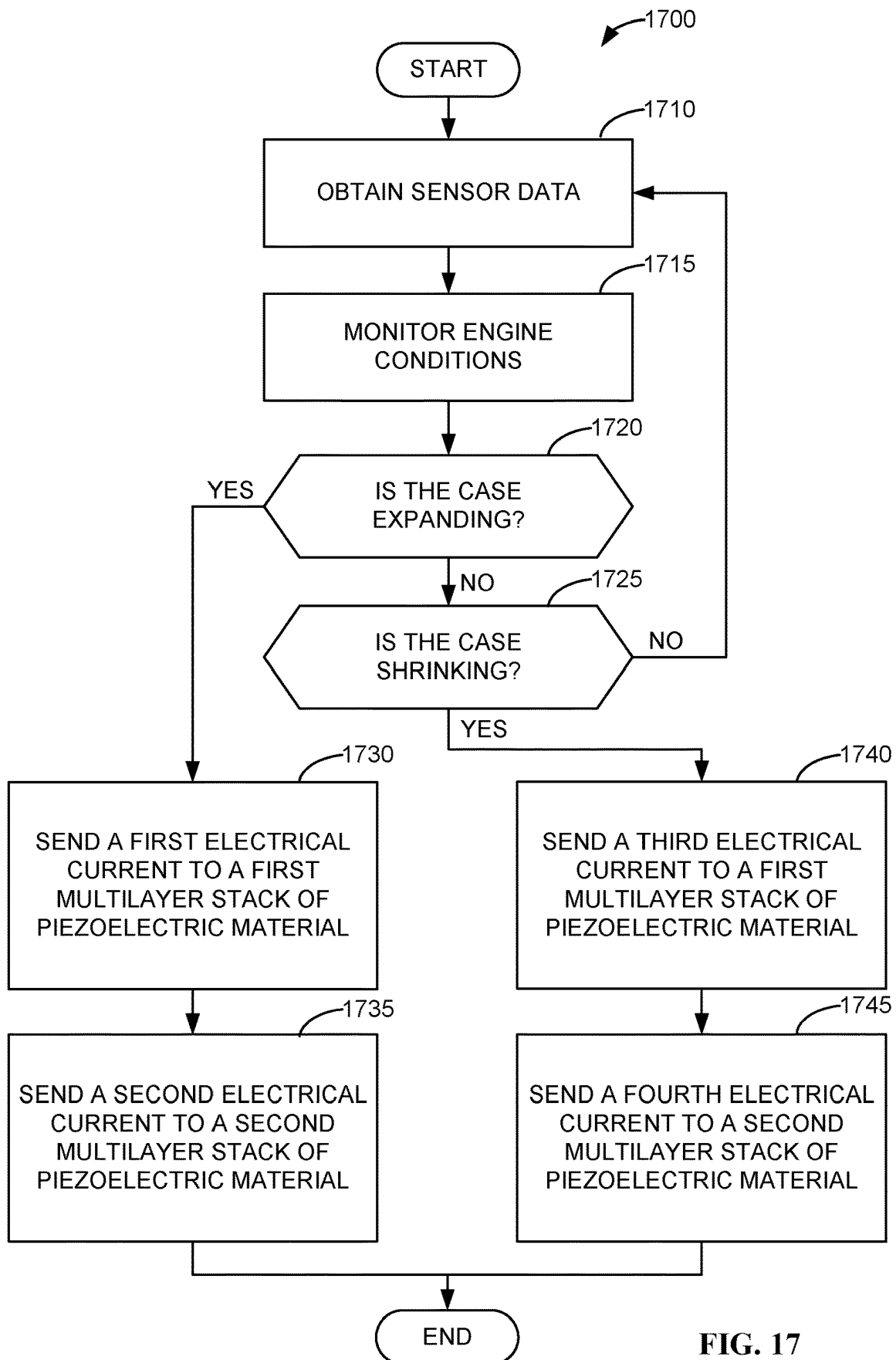


FIG. 17

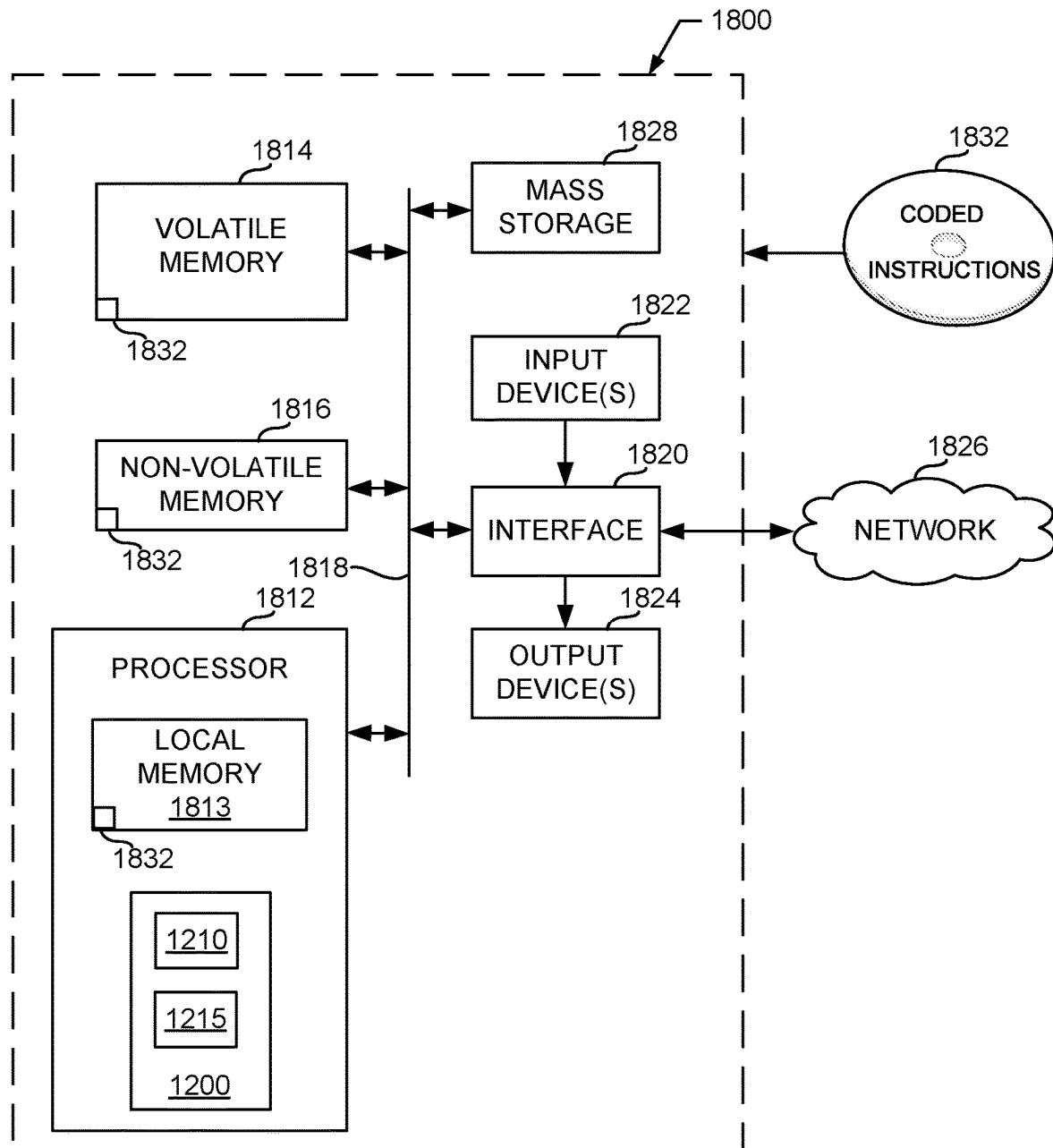


FIG. 18

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FAST RESPONSE ACTIVE CLEARANCE SYSTEMS WITH PIEZOELECTRIC ACTUATOR IN AXIAL, AXIAL/RADIAL COMBINED, AND CIRCUMFERENTIAL DIRECTIONS

REFERENCE TO RELATED APPLICATIONS

This patent claims the benefit of priority to U.S. patent application Ser. No. 17/739,766 (now U.S. Pat. No. 11,867,068), filed on May 9, 2022, entitled "FAST RESPONSE ACTIVE CLEARANCE SYSTEMS WITH PIEZOELECTRIC ACTUATOR IN AXIAL, AXIAL/RADIAL COMBINED, AND CIRCUMFERENTIAL DIRECTIONS," which is incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

This disclosure relates generally to a gas turbine engine and, more particularly, to fast response active clearance system with piezoelectric actuator in axial, axial/radial combined, and circumferential directions.

BACKGROUND

A gas turbine engine generally includes, in serial flow order, an inlet section, a compressor section, a combustion section, a turbine section, and an exhaust section. In operation, air enters the inlet section and flows to the compressor section where one or more axial compressors progressively compress the air until it reaches the combustion section. Fuel mixes with the compressed air and burns within the combustion section, thereby creating combustion gases. The combustion gases flow from the combustion section through a hot gas path defined within the turbine section and then exit the turbine section via the exhaust section.

In general, it is desirable for a gas turbine engine to maintain clearance between the tip of a blade in the gas turbine engine and the stationary parts of the gas turbine engine (e.g., the gas turbine engine casing, stator, etc.). During operation, the gas turbine engine is exposed to thermal (e.g., hot and cold air pumped into the gas turbine engine, etc.) and mechanical loads (e.g., centrifugal force on the blades on the gas turbine engine, etc.), which can expand and contract the gas turbine engine casing and rotor. The expansion and contraction of the gas turbine engine casing can change the clearance between the blade tip and the stationary parts of the gas turbine engine. There is a continuing need to control the clearance between the blade tip and the engine casing that fluctuates during normal operation for a gas turbine engine to avoid damage to the gas turbine engine (e.g., wear, breakage, etc. due to unintentional rub).

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic cross-sectional view of an example gas turbine engine in accordance with the examples disclosed herein.

FIG. 2 is a schematic cross-sectional view of an example gas turbine engine with a conventional active clearance control (ACC) system.

FIG. 3 is a schematic cross-sectional view of a prior ACC system for a gas turbine engine.

FIGS. 4A and 4B are schematic cross-sectional views of a first example ACC system with an example axial, outer-stack piezoelectric actuator in accordance with teachings disclosed herein.

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FIGS. 5A and 5B are schematic cross-sectional views of a second example ACC system with an example axial, inner-stack piezoelectric actuator in accordance with teachings disclosed herein.

FIGS. 6A and 6B are schematic cross-sectional views of a third example ACC system with an example axial, inner-stack piezoelectric actuator and spring load in accordance with teachings disclosed herein.

FIGS. 7A and 7B are schematic cross-sectional views of a fourth example ACC system with an example axial/radial combined piezoelectric actuator in accordance with teachings disclosed herein.

FIGS. 8A and 8B are schematic aft looking forward (ALF) and forward looking aft (FLA) views of a fifth example ACC system with an example circumferential piezoelectric actuator in accordance with teachings disclosed herein.

FIGS. 9A and 9B are schematic FLA views of the fifth example ACC system in a disassembled and an assembled state.

FIGS. 10A and 10B are schematic side views (Axi-sym) views of the fifth example ACC system in a disassembled and an assembled state.

FIGS. 11A-11C are schematic ALF views of the fifth example ACC system in cold conditions and hot conditions.

FIG. 12 is a block diagram of an example controller of the example ACC systems of FIGS. 4A-11C.

FIG. 13 is a flowchart representative of example machine readable instructions and/or example operations that may be executed by example processor circuitry to implement the example controller of FIG. 12 in conjunction with the first example ACC system of FIGS. 4A and 4B.

FIG. 14 is a flowchart representative of example machine readable instructions and/or example operations that may be executed by example processor circuitry to implement the example controller of FIG. 12 in conjunction with the second example ACC system of FIGS. 5A and 5B.

FIG. 15 is a flowchart representative of example machine readable instructions and/or example operations that may be executed by example processor circuitry to implement the example controller of FIG. 12 in conjunction with the third example ACC system of FIGS. 6A and 6B.

FIG. 16 is a flowchart representative of example machine readable instructions and/or example operations that may be executed by example processor circuitry to implement the example controller of FIG. 12 in conjunction with the fourth example ACC system of FIGS. 7A and 7B.

FIG. 17 is a flowchart representative of example machine readable instructions and/or example operations that may be executed by example processor circuitry to implement the example controller of FIG. 12 in conjunction with the fifth example ACC system of FIGS. 8A-11C.

FIG. 18 is a block diagram of an example processing platform including processor circuitry structured to execute the example machine readable instructions and/or the example operations of FIGS. 13-17 to implement the example controller of FIG. 12.

The figures are not to scale. Instead, the thickness of the layers or regions may be enlarged in the drawings. Although the figures show layers and regions with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular. In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. As used herein, unless otherwise stated, the term "above" describes the relationship of two parts relative to

Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another. As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween. As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name.

DETAILED DESCRIPTION

In the following detailed description, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration specific examples that may be practiced. These examples are described in sufficient detail to enable one skilled in the art to practice the subject matter, and it is to be understood that other examples may be utilized. The following detailed description is therefore, provided to describe example implementations and not to be taken limiting on the scope of the subject matter described in this disclosure. Certain features from different aspects of the following description may be combined to form yet new aspects of the subject matter discussed below.

When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “first,” “second,” and the like, do not denote any order, quantity, or importance, but rather are used to distinguish one element from another. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. As the terms “connected to,” “coupled to,” etc. are used herein, one object (e.g., a material, element, structure, member, etc.) can be connected to or coupled to another object regardless of whether the one object is directly connected or coupled to the other object or whether there are one or more intervening objects between the one object and the other object.

As used herein, the terms “system,” “unit,” “module,” “engine,” etc., may include a hardware and/or software system that operates to perform one or more functions. For example, a module, unit, or system may include a computer processor, controller, and/or other logic-based device that performs operations based on instructions stored on a tangible and non-transitory computer readable storage medium, such as a computer memory. Alternatively, a module, unit, engine, or system may include a hard-wired device that performs operations based on hard-wired logic of the device. Various modules, units, engines, and/or systems shown in the attached figures may represent the hardware that operates based on software or hardwired instructions, the software that directs hardware to perform the operations, or a combination thereof.

The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows. As used herein, “vertical” refers to the direction perpendicular to the ground. As used herein, “horizontal” refers to the direction parallel to the centerline of the gas turbine engine 100. As used herein, “lateral” refers to the direction perpendicular to the axial and vertical directions (e.g., into and out of the plane of FIGS. 1, 2, etc.).

In some examples used herein, the term “substantially” is used to describe a relationship between two parts that is within three degrees of the stated relationship (e.g., a substantially colinear relationship is within three degrees of being linear, a substantially perpendicular relationship is within three degrees of being perpendicular, a substantially parallel relationship is within three degrees of being parallel, etc.).

“Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A or B” is intended

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to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

A turbine engine, also called a combustion turbine or a gas turbine, is a type of internal combustion engine. Turbine engines are commonly utilized in aircraft and power-generation applications. As used herein, the terms “asset,” “aircraft turbine engine,” “gas turbine,” “land-based turbine engine,” and “turbine engine” are used interchangeably. A basic operation of the turbine engine includes an intake of fresh atmospheric air flow through the front of the turbine engine with a fan. In some examples, the air flow travels through an intermediate-pressure compressor or a booster compressor located between the fan and a high-pressure compressor. A turbine engine also includes a turbine with an intricate array of alternating rotating and stationary airfoil-section blades. As the hot combustion gas passes through the turbine, the hot combustion gas expands, causing the rotating blades to spin.

The components of the turbine engine (e.g., the fan, the booster compressor, the high-pressure compressor, the high-pressure turbine, the low-pressure turbine, etc.) can degrade over time due to demanding operating conditions such as extreme temperature and vibration. During operation, the turbine engine components are exposed to thermal (e.g., hot and cold air pumped into the turbine engine, etc.) and mechanical loads (e.g., centrifugal force on the blades on the turbine engine, etc.), which can expand and contract the turbine engine casing and/or compressor casing within the turbine engine along with other components of the turbine engine and/or its compressor. The expansion and contraction of the turbine engine casing and/or compressor casing within the turbine engine can change the clearance between the blades’ tips and the stationary components of the turbine engine. In some examples, if the clearance between the blades’ tips and the stationary components is not controlled, then the blades’ tips and stationary components can collide during operation and lead to further degradation of the components of the turbine engine.

The Active Clearance Control (ACC) System was developed to optimize blade tip clearance for engine performance improvement without unexpected rub events during flight and ground operations. A conventional ACC System includes using cooling air from a fan or compressor to control the clearance between the blade tip and an engine component that has shrunk (e.g., the stator, the case, etc.). The conventional ACC system is limited in that clearance is only modulated in one direction (e.g., engine component shrinkage). For a hot rotor condition (e.g., the engine component(s) are expanded), the conventional ACC system waits for rotor-stator thermal/mechanical growth matching to escape the hot rotor condition (e.g., modulate the blade tip clearance). Additionally, the conventional ACC system is limited in response time for modulating clearance (e.g., engine component shrinkage) due to the time delay for using a thermal response (e.g., cooling air). The response time for clearance modulation in the conventional ACC systems may be slow for certain systems that require faster responses for engine component shrinkage.

Examples disclosed herein optimize and/or otherwise improve an ACC system using piezoelectric actuator(s) that provide fast response clearance control without the mechanical delay seen in the conventional ACC system. Examples disclosed herein maintain desired clearances between the blade tip and other engine components without additional margin for various operating conditions, which will lead to performance improvement and provide better exhaust gas

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temperature (EGT) control capability. In certain examples, piezoelectric material generates linear displacement when an electric field is applied. The linear displacement can have a force, and examples disclosed herein apply the linear force of the piezoelectric material for the ACC system to achieve fast response clearance control. Examples disclosed herein apply the mechanical force from the linear displacement of the piezoelectric material on to modulating the ACC system. Examples disclosed herein can include other materials that generate linear displacement such as, shape memory alloy (SMA), etc. The range of displacement is increased by adding layers of piezoelectric material or SMA, called multilayer stacks, where more layers in a stack provides more axial movement and radial movement range and gives the ACC system more muscle capability.

Examples disclosed herein use an actuator to house the piezoelectric material. The actuator achieves clearance by converting the linear displacement of the piezoelectric material (e.g., axial displacement, axial/radial combined displacement, and circumferential displacement) into a radial directional clearance control and adjustment. In examples disclosed herein, the direction of the actuator (e.g., axial direction, axial/radial combined direction, and circumferential direction) achieves radial clearance control by mechanical conversion of the displacement of the piezoelectric material. Examples disclosed herein do not need additional clearance margin for maximum transient closure or hot-rotor condition like the conventional ACC system. Examples disclosed herein provide significant specific fuel consumption (SFC) improvement on tighter clearance and a better EGT control as there are no additional margins for transient closure or the hot rotor condition.

In the examples disclosed herein, the actuator for the piezoelectric material can provide a variety of design spaces with compact and simple piezo-stacks while providing the same high mechanical force as conventional ACC. Example disclosed herein propose five different mechanical design configurations for how to move the actuator and stack/locate the piezoelectric material in the actuator(s): (1) an axial actuator outside of a high pressure turbine (HPT) case or a compressor case (2) an axial actuator inside of hanger hooks (3) an axial actuator inside of hanger hooks with springs (4) an axial/radial combined actuator inside of hanger hooks, and (5) a circumferential actuator inside of the hanger. The example first mechanical design configuration includes an outer-stack piezoelectric actuator that generates a linear displacement from an applied electric field. The first mechanical design configuration includes the piezoelectric actuator in an axial direction that mechanically converts the linear displacement of the piezoelectric stack in the axial direction into a radial direction for clearance control (e.g., via a rod). The first mechanical design configuration has the benefit for easy access for maintenance and part replacement since the piezoelectric actuator is located outside the case (e.g., the HPT case, the compressor case, etc.). However, it also includes sealing concerns for the case. As the piezoelectric stack is located outside of the case, the first mechanical design configuration preserves the piezoelectric material in a cold condition, which reduces concern of temperature limitations for the piezoelectric material.

The example second mechanical design configuration includes an inner-stack piezoelectric actuator that applies two axial actuators on hanger hooks under the case. Each of the axial actuators of the second mechanical design configuration include two stacks of piezoelectric material on opposite sides of the hanger hooks to displace the hanger in an axial direction, which is mechanically converted to

clearance control in the radial direction. The piezoelectric stacks are positioned on left and right surfaces of the hanger hooks to achieve more accurate modulation, and the second mechanical design configuration relatively reduces the concern for sealing present in the first mechanical design configuration. However, this second mechanical design configuration does not allow easy access for maintenance or part replacement compared to the first mechanical design configuration.

The third mechanical design configuration includes an inner-stack piezoelectric actuator that applies two axial actuators on hanger hooks under the case. The two axial actuators include inner-stacks of piezoelectric materials on a right surface of the hanger hooks and springs on the left surface of the hanger hooks. Each of the axial actuators of the include a stack of piezoelectric material and a spring on opposite sides of the hanger hook to displace the hanger in an axial direction, which is mechanically converted to clearance control in the radial direction. The third mechanical design configuration is a similar design to the second mechanical design configuration except including springs. The third mechanical design configuration needs less piezoelectric material stacks for cost, but it may cause uncertainty of modulation accuracy. The third mechanical design configuration also has the disadvantage for maintenance or part replacement compared to the first mechanical design configuration.

The example fourth mechanical design configuration includes an inner-stack piezoelectric actuator that applies two axial/radial combined actuators on hanger hooks under the case. Each of the axial/radial combined actuators include four stacks of piezoelectric material on all surfaces of the hanger hooks (e.g., left surface, right surface, upper surface, and lower surface) to displace the hanger in an axial direction and a radial direction, which are combined and mechanically converted to clearance control in the radial direction. The piezoelectric stacks are positioned on the left, right, upper, and lower surfaces of the hanger hooks to achieve a stronger and more accurate modulation, and the fourth mechanical design configuration relatively reduces the concern for sealing present in the first mechanical design configuration. However, this fourth mechanical design configuration does not allow easy access for maintenance or part replacement compared to the first mechanical design configuration. Furthermore, the more piezoelectric material stacks included in the fourth mechanical design configuration increases the cost.

The example fifth mechanical design configuration includes an inner-stack piezoelectric actuator that applies circumferential displacement on the under the case. The circumferential actuator of the fifth mechanical design configuration is located inside of the hanger body below the hanger hooks, and the circumferential actuator includes two stacks of piezoelectric materials on a right surface and a left surface of a rod connected to between the circumferential actuator and the case. In the example fifth mechanical design configuration, the hanger includes slanted slots (slope angle) that are connected to pins in the case. The displacement of the piezoelectric materials in the circumferential actuator move the hanger along the slanted slots to mechanically convert the circumferential displacement of the hanger into clearance control in the radial direction. The fifth mechanical design configuration provides circumferential gap adjustment between the hanger/shroud and the blade tip while also providing radial clearance control using the piezoelectric material displacement combine with the slanted slots and pins of the hanger and case, respectively.

However, the fifth mechanical design configuration also has the disadvantage for maintenance or part replacement.

Certain examples provide an engine controller, referred to as a full authority digital engine (or electronics) control (FADEC). The FADEC includes a digital computer, referred to as an electronic engine controller (EEC) or engine control unit (ECU), and related accessories that control aspects of aircraft engine performance. The FADEC can be used with a variety of engines such as piston engines, jet engines, other aircraft engines, etc. In certain examples, the EEC/ECU is provided separate from the FADEC, allowing manual override or intervention by a pilot and/or other operator.

In examples disclosed herein, the engine controller receives values for a plurality of input variables relating to flight condition (e.g., air density, throttle lever position, engine temperatures, engine pressures, etc.). The engine controller computes engine operating parameters such as fuel flow, stator vane position, air bleed valve position, etc., using the flight condition data. The engine operating parameters can be used by the engine controller to control operation of the piezoelectric actuator(s) to modulate blade tip clearance in the turbine engine.

Reference now will be made in detail to examples of the present disclosure, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation only, not limitation of the present disclosure. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the preferred embodiments without departing from the scope or spirit of the present disclosure. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the examples disclosed and described herein cover such modifications and variations as come within the scope of the appended claims and their equivalents.

FIG. 1 is a schematic cross-sectional view of a conventional turbofan-type gas turbine engine **100** ("turbofan **100**"). As shown in FIG. 1, the turbofan **100** defines a longitudinal or axial centerline axis **102** extending there-through for reference. In general, the turbofan **100** may include a core turbine or gas turbine engine **104** disposed downstream from a fan section **106**.

The core turbine **104** generally includes a substantially tubular outer casing **108** that defines an annular inlet **110**. The outer casing **108** can be formed from a single casing or multiple casings. The outer casing **108** encloses, in serial flow relationship, a compressor section having a booster or low pressure compressor **112** ("LP compressor **112**") and a high pressure compressor **114** ("HP compressor **114**"), a combustion section **116**, a turbine section having a high pressure turbine **118** ("HP turbine **118**") and a low pressure turbine **120** ("LP turbine **120**"), and an exhaust section **122**. A high pressure shaft or spool **124** ("HP shaft **124**") drivingly couples the HP turbine **118** and the HP compressor **114**. A low pressure shaft or spool **126** ("LP shaft **126**") drivingly couples the LP turbine **120** and the LP compressor **112**. The LP shaft **126** may also couple to a fan spool or shaft **128** of the fan section **106**. In some examples, the LP shaft **126** may couple directly to the fan shaft **128** (i.e., a direct-drive configuration). In alternative configurations, the LP shaft **126** may couple to the fan shaft **128** via a reduction gearbox **130** (i.e., an indirect-drive or geared-drive configuration).

As shown in FIG. 1, the fan section **106** includes a plurality of fan blades **132** coupled to and extending radially outwardly from the fan shaft **128**. An annular fan casing or

nacelle **134** circumferentially encloses the fan section **106** and/or at least a portion of the core turbine **104**. The nacelle **134** is supported relative to the core turbine **104** by a plurality of circumferentially-spaced apart outlet guide vanes **136**. Furthermore, a downstream section **138** of the nacelle **134** can enclose an outer portion of the core turbine **104** to define a bypass airflow passage **140** therebetween.

As illustrated in FIG. 1, air **142** enters an inlet portion **144** of the turbofan **100** during operation thereof. A first portion **146** of the air **142** flows into the bypass airflow passage **140**, while a second portion **148** of the air **142** flows into the inlet **110** of the LP compressor **112**. One or more sequential stages of LP compressor stator vanes **150** and LP compressor rotor blades **152** coupled to the LP shaft **126** progressively compress the second portion **148** of the air **142** flowing through the LP compressor **112** en route to the HP compressor **114**. Next, one or more sequential stages of HP compressor stator vanes **154** and HP compressor rotor blades **156** coupled to the HP shaft **124** further compress the second portion **148** of the air **142** flowing through the HP compressor **114**. This provides compressed air **158** to the combustion section **116** where it mixes with fuel and burns to provide combustion gases **160**.

The combustion gases **160** flow through the HP turbine **118** in which one or more sequential stages of HP turbine stator vanes **162** and HP turbine rotor blades **164** coupled to the HP shaft **124** extract a first portion of kinetic and/or thermal energy from the combustion gases **160**. This energy extraction supports operation of the HP compressor **114**. The combustion gases **160** then flow through the LP turbine **120** where one or more sequential stages of LP turbine stator vanes **166** and LP turbine rotor blades **168** coupled to the LP shaft **126** extract a second portion of thermal and/or kinetic energy therefrom. This energy extraction causes the LP shaft **126** to rotate, thereby supporting operation of the LP compressor **112** and/or rotation of the fan shaft **128**. The combustion gases **160** then exit the core turbine **104** through the exhaust section **122** thereof.

Along with the turbofan **100**, the core turbine **104** serves a similar purpose and operates in a similar environment in land-based gas turbines, turbojet engines in which the ratio of the first portion **146** of the air **142** to the second portion **148** of the air **142** is less than that of a turbofan, and unducted fan engines in which the fan section **106** is devoid of the nacelle **134**. In each of the turbofan, turbojet, and unducted engines, a speed reduction device (e.g., the reduction gearbox **130**) may be included between any shafts and spools. For example, the reduction gearbox **130** may be disposed between the LP shaft **126** and the fan shaft **128** of the fan section **106**.

FIG. 2 is a schematic cross-sectional view of an example gas turbine engine (such as the example turbofan **100**) with a conventional active clearance control (ACC) system **200**. The ACC system **200** includes an example main pipe **205**, an example high pressure turbine **210**, an example low pressure turbine **215**, example manifolds **220A**, **220B**, **220C**, example flanges **225A**, **225B**, and example mid-rings **230A**, **230B**. In the illustrated example of FIG. 2, air from a fan (e.g., from the fan section **106**) enters the main pipe **205**, where the airflow in the main pipe **205** is shown by the arrows in FIG. 2. In some examples, the inlet of the main pipe **205** is located at the fan (e.g., the fan section **106** of FIG. 1) or upstream of a compressor (e.g., the HP compressor **114** of FIG. 1) for the high pressure turbine **210**. In some examples, the ACC system **200** is applicable for a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. 1) and the low pressure turbine **215**. The main pipe **205**

delivers the air from the fan to the manifolds **220A**, **220B**, **220C**. The manifolds **220A**, **220B**, **220C** evenly distribute the air from the fan to the high pressure turbine **210** and the low pressure turbine **215**. In some examples, the high pressure turbine **210** is substantially similar to the HP turbine **118** and the low pressure turbine **215** is substantially similar to the LP turbine **120**. The flanges **225A**, **225B** and mid-rings **230A**, **230B** are joined to the outer surfaces of the high pressure turbine **210** case and the low pressure turbine **215** case. The flanges **225A**, **225B** and mid-rings **230A**, **230B** are configured to contract radially inward and/or expand radially outward in responses to changes in temperature (e.g., changes in temperature caused by the air from the manifolds **220A**, **220B**, **220C**). In some examples, at least some of the air is directed to impinge on the surfaces of the flanges **225A**, **225B** and mid-rings **230A**, **230B**. In some examples, the contraction inward and expansion outward of the flanges **225A**, **225B** and the mid-rings **230A**, **230B** can change blade tip clearances in the high pressure turbine **210** and the low pressure turbine **215**.

FIG. 3 is a schematic cross-sectional view of a prior ACC system **300** for the example gas turbine engine **100** of FIG. 1. The prior ACC system **300** includes a case **305**, guiding hooks **310A**, **310B**, a hanger **315**, a shroud **320**, and a blade **325**. In the illustrated example of FIG. 3, the case **305** is the casing surrounding either the HP turbine **118**, the LP turbine **120**, and/or the compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. 1). The case **305** includes the guiding hooks **310A**, **310B**, wherein the guiding hooks **310A**, **310B** connect the case **305** to the hanger **315**. The hanger **315** is connected to the shroud **320**.

In the illustrated example of FIG. 3, the prior ACC system **300** determines the clearance between the shroud **320** and the blade **325**. The arrows **330A-330D** in the prior ACC system **300** are representative of the cooling airflow from the main pipe **205** and manifolds **220A**, **220B**, **220C** of the example FIG. 2. The prior ACC system **300** controls the movement of the shroud **320** in only one direction (e.g., inward towards the blade **325**). The prior ACC system **300** uses the cooling airflow from the compressor or fan to cool the case **305**. The case **305** shrinks (e.g., moves inward) as it is cooled by the airflow. The case **305** moves the hanger **315** and shroud **320** inward towards the blade **325**. The prior ACC system **300** is unable to move the case **305**, the hanger **315**, and the shroud **320** for expansion. For example, the ACC system **300** is unable to expand the case **305** (e.g., move outward) to increase the clearance between the shroud **320** and the blade **325**. In such examples, the prior ACC system **300** waits for clearance between the shroud **320** and the blade **325** to open (e.g., increase). The prior ACC system **300** does not provide bi-directional control of the clearance between the shroud **320** and the blade **325**.

In some examples (e.g., the prior ACC system **300** of FIG. 3), an ACC system directs airflow around the case of an engine to control clearance between the case and the blade tip. For example, the ACC system controls the cooling airflow (represented as arrows **330A-330D** in FIG. 3) from a compressor or fan to the case **305**. In some examples, the ACC system mixes hot and cold air from a compressor and a bypass duct (contains turbofan airflow that bypassed the engine core) respectively to a desired temperature. In some examples, the ACC system helps to maintain and adjust the clearance between the engine case and the blade tip in prior ACC systems. In prior ACC systems (e.g., the prior ACC system **300** of FIG. 3), cooling airflow around the engine case (e.g., case **305**) adjusts the clearance by controlling the thermal expansion and contraction of the case. In some

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examples, the ACC system controls the cooling airflow to either contract or expand the turbine engine case. For example, the prior ACC system 300 directs cooling airflow to the case 305 to contract the case 305 and restricts the cooling airflow to the case 305 to expand the case 305. The ACC system controls the cooling airflow to adjust the clearance to compensate any changes in the blade of the turbine engine. In some examples, the ACC system is controlled by a controller in the turbine engine (e.g., the FADEC). The FADEC sends electrical control signals to the ACC system to signal the ACC system to modulate the airflow to control the case thermal expansion. The ACC system ultimately controls the amount of cooling airflow to manage the turbine engine casing temperatures, thereby adjusting the blade tip clearance.

FIGS. 4A and 4B are schematic cross-sectional views of a first example ACC system 400, 465 including an example axial, outer-stack piezoelectric actuator in accordance with teachings disclosed herein. An example ACC system 400 of FIG. 4A illustrates an example actuator 405 in an open clearance state. The example ACC system 400 of FIG. 4A includes an actuator 405, a first rod 410, a second rod 415, a case 420, outer sealing 430A, 430B, a hanger 435, a shroud 440, inner sealing 445A, 445B, and a blade 450. The actuator 405 includes a multilayer piezoelectric stack 455, for example. The example ACC system 400 of FIG. 4A includes an open clearance 460 between the shroud 440 and the blade 450.

FIG. 4B shows an alternative implementation of an ACC system 465. The example ACC system 465 of FIG. 4B illustrates the example actuator 405 in a closed/tight clearance state. The example ACC system 465 includes the actuator 405, the first rod 410, the second rod 415, the case 420, the outer sealing 430A, 430B, the hanger 435, the shroud 440, the inner sealing 445A, 445B, and the blade 450 of FIG. 4A. The actuator 405 of FIG. 4B includes the multilayer piezoelectric stack 455 which is expanded (or elongated) in the axial direction and contracted in the radial direction. The ACC system 465 of FIG. 4B includes a tight/closed clearance 470 between the shroud 440 and the blade 450. In examples disclosed herein, the case 420 includes the guiding hooks 425A, 425B, 425C, wherein the guiding hooks 425A, 425B, 425C connect the case 420 to the hanger 435. The hanger 435 is connected to the shroud 440.

In the illustrated examples of FIGS. 4A and 4B, the actuator 405 is located outside of the case 420 and oriented in an axial direction. In some examples, the case 420 is a case surrounding a high pressure turbine (e.g., the HP turbine 118 of FIG. 1), a low pressure turbine (e.g., the LP turbine 120 of FIG. 1), and/or a compressor (e.g., the HP compressor 114 and LP compressor 112 of FIG. 1). In some examples, locating the actuator 405 outside of the case 420 prevents material temperature limitations from affecting the actuator 405. For example, hot gas temperatures in a high pressure turbine such as the HP turbine 118 of FIG. 1, could cause material limitations for the actuator 405 if the actuator 405 was located inside the case 420. In the example ACC systems 400 and 465, the actuator 405 includes a multilayer stack of piezoelectric material 455. In some examples, the piezoelectric material of the multilayer stack of piezoelectric material 455 includes quartz, topaz, etc. However, other piezoelectric materials or other materials that generate linear displacement, such as shape memory alloy (SMA) materials, etc., can be additionally and/or alternatively included. In some examples, locating the actuator 405 and the multilayer stack of piezoelectric material 455 outside of the case 420

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helps to preserve the piezoelectric material in a cold condition without concern of temperature limitations. The location of the actuator 405 and the multilayer stack of piezoelectric material 455 provide a benefit of easy access for maintenance and part replacement, for example.

In the illustrated examples of FIGS. 4A and 4B, the multilayer stack of piezoelectric material 455 is connected to the first rod 410. The first rod 410 is connected to the multilayer stack of piezoelectric material 455 and the second rod 415. The second rod 415 is connected to the hanger 435 through the case 420. Since the actuator 405 and the multilayer stack of piezoelectric material 455 are located outside of the case 420, the second rod 415 is inserted through the case to connect to the first rod 410 (which is connected to the multilayer stack of piezoelectric material 455) and the hanger 435. In some examples, the opening in the case 420 for the second rod 415 to be inserted through introduces possible leakage through the case 420. In such examples, the second rod 415 is surrounded by the outer sealing 430A, 430B and the inner sealing 445A, 445B to seal the opening in the case 420 that the second rod 415 is inserted through. For example, the second rod 415 is surrounded by the outer sealing 430A, 430B to seal the opening in the case 420, and the second rod 415 is surrounded by the inner sealing 445A, 445B to seal the opening between the guiding hooks 425B and 425C.

In the illustrated examples of FIGS. 4A and 4B, the multilayer stack of piezoelectric material 455 generates a linear displacement of the first rod 410 and the second rod 415 from an electrical signal generated by an example controller (not shown). An example implementation of the controller that generates the electrical signal is illustrated in FIG. 12, which is described in further detail below. In some examples, the first rod 410 moves the second rod 415 in an axial direction using the linear displacement generated by the multilayer stack of piezoelectric material 455. In some examples, the second rod 415 moves the hanger 435 in an axial direction using the linear displacement from the first rod 410 (generated by the multilayer stack of piezoelectric material 455). In the illustrated example, the hanger 435 and the shroud 440 are connected and move together. Therefore, in the illustrated example, the second rod 415 moves the hanger 435 and the shroud 440 using the linear displacement generated by the multilayer stack of piezoelectric material 455. In some examples, the ACC system 400 includes the shroud 440 without the hanger 435. In such examples, the second rod 415 moves the shroud 440 using the linear displacement from the first rod 410 (generated by the multilayer stack of piezoelectric material 455). In some examples, the range of the linear displacement is increased by adding more layers of piezoelectric material to the multilayer stack of piezoelectric material 455. For example, adding layers in the multilayer stack of piezoelectric material 455, increase the axial movement range and muscle capability for the ACC system 400, 465.

In the illustrated example of FIG. 4A, the ACC system 400 has an open clearance represented by the open clearance 460 between the shroud 440 and the blade 450. The multilayer stack of piezoelectric material 455 included in the actuator 405 controls the open clearance 460. In the ACC system 400, the actuator 405 receives a first electrical signal from an example controller, and the actuator 405 provides the first electrical signal to the multilayer stack of piezoelectric material 455. The first electrical signal causes a linear displacement of the multilayer stack of piezoelectric material 455 (e.g., each stack in the multilayer stack of piezoelectric material 455 is long and thin as seen in the

example FIG. 4A). The linear displacement of the multilayer stack of piezoelectric material **455** moves the first rod **410** in an axial direction (e.g., leftward). The first rod **410** moves the second rod **415** in the axial direction (e.g., leftward), which moves the hanger **435** and the shroud **440** in the axial direction (e.g., leftward) away from the blade **450**. In examples disclosed herein, and edge of the blade **450** near the shroud **440** is angled to control the clearance. In the illustrated example of FIG. 4A, the second rod **415** moves the hanger **435** and the shroud **440** in the axial direction (e.g., leftward) away from the edge of the blade **450**, which increases the open clearance **460** in the radial direction.

The example ACC system **465** includes a tight/closed clearance, indicated by the tight clearance **470** between the shroud **440** and the blade **450** shown in FIG. 4B. The multilayer stack of piezoelectric material **455** included in the actuator **405** controls the tight clearance **470**. In the ACC system **465**, the actuator **405** receives a second electrical signal from an example controller, and the actuator **405** provides the second electrical signal to the multilayer stack of piezoelectric material **455**. The second electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **455** (e.g., each stack in the multilayer stack of piezoelectric material **455** is short and thick as seen in the example FIG. 4B). The linear displacement of the multilayer stack of piezoelectric material **455** moves the first rod **410** in an axial direction (e.g., rightward). The first rod **410** moves the second rod **415** in the axial direction (e.g., rightward), which moves the hanger **435** and the shroud **440** in the axial direction (e.g., rightward) toward the blade **450**. In the illustrated example of FIG. 4B, the second rod **415** moves the hanger **435** and the shroud **440** in the axial direction (e.g., rightward) toward the edge of the blade **450**, which decreases the tight clearance **470** in the radial direction.

In the illustrated examples of FIGS. 4A and 4B, the actuator **405** adjusts the clearance in two directions (e.g., shrinkage and expansion). The actuator **405** can be installed for an individual shroud (e.g., the shroud **440**), partial groups of shrouds (e.g., for groups of three shrouds, for groups of five shrouds, etc.), or for an entire group of shrouds in a turbine (e.g., the shrouds surrounding the 360 degree inner surface of the case **420**).

FIGS. 5A and 5B are schematic cross-sectional views of a second example ACC system **500**, **575** with an example axial, inner-stack piezoelectric actuator in accordance with teachings disclosed herein. The example ACC system **500** of FIG. 5A includes a case **505**, guiding hooks **510A**, **510B**, an actuator **515**, an actuator **520**, a hanger **525**, hanger extensions **530A**, **530B**, sealant **535A**, **535B**, a shroud **540**, and a blade **545**. The actuator **515** includes a multilayer stack of piezoelectric material **550** and a multilayer stack of piezoelectric material **555**. The actuator **520** includes a multilayer stack of piezoelectric material **560** and a multilayer stack of piezoelectric material **565**. The ACC system **500** includes an open clearance **570** between the shroud **540** and the blade **545**. The example ACC system **575** of FIG. 5B includes the case **505**, the guiding hooks **510A**, **510B**, the actuator **515**, the actuator **520**, the hanger **525**, the hanger extensions **530A**, **530B**, the sealant **535A**, **535B**, the shroud **540**, and the blade **545** of FIG. 5A. The actuator **515** of FIG. 5B includes the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **555**. The actuator **520** of FIG. 5B includes the multilayer stack of piezoelectric material **560** and the multilayer stack of piezoelectric material **565**. The example ACC system **575** includes a tight clearance **580** between the shroud **540** and

the blade **545**. The case **505** includes the guiding hooks **510A**, **510B**, wherein the guiding hooks **510A**, **510B** connect the actuator **515** and the actuator **520** to the hanger **525**. The hanger **525** is connected to the shroud **540**.

In the illustrated examples of FIGS. 5A and 5B, the actuator **515** is located under the case **505** on the guiding hook **510A**, and the actuator **520** is located under the case **505** on the guiding hook **510B**. In some examples, the case **505** is a case surrounding a high pressure turbine (e.g., the HP turbine **118** of FIG. 1), a low pressure turbine (e.g., the LP turbine **120** of FIG. 1) or a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. 1). In some examples, locating the actuator **515** and the actuator **520** under the case **505** reduces sealing concerns prevalent in the example ACC systems **400** and **465** of FIGS. 4A and 4B respectively, as described above. In the illustrated examples of FIGS. 5A and 5B, the actuator **515** and the actuator **520** are surrounded by the sealant **535A**, **535B**, respectively, to seal the opening in the guiding hooks **510A**, **510B** that the hanger extensions **530A**, **530B** are inserted through. However, the location of the actuator **515** and the actuator **520** prevents easy access for maintenance and part replacement.

In the illustrated examples of FIGS. 5A and 5B, the hanger **525** extends into the actuator **515** and the actuator **520** via the hanger extensions **530A**, **530B**. The multilayer stacks of piezoelectric material **550**, **555**, **560**, **565** are connected to the hanger extensions **530A**, **530B**. The multilayer stack of piezoelectric material **550** is connected to a side (left) surface of the hanger extension **530A** in the actuator **515**. The multilayer stack of piezoelectric material **555** is connected to an opposite side (right) surface of the hanger extension **530A** in the actuator **515**. The multilayer stack of piezoelectric material **560** is connected to a side (left) surface of the hanger extension **530B** in the actuator **520**. The multilayer stack of piezoelectric material **565** is connected to an opposite side (right) surface of the hanger extension **530B** in the actuator **520**. In some examples, the piezoelectric material of the multilayer stacks of piezoelectric material **550**, **555**, **560**, **565** can include quartz, topaz, etc. However, other piezoelectric materials or other materials that generate linear displacement, such as shape memory alloy (SMA) materials, etc., can be additionally and/or alternatively included.

In the illustrated examples of FIGS. 5A and 5B, the multilayer stacks of piezoelectric material **550**, **555**, **560**, **565** generate a linear displacement of the hanger **525** from electrical signals generated by an example controller. An example controller that generates the electrical signal is illustrated in FIG. 12, which is described in further detail below. In the examples of FIGS. 5A and 5B, the hanger **525** and the shroud **540** are connected and move together. As such, the hanger **525** moves the shroud **540** in an axial direction using the linear displacement generated by the multilayer stacks of piezoelectric material **550**, **555**, **560**, **565**. In some examples, the ACC system **500** includes the shroud **540** without the hanger **525**. In such examples, the shroud **540** moves using the linear displacement generated by the multilayer stacks of piezoelectric material **550**, **555**, **560**, **565**. The multilayer stacks of piezoelectric material **550**, **555**, **560**, **565** are positioned on side surfaces (left and right) of the hanger extensions **530A**, **530B** in the actuator **515** and the actuator **520** to accurately modulate the linear displacement in the axial direction. In some examples, the range of the linear displacement is increased by adding more layers of piezoelectric material to the multilayer stacks of piezoelectric material **550**, **555**, **560**, **565**. For example, the more layers added in the multilayer stacks of piezoelectric

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material **550**, **555**, **560**, **565**, the more axial movement range and muscle capability for the ACC system.

The example ACC system **500** has an open clearance represented by the open clearance **570** between the shroud **540** and the blade **545**. The multilayer stacks of piezoelectric material **550**, **555**, **560**, **565** control the open clearance **570**. In the ACC system **500**, the actuator **515** and the actuator **520** receive a first electrical signal from an example controller. The actuator **515** provides the first electrical signal to the multilayer stack of piezoelectric material **550**, and actuator **520** provides the first electrical signal to the multilayer stack of piezoelectric material **560**. The first electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **550** (e.g., each stack in the multilayer stack of piezoelectric material **550** is long and thin as seen in the example FIG. 5A) and the multilayer stack of piezoelectric material **560** (e.g., each stack in the multilayer stack of piezoelectric material **560** is long and thin as seen in the example FIG. 5A).

In the ACC system **500**, the actuator **515** and the actuator **520** receive a second electrical signal from an example controller. In some examples, the actuator **515** and the actuator **520** receive the first electrical signal and the second electrical signal at the same time or at substantially the same time given transmission delay (e.g., in parallel). The actuator **515** provides the second electrical signal to the multilayer stack of piezoelectric material **555**, and actuator **520** provides the second electrical signal to the multilayer stack of piezoelectric material **565**. The second electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **555** (e.g., each stack in the multilayer stack of piezoelectric material **555** is short and thick as seen in the example FIG. 5A) and the multilayer stack of piezoelectric material **565** (e.g., each stack in the multilayer stack of piezoelectric material **565** is short and thick as seen in the example FIG. 5A). The linear displacement of the multilayer stacks of piezoelectric material **550**, **555**, **560**, **565** in the axial direction move the hanger **525** and shroud **540** the axial direction (e.g., leftward), which increases the distance between the shroud **540** and the blade **545** in the radial direction and increases the open clearance **570**.

In the example of FIG. 5B, the ACC system **575** has a tight/closed clearance represented by the tight clearance **580** between the shroud **540** and the blade **545**. The multilayer stacks of piezoelectric material **550**, **555**, **560**, **565** control the tight clearance **580**. In the ACC system **575**, the actuator **515** and the actuator **520** receive a third electrical signal from an example controller. The actuator **515** provides the third electrical signal to the multilayer stack of piezoelectric material **550**, and actuator **520** provides the third electrical signal to the multilayer stack of piezoelectric material **560**. The third electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **550** (e.g., each stack in the multilayer stack of piezoelectric material **550** is short and thick as seen in the example FIG. 5B) and the multilayer stack of piezoelectric material **560** (e.g., each stack in the multilayer stack of piezoelectric material **560** is short and thick as seen in the example FIG. 5B).

In the ACC system **575**, the actuator **515** and the actuator **520** receive a fourth electrical signal from an example controller. In some examples, the actuator **515** and the actuator **520** receive the third electrical signal and the fourth electrical signal at the same time or at substantially the same time given transmission delay (e.g., in parallel). The actuator **515** provides the fourth electrical signal to the multilayer stack of piezoelectric material **555**, and actuator **520** provides the fourth electrical signal to the multilayer stack of

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piezoelectric material **565**. The fourth electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **555** (e.g., each stack in the multilayer stack of piezoelectric material **555** is long and thin as seen in the example FIG. 5B) and the multilayer stack of piezoelectric material **565** (e.g., each stack in the multilayer stack of piezoelectric material **565** is long and thin as seen in the example FIG. 5B). The linear displacement of the multilayer stacks of piezoelectric material **550**, **555**, **560**, **565** in the axial direction move the hanger **525** and shroud **540** the axial direction (e.g., rightward), which decreases the distance between the shroud **540** and the blade **545** in the radial direction and decreases the tight clearance **580**.

In the illustrated examples of FIGS. 5A and 5B, the actuator **515** and the actuator **520** adjust the clearance between the shroud **540** and the blade **545** in two directions (e.g., shrinkage and expansion). The actuator **515** and the actuator **520** can be installed for an individual shroud (e.g., the shroud **540**), partial groups of shrouds (e.g., for groups of three shrouds, for groups of five shrouds, etc.), or for an entire group of shrouds in a turbine (e.g., the shrouds surrounding the 360 degree inner surface of the case **505**).

FIGS. 6A and 6B are schematic cross-sectional views of a third example ACC system **600**, **675** with an example axial, inner-stack piezoelectric actuator and spring load in accordance with teachings disclosed herein. The example ACC system **600** of FIG. 6A includes an example case **605**, example guiding hooks **610A**, **610B**, an example actuator **615**, an example actuator **620**, an example hanger **625**, example hanger extensions **630A**, **630B**, example sealant **635A**, **635B**, an example shroud **640**, and an example blade **645**. The actuator **615** includes an example spring **650** and an example stack of piezoelectric material **655**. The actuator **620** includes an example spring **660** and an example stack of piezoelectric material **665**. The ACC system **600** includes an example open clearance **670** between the shroud **640** and the blade **645**. The example ACC system **675** of FIG. 6B includes the case **605**, the guiding hooks **610A**, **610B**, the actuator **615**, the actuator **620**, the hanger **625**, the hanger extensions **630A**, **630B**, the sealant **635A**, **635B**, the shroud **640**, and the blade **645** of FIG. 6A. The actuator **615** of FIG. 6B includes the spring **650** and an example stack of piezoelectric material **655**. The example actuator **620** of FIG. 6B includes the spring **660** and an example stack of piezoelectric material **665**. The ACC system **675** includes an example tight clearance **680** between the shroud **640** and the blade **645**.

In the illustrated examples of FIGS. 6A and 6B, the actuator **615** is located under the case **605** on the guiding hook **610A**, and the actuator **620** is located under the case **605** on the guiding hook **610B**. In some examples, the case **605** is a case surrounding a high pressure turbine (e.g., the HP turbine **118** of FIG. 1), a low pressure turbine (e.g., the LP turbine **120** of FIG. 1), or a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. 1). In some examples, locating the actuator **615** and the actuator **620** under the case **605** reduces sealing concerns prevalent in the example ACC systems **400** and **465** of FIGS. 4A and 4B respectively, as described above. In the illustrated examples of FIGS. 6A and 6B, the actuator **615** and the actuator **620** are surrounded by the sealant **635A**, **635B**, respectively, to seal the opening in the guiding hooks **610A**, **610B** that the hanger extensions **630A**, **630B** are inserted through. However, the location of the actuator **615** and the actuator **620** prevents easy access for maintenance and part replacement.

In the example ACC systems **600** and **675**, the actuator **615** includes the spring **650** and the multilayer stack of

piezoelectric material **655**. In the example ACC systems **600** and **675**, the actuator **620** includes the spring **660** and the multilayer stack of piezoelectric material **665**. In some examples, the piezoelectric material of the multilayer stacks of piezoelectric material **655**, **665** can include quartz, topaz, etc. However, other piezoelectric materials or other materials that generate linear displacement, such as shape memory alloy (SMA) materials, etc., can be additionally and/or alternatively included. The multilayer stack of piezoelectric material **655** and the multilayer stack of piezoelectric material **665** each receive control electrical signals to operate in the ACC systems **600** and **675**. The actuator **615** and the actuator **620** include springs **650**, **660** instead of additional multilayer stacks of piezoelectric material because the springs reduce the controls complexity for the actuators **615**, **620** (e.g., including the springs **650**, **660** allows for the actuator **615** and the actuator **620** to only have to receive one electrical control signal each). However, the springs **650**, **660** may cause uncertainty in linear displacement modulation in the example ACC system **600**, **675** as compared to the example ACC system **500**, **575**.

In the illustrated examples of FIGS. **6A** and **6B**, the hanger **625** extends into the actuator **615** and the actuator **620** via the hanger extensions **630A**, **630B**. The multilayer stacks of piezoelectric material **655**, **665** and the springs **650**, **660** are connected to the hanger extensions **630A**, **630B**. The spring **650** is connected to a side (left) surface of the hanger extension **630A** in the actuator **615**. The multilayer stack of piezoelectric material **655** is connected to an opposite side (right) surface of the hanger extension **630A** in the actuator **615**. The spring **660** is connected to a side (left) surface of the hanger extension **630B** in the actuator **620**. The multilayer stack of piezoelectric material **665** is connected to an opposite side (right) surface of the hanger extension **630B** in the actuator **620**.

In the illustrated examples of FIGS. **6A** and **6B**, the multilayer stacks of piezoelectric material **655**, **665** generate a linear displacement of the hanger **625** in an axial direction from electrical signals generated by an example controller. An example controller that generates the electrical signal is illustrated in FIG. **12**, which is described in further detail below. The springs **650**, **660** provide a load for the side (left) surfaces of the hanger extensions **630A**, **630B** based on the linear displacement of the multilayer stacks of piezoelectric material **655**, **665**. In the illustrated example of FIGS. **6A** and **6B**, the hanger **625** and the shroud **640** are connected and move together. As such, the hanger **625** moves the shroud **640** in an axial direction using the linear displacement generated by the multilayer stacks of piezoelectric material **655**, **665**. In some examples, the ACC system **600** includes the shroud **640** without the hanger **625**. In such examples, the shroud **640** moves using the linear displacement generated by the multilayer stacks of piezoelectric material **655**, **665**. The multilayer stacks of piezoelectric material **655**, **665** are positioned on side (right) surfaces of the hanger extensions **630A**, **630B** in the actuator **615** and the actuator **620** to accurately modulate the linear displacement in the axial direction. The springs **650**, **660** are positioned on the side (left) surfaces of the hanger extensions **630A**, **630B** in the actuator **615** and the actuator **620** to provide a spring load to the hanger **625** based on the linear displacement generated by the multilayer stacks of piezoelectric material **655**, **665**. In some examples, the range of the linear displacement is increased by adding more layers of piezoelectric material to the multilayer stacks of piezoelectric material **655**, **665**. For example, the more layers

added in the multilayer stacks of piezoelectric material **655**, **665** the more axial movement range and muscle capability for the ACC system.

In the illustrated example of FIG. **6A**, the ACC system **600** has an open clearance represented by the open clearance **670** between the shroud **640** and the blade **645**. The multilayer stacks of piezoelectric material **655**, **665** control the open clearance **670**. In the ACC system **600**, the actuator **615** and the actuator **620** receive a first electrical signal from an example controller. The actuator **615** provides the first electrical signal to the multilayer stack of piezoelectric material **655**, and actuator **620** provides the first electrical signal to the multilayer stack of piezoelectric material **665**. The first electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **655** (each stack in the multilayer stack of piezoelectric material **655** is short and thick as seen in the example FIG. **6A**) and the multilayer stack of piezoelectric material **665** (each stack in the multilayer stack of piezoelectric material **665** is short and thick as seen in the example FIG. **6A**). The springs **650**, **660** provide a spring load to match the linear displacement of the multilayer stacks of piezoelectric material **655**, **665**. For example, the springs **650**, **660** compress to provide a load to match the change in linear displacement from the multilayer stacks of piezoelectric material **655**, **665**. The linear displacement of the multilayer stacks of piezoelectric material **655**, **665** and the loads from the springs **650**, **660** in the axial direction move the hanger **625** and shroud **640** in the axial direction (e.g., leftward), which increases the distance between the shroud **640** and the blade **645** in the radial direction and increases the open clearance **670**.

In the illustrated example of FIG. **6B**, the ACC system **675** has a tight/closed clearance represented by the tight clearance **680** between the shroud **640** and the blade **645**. The multilayer stacks of piezoelectric material **655**, **665** and the springs **650**, **660** control the tight clearance **680**. In the ACC system **675**, the actuator **615** and the actuator **620** receive a second electrical signal from an example controller. The actuator **615** provides the second electrical signal to the multilayer stack of piezoelectric material **655**, and actuator **620** provides the second electrical signal to the multilayer stack of piezoelectric material **665**. The second electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **655** (e.g., each stack in the multilayer stack of piezoelectric material **655** is long and thin as seen in the example FIG. **6B**) and the multilayer stack of piezoelectric material **665** (e.g., each stack in the multilayer stack of piezoelectric material **665** is long and thin as seen in the example FIG. **6B**). The springs **650**, **660** provide a spring load to match the linear displacement of the multilayer stacks of piezoelectric material **655**, **665**. For example, the springs **650**, **660** extend to provide a load to match the change in linear displacement from the multilayer stacks of piezoelectric material **655**, **665**. The linear displacement of the multilayer stacks of piezoelectric material **655**, **665** and the loads from the springs **650**, **660** in the axial direction move the hanger **625** and shroud **640** in the axial direction (e.g., rightward), which decreases the distance between the shroud **640** and the blade **645** in the radial direction and decreases the tight clearance **680**.

In the illustrated examples of FIGS. **6A** and **6B**, the actuator **615** and the actuator **620** adjust the clearance between the shroud **640** and the blade **645** in two directions (e.g., shrinkage and expansion). The actuator **615** and the actuator **620** can be installed for an individual shroud (e.g., the shroud **640**), partial groups of shrouds (e.g., for groups of three shrouds, for groups of five shrouds, etc.), or for an

entire group of shrouds in a turbine (e.g., the shrouds surrounding the 360 degree inner surface of the case 605).

FIGS. 7A and 7B are schematic cross-sectional views of a fourth example ACC system 700, 790 with an example axial/radial combined piezoelectric actuator in accordance with teachings disclosed herein. The example ACC system 700 of FIG. 7A includes a case 705, guiding hooks 510A, 510B, an actuator 715, an actuator 720, a hanger 725, hanger extensions 730A, 730B, sealant 735A, 735B, a shroud 740, and a blade 745. The actuator 715 includes a multilayer stack of piezoelectric material 750, a multilayer stack of piezoelectric material 755, a multilayer stack of piezoelectric material 760, and a multilayer stack of piezoelectric material 765. The actuator 720 includes a multilayer stack of piezoelectric material 770, a multilayer stack of piezoelectric material 775, a multilayer stack of piezoelectric material 780, and a multilayer stack of piezoelectric material 782. The ACC system 700 includes an open clearance 785 between the shroud 740 and the blade 745. The example ACC system 790 of FIG. 7B includes the case 705, the guiding hooks 510A, 510B, the actuator 715, the actuator 720, the hanger 725, the hanger extensions 730A, 730B, the sealant 735A, 735B, the shroud 740, and the blade 745 of FIG. 7A. The actuator 715 of FIG. 7B includes the multilayer stack of piezoelectric material 750, the multilayer stack of piezoelectric material 755, the multilayer stack of piezoelectric material 760, and the multilayer stack of piezoelectric material 765. The actuator 720 of FIG. 7B includes the multilayer stack of piezoelectric material 770, the multilayer stack of piezoelectric material 775, the multilayer stack of piezoelectric material 780, and the multilayer stack of piezoelectric material 782. The example ACC system 790 includes a tight/closed clearance 795 between the shroud 740 and the blade 745. The case 705 includes the guiding hooks 710A, 710B, wherein the guiding hooks 710A, 710B connect the actuator 715 and the actuator 720 to the hanger 725. The hanger 725 is connected to the shroud 740.

In the illustrated examples of FIGS. 7A and 7B, the actuator 715 is located under the case 705 on the guiding hook 710A, and the actuator 720 is located under the case 705 on the guiding hook 710B. In some examples, the case 705 is a case surrounding a high pressure turbine (e.g., the HP turbine 118 of FIG. 1), a low pressure turbine (e.g., the LP turbine 120 of FIG. 1) or a compressor (e.g., the HP compressor 114 and LP compressor 112 of FIG. 1). In some examples, locating the actuator 715 and the actuator 720 under the case 705 reduces sealing concerns prevalent in the example ACC systems 400 and 465 of FIGS. 4A and 4B respectively, as described above. In the illustrated examples of FIGS. 7A and 7B, the actuator 715 and the actuator 720 are surrounded by the sealant 735A, 735B, respectively, to seal the opening in the guiding hooks 710A, 710B that the hanger extensions 730A, 730B are inserted through. However, the location of the actuator 715 and the actuator 720 prevents easy access for maintenance and part replacement.

In the illustrated examples of FIGS. 7A and 7B, the hanger 725 extends into the actuator 715 and the actuator 720 via the hanger extensions 730A, 730B. The multilayer stacks of piezoelectric material 750, 755, 760, 765, 770, 775, 780, 782 are connected to the hanger extensions 730A, 730B. The multilayer stack of piezoelectric material 750 is connected to a side (left) surface of the hanger extension 730A in the actuator 715. The multilayer stack of piezoelectric material 755 is connected to an opposite side (right) surface of the hanger extension 730A in the actuator 715. The multilayer stack of piezoelectric material 760 is connected to a top surface of the hanger extension 730A in the

actuator 715. The multilayer stack of piezoelectric material 765 is connected to a bottom surface of the hanger extension 730A in the actuator 715. The multilayer stack of piezoelectric material 770 is connected to a side (left) surface of the hanger extension 730B in the actuator 720. The multilayer stack of piezoelectric material 775 is connected to an opposite side (right) surface of the hanger extension 730B in the actuator 720. The multilayer stack of piezoelectric material 780 is connected to a top surface of the hanger extension 730B in the actuator 720. The multilayer stack of piezoelectric material 782 is connected to a bottom surface of the hanger extension 730B in the actuator 720. In some examples, the piezoelectric material of the multilayer stacks of piezoelectric material 750, 755, 760, 765, 770, 775, 780, 782 can include quartz, topaz, etc. However, other piezoelectric materials or other materials that generate linear displacement, such as shape memory alloy (SMA) materials, etc., can be additionally and/or alternatively included.

In the illustrated examples of FIGS. 7A and 7B, the multilayer stacks of piezoelectric material 750, 755, 760, 765, 770, 775, 780, 782 generate a linear displacement of the hanger 725 from electrical signals generated by an example controller. An example controller that generates the electrical signals is illustrated in FIG. 12, which is described in further detail below. In the examples of FIGS. 7A and 7B, the hanger 725 and the shroud 740 are connected and move together. As such, the hanger 725 moves the shroud 740 in an axial direction and a radial direction using the linear displacement generated by the multilayer stacks of piezoelectric material 750, 755, 760, 765, 770, 775, 780, 782. In some examples, the ACC system 700 includes the shroud 740 without the hanger 725. In such examples, the shroud 740 moves using the linear displacement generated by the multilayer stacks of piezoelectric material 750, 755, 760, 765, 770, 775, 780, 782. In the illustrated examples of FIGS. 7A and 7B, the actuator 715 and the actuator 720 include four stacks of piezoelectric material on all surfaces of the hanger extensions 730A, 730B (e.g., left surface, right surface, upper/top surface, and lower/bottom surface) to displace the hanger 725 in an axial direction and a radial direction. The piezoelectric stacks (e.g., the multilayer stacks of piezoelectric material 750, 755, 760, 765 in the actuator 715 and the multilayer stacks of piezoelectric material 770, 775, 780, 782 in the actuator 720) are positioned on the left, right, upper, and lower surfaces of the hanger extensions 730A, 730B to achieve a stronger and more accurate clearance modulation. In some examples, the range of the linear displacement is increased by adding more layers of piezoelectric material to the multilayer stacks of piezoelectric material 750, 755, 760, 765, 770, 775, 780, 782. For example, the more layers added in the multilayer stacks of piezoelectric material 750, 755, 760, 765, 770, 775, 780, 782, the more axial and radial movement range and muscle capability for the ACC system.

The example ACC system 700 has an open clearance represented by the open clearance 785 between the shroud 740 and the blade 745. The multilayer stacks of piezoelectric material 750, 755, 760, 765, 770, 775, 780, 782 control the open clearance 785. In the ACC system 700, the actuator 715 and the actuator 720 receive a first electrical signal from an example controller. The actuator 715 provides the first electrical signal to the multilayer stack of piezoelectric material 750 and the multilayer stack of piezoelectric material 760, and actuator 720 provides the first electrical signal to the multilayer stack of piezoelectric material 770 and the multilayer stack of piezoelectric material 780. The first electrical signal causes a linear displacement of the multi-

layer stack of piezoelectric material **750**, the multilayer stack of piezoelectric material **760**, the multilayer stack of piezoelectric material **770**, and the multilayer stack of piezoelectric material **780** (e.g., each stack in the multilayer stack of piezoelectric material **750**, the multilayer stack of piezoelectric material **760**, the multilayer stack of piezoelectric material **770**, and the multilayer stack of piezoelectric material **780** are long and thin as seen in the example FIG. 7A).

In the ACC system **700**, the actuator **715** and the actuator **720** receive a second electrical signal from an example controller. In some examples, the actuator **715** and the actuator **720** receive the first electrical signal and the second electrical signal at the same time or at substantially the same time given transmission delay (e.g., in parallel). The actuator **715** provides the second electrical signal to the multilayer stack of piezoelectric material **755** and the multilayer stack of piezoelectric material **765**, and actuator **720** provides the second electrical signal to the multilayer stack of piezoelectric material **775** and the multilayer stack of piezoelectric material **782**. The second electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **755**, the multilayer stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782** (e.g., each stack in the multilayer stack of piezoelectric material **755**, the multilayer stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782** are short and thick as seen in the example FIG. 7A). The linear displacement of the multilayer stacks of piezoelectric material **750**, **755**, **770**, **775** in the axial direction and the linear displacement of the multilayer stacks of piezoelectric material **760**, **765**, **780**, **782** in the radial direction move the hanger **525** and shroud **540** is the axial direction (e.g., leftward) and the radial direction (e.g., upward), which increases the distance between the shroud **740** and the blade **745** in the radial direction and increases the open clearance **785**.

In the example of FIG. 7B, example ACC system **790** has a tight/closed clearance represented by the tight clearance **795** between the shroud **740** and the blade **745**. The multilayer stacks of piezoelectric material **750**, **755**, **760**, **765**, **770**, **775**, **780**, **782** control the tight clearance **795**. In the ACC system **790**, the actuator **715** and the actuator **720** receive a third electrical signal from an example controller. The actuator **715** provides the third electrical signal to the multilayer stack of piezoelectric material **750** and the multilayer stack of piezoelectric material **760**, and actuator **720** provides the third electrical signal to the multilayer stack of piezoelectric material **770** and the multilayer stack of piezoelectric material **780**. The third electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **750**, the multilayer stack of piezoelectric material **760**, the multilayer stack of piezoelectric material **770**, and the multilayer stack of piezoelectric material **780** (e.g., each stack in the multilayer stack of piezoelectric material **750**, the multilayer stack of piezoelectric material **760**, the multilayer stack of piezoelectric material **770**, and the multilayer stack of piezoelectric material **780** are short and thick as seen in the example FIG. 7B).

In the ACC system **790**, the actuator **715** and the actuator **720** receive a fourth electrical signal from an example controller. In some examples, the actuator **715** and the actuator **720** receive the third electrical signal and the fourth electrical signal at the same time or at substantially the same time given transmission delay (e.g., in parallel). The actuator **715** provides the fourth electrical signal to the multilayer stack of piezoelectric material **755** and the multilayer stack

of piezoelectric material **765**, and actuator **720** provides the fourth electrical signal to the multilayer stack of piezoelectric material **775** and the multilayer stack of piezoelectric material **782**. The fourth electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **755**, the multilayer stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782** (e.g., each stack in the multilayer stack of piezoelectric material **755**, the multilayer stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782** are long and thin as seen in the example FIG. 7B). The linear displacement of the multilayer stacks of piezoelectric material **750**, **755**, **770**, **775** in the axial direction and the linear displacement of the multilayer stacks of piezoelectric material **760**, **765**, **780**, **782** in the radial direction move the hanger **525** and shroud **540** is the axial direction (e.g., rightward) and the radial direction (e.g., downward), which decreases the distance between the shroud **740** and the blade **745** in the radial direction and decreases the tight clearance **795**.

In the illustrated examples of FIGS. 7A and 7B, the actuator **715** and the actuator **720** adjust the clearance between the shroud **740** and the blade **745** in two directions (e.g., shrinkage and expansion). The actuator **715** and the actuator **720** can be installed for an individual shroud (e.g., the shroud **740**), partial groups of shrouds (e.g., for groups of three shrouds, for groups of five shrouds, etc.), or for an entire group of shrouds in a turbine (e.g., the shrouds surrounding the 360 degree inner surface of the case **705**).

FIGS. 8A and 8B are schematic aft looking forward (ALF) and forward looking aft (FLA) views of a fifth example ACC system **800**, **890** with an example circumferential piezoelectric actuator in accordance with teachings disclosed herein. An example ACC system **800** of FIG. 8A includes a case **805**, a hanger **810**, a shroud **815**, an actuator **820**, and a blade **870**. The actuator **820** includes a multilayer stack of piezoelectric material **825** and a multilayer stack of piezoelectric material **830**. The case **805** includes a rod **835**, a case extension **850**, a case extension **855**, and example holding bars **860**, **865**. The shroud hanger **810** includes example slots **840**, **845**. The ACC system **800** includes an open clearance **875** between the shroud **815** and the blade **870**. The example ACC system **890** of FIG. 8B includes the case **805**, the hanger **810**, the shroud **815**, the actuator **820**, and the blade **870** of FIG. 8A. The actuator **820** of FIG. 8B includes the multilayer stack of piezoelectric material **825** and the multilayer stack of piezoelectric material **830**. The case **805** of FIG. 8B includes the rod **835**, the case extension **850**, the case extension **855**, and the holding bars **860**, **865**, and the hanger **810** includes the slots **840**, **845** of FIG. 8A. The ACC system **890** includes a tight/closed clearance **895** between the shroud **815** and the blade **870**. The case **805** includes the case extension **850**, the case extension **855**, and the holding bars **860**, **865** to connect to the hanger **810**. The hanger **810** is connected to the shroud **815**.

In the illustrated examples of FIGS. 8A and 8B, the actuator **820** is located under the case **805** inside of the hanger **810**. In some examples, the case **805** is a case surrounding a high pressure turbine (e.g., the HP turbine **118** of FIG. 1), a low pressure turbine (e.g., the LP turbine **120** of FIG. 1) or a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. 1). In some examples, locating the actuator **820** under the case **805** reduces sealing concerns prevalent in the example ACC systems **400** and **465** of FIGS. 4A and 4B respectively, as described above. However, the

location of the actuator **820** prevents easy access for maintenance and part replacement.

In the illustrated examples of FIGS. **8A** and **8B**, the case **805** extends into the actuator **820** via the rod **835**. The multilayer stacks of piezoelectric material **825**, **830** are connected to the rod **835**. The multilayer stack of piezoelectric material **825** is connected to a side (left) surface of the rod **835** in the actuator **820**, and the multilayer stack of piezoelectric material **830** is connected to an opposite side (right) surface of the rod **835** in the actuator **820**. In some examples, the piezoelectric material of multilayer stacks of piezoelectric material **825**, **830** can include quartz, topaz, etc. However, other piezoelectric materials or other materials that generate linear displacement, such as shape memory alloy (SMA) materials, etc., can be additionally and/or alternatively included. In the illustrated examples of FIGS. **8A** and **8B**, the case **805** also overlaps the hanger **810** via the case extension **850** and the case extension **855**. The case extension **850** connects to the slot **840** in the hanger **810** via the holding bar **860**, and the case extension **855** connects to the slot **845** in the hanger **810** via the holding bar **865**.

In the illustrated examples of FIGS. **8A** and **8B**, the multilayer stacks of piezoelectric material **825**, **830** generate a linear displacement of the hanger **810** from electrical signals generated by an example controller. An example controller that generates the electrical signals is illustrated in FIG. **12**, which is described in further detail below. In some examples, the linear displacement of the multilayer stacks of piezoelectric material **825**, **830** moves the hanger **810** circumferentially along the slanted slots **840** and **845** to mechanically convert the circumferential displacement of the hanger into clearance control in the radial direction. The example actuator **820** of FIGS. **8A** and **8B** provides circumferential gap adjustment between the hanger **810**/shroud **815** and the blade **870** while also providing radial clearance control using the piezoelectric material displacement from the multilayer stacks of piezoelectric material **825**, **830** combined with the slanted slots **840**, **845** and holding bars **860**, **865** of the hanger **810** and the case **805**, respectively. In the examples of FIGS. **8A** and **8B**, the hanger **810** and the shroud **815** are connected and move together. As such, the hanger **810** moves the shroud **815** in a circumferential direction using the linear displacement generated by the multilayer stacks of piezoelectric material **825**, **830**. In some examples, the ACC system **800** includes the shroud **815** without the hanger **810**. In such examples, the shroud **815** moves using the linear displacement generated by the multilayer stacks of piezoelectric material **825**, **830**. The multilayer stacks of piezoelectric material **825**, **830** are positioned on side surfaces (left and right) of the rod **835** to accurately modulate the linear displacement in the circumferential direction. In some examples, the range of the linear displacement is increased by adding more layers of piezoelectric material to the multilayer stacks of piezoelectric material **825**, **830**. For example, the more layers added in the multilayer stacks of piezoelectric material **825**, **830**, the more circumferential movement range and muscle capability for the ACC system.

The example ACC system **800** has an open clearance represented by the open clearance **875** between the shroud **815** and the blade **870**. The multilayer stacks of piezoelectric material **825**, **830** control the open clearance **875**. In the ACC system **800**, the actuator **820** receives a first electrical signal from an example controller. The actuator **820** provides the first electrical signal to the multilayer stack of piezoelectric material **825**. The first electrical signal causes a linear displacement of the multilayer stack of piezoelectric

material **825** (e.g., each stack in the multilayer stack of piezoelectric material **825** is short and thick as seen in the example FIG. **8A**). In the ACC system **800**, the actuator **820** receives a second electrical signal from an example controller. In some examples, the actuator **820** receives the first electrical signal and the second electrical signal at the same time or at substantially the same time given transmission delay (e.g., in parallel). The actuator **820** provides the second electrical signal to the multilayer stack of piezoelectric material **830**. The second electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **830** (e.g., each stack in the multilayer stack of piezoelectric material **830** is long and thin as seen in the example FIG. **8A**).

In the example ACC system **800**, the linear displacement of the multilayer stacks of piezoelectric material **825**, **830** in a first axial direction (e.g., to the left) moves the hanger **810** along the slanted slots **840**, **845** with respect to the stationary case **805**. The case extension **850** includes the holding bar **860** to connect to the slot **840**, and the case extension **855** includes the holding bar **865** to connect to the slot **845**. When the hanger **810** is displaced by the linear displacement of the multilayer stacks of piezoelectric material **825**, **830**, the hanger **810** moves with the slots **840**, **845** along the holding bars **860**, **865**, respectively, in the first axial direction to displace the hanger **810** and the shroud **815** circumferentially in relation to the case **805**. The linear displacement of the multilayer stacks of piezoelectric material **825**, **830** in the first axial direction moves the hanger **810** and shroud **815** in the circumferential direction away from the blade **870**, which increases the distance between the shroud **815** and the blade **870** in the radial direction and increases the open clearance **875**.

In the example of FIG. **8B**, the ACC system **890** has a tight/closed clearance represented by the tight clearance **895** between the shroud **815** and the blade **870**. The multilayer stacks of piezoelectric material **825**, **830** control the tight clearance **895**. In the ACC system **890**, the actuator **820** receives a third electrical signal from an example controller. The actuator **820** provides the third electrical signal to the multilayer stack of piezoelectric material **825**. The third electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **825** (e.g., each stack in the multilayer stack of piezoelectric material **825** is long and thin as seen in the example FIG. **8B**). In the ACC system **890**, the actuator **820** receives a fourth electrical signal from an example controller. In some examples, the actuator **820** receives the third electrical signal and the fourth electrical signal at the same time or at substantially the same time given transmission delay (e.g., in parallel). The actuator **820** provides the fourth electrical signal to the multilayer stack of piezoelectric material **830**. The fourth electrical signal causes a linear displacement of the multilayer stack of piezoelectric material **830** (e.g., each stack in the multilayer stack of piezoelectric material **830** is short and thick as seen in the example FIG. **8B**).

In the example ACC system **890**, the linear displacement of the multilayer stacks of piezoelectric material **825**, **830** in a second axial direction (e.g., to the right) moves the hanger **810** along the slanted slots **840**, **845** with respect to the stationary case **805**. The case extension **850** includes the holding bar **860** to connect to the slot **840**, and the case extension **855** includes the holding bar **865** to connect to the slot **845**. When the hanger **810** is displaced by the linear displacement of the multilayer stacks of piezoelectric material **825**, **830**, the hanger **810** moves with the slots **840**, **845** along the holding bars **860**, **865**, respectively, in the second

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axial direction to displace the hanger **810** and the shroud **815** circumferentially in relation to the case **805**. The linear displacement of the multilayer stacks of piezoelectric material **825**, **830** in the second axial direction moves the hanger **810** and shroud **815** in the circumferential direction toward the blade **870**, which decreases the distance between the shroud **815** and the blade **870** in the radial direction and decreases the tight clearance **895**.

In the illustrated examples of FIGS. **8A** and **8B**, the actuator **820** adjusts the clearance between the shroud **815** and the blade **870** in two directions (e.g., shrinkage and expansion). The actuator **820** can be installed for an individual shroud (e.g., the shroud **815**), partial groups of shrouds (e.g., for groups of three shrouds, for groups of five shrouds, etc.), or for an entire group of shrouds in a turbine (e.g., the shrouds surrounding the 360 degree inner surface of the case **805**).

FIG. **9A** illustrates an example FLA view **900** of the ACC system **800**, **890** of FIGS. **8A**, **8B** in a disassembled state. FIG. **9B** illustrates an example FLA view **910** of the ACC system **800**, **890** of FIGS. **8A**, **8B** in an assembled state. In the FLA view **900** and the FLA view **910**, the example ACC system **800**, **890** includes a thermal shield **905**. In the illustrated examples, the thermal shield **905** is positioned between the case **805** and the hanger **810**. In some examples, the thermal shield **905** seals the connection between the case **805** and the hanger **810** via the case extensions **850**, **855**, the holding bars **860**, **865**, and the slots **840**, **845**.

FIG. **10A** illustrates an example side view (Axi-sym) view **1000** of the ACC system **800**, **890** of FIGS. **8A**, **8B** in a disassembled state. FIG. **10B** illustrates an example Axi-sym view **1005** of the ACC system **800**, **890** of FIGS. **8A**, **8B** in an assembled state. The illustrated examples of FIGS. **10A** and **10B** illustrate the case **805**, the hanger **810**, the shroud **815**, the actuator **820**, the holding bars **860**, **865**, and the thermal shield **905** in the Axi-sym views in the disassembled and assembled states.

FIG. **11A** is a schematic ALF view **1100** of the fifth example ACC system **800**, **890** of FIGS. **8A-10B** in the cold condition. The ALF view **1100** illustrates the assembled ACC system **800**, **890** in a cold condition (e.g., when the actuator **820** is not operating to control clearance and the gas turbine engine is not operating). In the illustrated example of FIG. **11A**, the multilayer stack of piezoelectric material **825** and the multilayer stack of piezoelectric material **830** located in the actuator **820** are not linearly displaced by any electrical signals.

FIG. **11B** illustrates an ALF view **1105** of the fifth example ACC system **800**, **890** of FIGS. **8A-10B** in a hot condition with open clearance similar to the state of the example ACC system **800** of FIG. **8A**. The ALF view **1105** illustrates the assembled ACC system **800** in a hot condition (e.g., when the actuator **820** is operating to control clearance and the gas turbine engine is operating). In the illustrated example of FIG. **11B**, the multilayer stack of piezoelectric material **825** and the multilayer stack of piezoelectric material **830** are linearly displaced in a first axial direction (e.g., to the left) by a first electrical signal and a second electrical signal, respectively. In the illustrated example of FIG. **11B**, the linear displacement of the multilayer stacks of piezoelectric material **825**, **830** in the first axial direction moves the hanger **810** along the slanted slots **840**, **845** with respect to the stationary case **805**. When the hanger **810** is displaced by the linear displacement of the multilayer stacks of piezoelectric material **825**, **830**, the hanger **810** moves with the slots **840**, **845** (illustrated in FIGS. **9A** and **9B**) along the holding bars **860**, **865** (illustrated in FIGS. **9A-10B**), respec-

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tively, in the first axial direction to displace the hanger **810** and the shroud **815** circumferentially in relation to the case **805**. The linear displacement of the multilayer stacks of piezoelectric material **825**, **830** in the first axial direction moves the hanger **810** and shroud **815** in the circumferential direction away from a blade (e.g., the blade **870**).

FIG. **11C** illustrates an ALF view **1110** of the fifth example ACC system **800**, **890** of FIGS. **8A-10B** in a hot condition with tight clearance similar to the state of the example ACC system **890** of FIG. **8B**. The ALF view **1110** illustrates the assembled ACC system **890** in a hot condition (e.g., when the actuator **820** is operating to control clearance and the gas turbine engine is operating). In the illustrated example of FIG. **11C**, the multilayer stack of piezoelectric material **825** and the multilayer stack of piezoelectric material **830** are linearly displaced in a second axial direction (e.g., to the right) by a third electrical signal and a fourth electrical signal, respectively. In the illustrated example of FIG. **11C**, the linear displacement of the multilayer stacks of piezoelectric material **825**, **830** in the second axial direction moves the hanger **810** along the slanted slots **840**, **845** with respect to the stationary case **805**. When the hanger **810** is displaced by the linear displacement of the multilayer stacks of piezoelectric material **825**, **830**, the hanger **810** moves with the slots **840**, **845** along the holding bars **860**, **865**, respectively, in the second axial direction to displace the hanger **810** and the shroud **815** circumferentially in relation to the case **805**. The linear displacement of the multilayer stacks of piezoelectric material **825**, **830** in the second axial direction moves the hanger **810** and shroud **815** in the circumferential direction toward a blade (e.g., the blade **870**).

FIG. **12** is a block diagram of an example controller **1200** of the example ACC systems of FIGS. **4A-11C**. The controller **1200** of FIG. **12** may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by processor circuitry such as a central processing unit executing instructions. Additionally or alternatively, the controller **1200** of FIG. **12** may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by an Application Specific Integrated Circuit (ASIC) or a Field Programmable Gate Array (FPGA) structured to perform operations corresponding to the instructions. It should be understood that some or all of the circuitry of FIG. **12** may, thus, be instantiated at the same or different times. Some or all of the circuitry may be instantiated, for example, in one or more threads executing concurrently on hardware and/or in series on hardware. Moreover, in some examples, some or all of the circuitry of FIG. **12** may be implemented by one or more virtual machines and/or containers executing on the microprocessor.

In FIG. **12**, the controller **1200** can be a full-authority digital engine control (FADEC) unit, an engine control unit (ECU), an electronic engine control (EEC) unit, etc., or any other type of data acquisition and/or control computing device, processor platform (e.g., processor-based computing platform), etc. The controller **1200** communicates with the example engine sensor(s) **1205**. The controller **1200** includes an example sensor(s) processor **1210** and an example actuator controller **1215**.

In the illustrated example of FIG. **12**, the controller **1200** receives values for a plurality of input variables relating to flight condition (e.g., air density, throttle lever position, engine temperatures, engine pressures, direct clearance measurements, indirect clearance measurements, etc.). The controller **1200** receives the flight condition data from the

engine sensor(s) **1205**. The engine sensor(s) **1205** can be mounted on the gas turbine engine **100** and/or positioned elsewhere in the aircraft (e.g., on wing, in cockpit, in main cabin, in engine compartment, in cargo, etc.). The communication between the controller **1200** and the engine sensor(s) **1205** can be one-way communication and/or two-way communication, for example. The controller **1200** computes engine operating parameters such as fuel flow, stator vane position, air bleed valve position, etc., using the flight condition data.

In the illustrated example of FIG. **12**, the sensor(s) processor **1210** obtains the sensor data from the example engine sensor(s) **1205**. The sensor data includes the flight condition data obtained from the gas turbine engine **100**. The sensor(s) processor **1210** monitors engine conditions based on the sensor data from the engine sensor(s) **1205**. For example, the sensor(s) processor **1210** can calculate and monitor the fuel flow, stator vane position, air bleed valve position, etc. In some examples, the sensor(s) processor **1210** determines if the turbine case is expanding or shrinking based on the engine conditions determined from the obtained flight condition data. In the illustrated example of FIG. **12**, the actuator controller **1215** generates electrical signals and supplies the electrical signals to the actuator(s) of an ACC system. In some examples, the actuator controller **1215** generates an electrical control signal to the actuator(s) of an ACC system **400-890** based on the results from sensor(s) processor **1210**.

For the example ACC systems **400** and **465** of FIGS. **4A** and **4B** respectively, the actuator controller **1215** generates and sends a first electrical current via a first electrical signal to the multilayer stack of piezoelectric material **455** located in the actuator **405**. In some examples, the actuator controller **1215** sends the first electrical current to the actuator **405** when the sensor(s) processor **1210** determines that the turbine case is expanding. In some examples, the first electrical current causes a linear displacement in the multilayer stack of piezoelectric material **455** that moves the shroud **440** in an axial direction (e.g., rightward) toward the edge of the blade **450**, which decreases the tight clearance **470** in the radial direction (similar to the example ACC system **465** of FIG. **4B**). However, the actuator controller **1215** can send the first electrical current to the actuator **405** for additional and/or alternative flight conditions (e.g., flight conditions other than those indicative of turbine case expansion) determined by the sensor(s) processor **1210**. In other examples, the actuator controller **1215** generates and sends a second electrical current via a second electrical signal to the multilayer stack of piezoelectric material **455** located in the actuator **405**. In some examples, the actuator controller **1215** sends the second electrical current to the actuator **405** when the sensor(s) processor **1210** determines that the turbine case is shrinking. In some examples, the second electrical current causes a linear displacement in the multilayer stack of piezoelectric material **455** that moves the shroud **440** in the axial direction (e.g., leftward) away from the edge of the blade **450**, which increases the open clearance **460** in the radial direction (similar to the example ACC system **400** of FIG. **4A**). However, the actuator controller **1215** can send the second electrical current to the actuator **405** for additional and/or alternative flight conditions (e.g., flight conditions other than those indicative of turbine case shrinkage) determined by the sensor(s) processor **1210**.

For the example ACC systems **500** and **575** of FIGS. **5A** and **5B** respectively, the actuator controller **1215** generates and sends a first electrical current via a first electrical signal to the multilayer stack of piezoelectric material **550** and the

multilayer stack of piezoelectric material **560** located in the actuator **515** and the actuator **520** respectively. The actuator controller **1215** also generates and sends a second electrical current via a second electrical signal to the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565** located in the actuator **515** and the actuator **520**, respectively. In some examples, the actuator controller sends the first electrical current and the second electrical current to the actuator **515** and the actuator **520** when the sensor(s) processor **1210** determines that the turbine case is expanding. In some examples, the first electrical current causes a first linear displacement in the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560** in a first axial direction (e.g., rightward). In some examples, the second electrical current causes a second linear displacement in the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565** in the first axial direction (e.g., rightward). In some examples, the second linear displacement is opposite of the first linear displacement. For example, if the first linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560**, then the second linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565**. The first linear displacement and the second linear displacement move the shroud **540** in the first axial direction (e.g., rightward), which decreases the distance between the shroud **540** and the blade **545** in the radial direction and decreases the tight clearance **580** (similar to the example ACC system **575** of FIG. **5B**). However, the actuator controller **1215** can send the first electrical current and the second electrical current to the actuator **515** and the actuator **520** for additional and/or alternative flight conditions (e.g., flight conditions other than those indicative of case shrinkage) determined by the sensor(s) processor **1210**.

In other examples, the actuator controller **1215** generates and sends a third electrical current via a third electrical signal to the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560**. The actuator controller **1215** also generates and sends a fourth electrical current via a fourth electrical signal to the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565**. In some examples, the actuator controller sends the third electrical current and the fourth electrical current to the actuator **515** and the actuator **520** when the sensor(s) processor **1210** determines that the case is shrinking. In some examples, the third electrical current causes a third linear displacement in the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560** in a second axial direction (e.g., leftward). In some examples, the fourth electrical current causes a fourth linear displacement in the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565** in the second axial direction (e.g., leftward). In some examples, the fourth linear displacement is opposite of the third linear displacement. For example, if the third linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560**, then the fourth linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565**. The third linear displacement and the fourth linear displacement move the

shroud 540 in the axial direction (e.g., leftward), which increases the distance between the shroud 540 and the blade 545 in the radial direction and increases the open clearance 570 (similar to the example ACC system 500 of FIG. 5A). However, the actuator controller 1215 can send the third electrical current and the fourth electrical current to the actuator 515 and the actuator 520 for additional and/or alternative flight conditions (e.g., flight conditions other than those indicative of case shrinkage) determined by the sensor(s) processor 1210.

For the example ACC systems 600 and 675 of FIGS. 6A and 6B respectively, the actuator controller 1215 generates and sends a first electrical current via a first electrical signal to the multilayer stack of piezoelectric material 655 and the multilayer stack of piezoelectric material 665 located in the actuator 615 and the actuator 620 respectively. In some examples, the actuator controller 1215 sends the first electrical current to the actuator 615 and the actuator 620 when the sensor(s) processor 1210 determines that the case is expanding. In some examples, the first electrical current causes a linear displacement in the multilayer stack of piezoelectric material 655 and the multilayer stack of piezoelectric material 665 that moves the shroud 640 in the axial direction (e.g., rightward), which decreases the distance between the shroud 640 and the blade 645 in the radial direction and decreases the tight clearance 680 (similar to the example ACC system 675 of FIG. 6B). However, the actuator controller 1215 can send the first electrical current to the actuator 615 and the actuator 620 for additional and/or alternative flight conditions (e.g., flight conditions other than those indicative of case expansion) determined by the sensor(s) processor 1210. In some examples, the actuator controller 1215 generates and sends a second electrical current via a second electrical signal to the multilayer stack of piezoelectric material 655 and the multilayer stack of piezoelectric material 665 located in the actuator 615 and the actuator 620, respectively. In some examples, the actuator controller 1215 sends the second electrical current to the actuator 615 and the actuator 620 when the sensor(s) processor 1210 determines that the case is shrinking. In some examples, the second electrical current causes a linear displacement in the multilayer stack of piezoelectric material 655 and the multilayer stack of piezoelectric material 665 that moves the shroud 640 in the axial direction (e.g., leftward), which increases the distance between the shroud 640 and the blade 645 in the radial direction and increases the open clearance 670 (similar to the example ACC system 600 of FIG. 6A). However, the actuator controller 1215 can send the second electrical current to the actuator 615 and the actuator 620 for additional and/or alternative flight conditions (e.g., flight conditions other than those indicative of turbine case shrinkage) determined by the sensor(s) processor 1210.

For the example ACC systems 700 and 790 of FIGS. 7A and 7B respectively, the actuator controller 1215 generates and sends a first electrical current via a first electrical signal to the multilayer stack of piezoelectric material 750 and the multilayer stack of piezoelectric material 760 located in the actuator 715 and the multilayer stack of piezoelectric material 770 and the multilayer stack of piezoelectric material 780 located in the actuator 720. The actuator controller 1215 also generates and sends a second electrical current via a second electrical signal to the multilayer stack of piezoelectric material 755 and the multilayer stack of piezoelectric material 765 located in the actuator 715 and the multilayer stack of piezoelectric material 775 and the multilayer stack of piezoelectric material 782 located in the actuator 720. In

some examples, the actuator controller 1215 sends the first electrical current and the second electrical current to the actuator 715 and the actuator 720 when the sensor(s) processor 1210 determines that the turbine case is expanding. In some examples, the first electrical current causes a first linear displacement in the multilayer stack of piezoelectric material 750, the multilayer stack of piezoelectric material 760, the multilayer stack of piezoelectric material 770, and the multilayer stack of piezoelectric material 780 in a first axial direction (e.g., rightward) and a first radial direction (e.g., downward). In some examples, the second electrical current causes a second linear displacement in the multilayer stack of piezoelectric material 755, the multilayer stack of piezoelectric material 765, the multilayer stack of piezoelectric material 775, and the multilayer stack of piezoelectric material 782 in the first axial direction (e.g., rightward) and the first radial direction (e.g., downward). In some examples, the second linear displacement is opposite of the first linear displacement. For example, if the first linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material 750, the multilayer stack of piezoelectric material 760, the multilayer stack of piezoelectric material 770, and the multilayer stack of piezoelectric material 780, then the second linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material 755, the multilayer stack of piezoelectric material 765, the multilayer stack of piezoelectric material 775, and the multilayer stack of piezoelectric material 782. The first linear displacement and the second linear displacement move the shroud 740 in the first axial direction (e.g., rightward) and the first radial direction (e.g., downward), which decreases the distance between the shroud 740 and the blade 745 in the radial direction and decreases the tight clearance 795 (similar to the example ACC system 790 of FIG. 7B). However, the actuator controller 1215 can send the first electrical current and the second electrical current to the actuator 715 and the actuator 720 for additional and/or alternative flight conditions (e.g., flight conditions other than those indicative of case shrinkage) determined by the sensor(s) processor 1210.

In other examples, the actuator controller 1215 generates and sends a third electrical current via a third electrical signal to the multilayer stack of piezoelectric material 750 and the multilayer stack of piezoelectric material 760 located in the actuator 715 and the multilayer stack of piezoelectric material 770 and the multilayer stack of piezoelectric material 780 located in the actuator 720. The actuator controller 1215 also generates and sends a fourth electrical current via a fourth electrical signal to the multilayer stack of piezoelectric material 755 and the multilayer stack of piezoelectric material 765 located in the actuator 715 and the multilayer stack of piezoelectric material 775 and the multilayer stack of piezoelectric material 782 located in the actuator 720. In some examples, the actuator controller 1215 sends the third electrical current and the fourth electrical current to the actuator 715 and the actuator 720 when the sensor(s) processor 1210 determines that the case is shrinking. In some examples, the third electrical current causes a third linear displacement in the multilayer stack of piezoelectric material 750, the multilayer stack of piezoelectric material 760, the multilayer stack of piezoelectric material 770, and the multilayer stack of piezoelectric material 780 in a second axial direction (e.g., leftward) and a second radial direction (e.g., upward). In some examples, the fourth electrical current causes a fourth linear displacement in the multilayer stack of piezoelectric material 755, the multilayer

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stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782** in the second axial direction (e.g., leftward) and the second radial direction (e.g., upward). In some examples, the fourth linear displacement is opposite of the third linear displacement. For example, if the third linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **750**, the multilayer stack of piezoelectric material **760**, the multilayer stack of piezoelectric material **770**, and the multilayer stack of piezoelectric material **780**, then the fourth linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **755**, the multilayer stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782**. The third linear displacement and the fourth linear displacement move the shroud **740** in the second axial direction (e.g., leftward) and the second radial direction (e.g., upward), which increases the distance between the shroud **740** and the blade **745** in the radial direction and increases the open clearance **785** (similar to the example ACC system **700** of FIG. 7A). However, the actuator controller **1215** can send the third electrical current and the fourth electrical current to the actuator **715** and the actuator **720** for additional and/or alternative flight conditions (e.g., flight conditions other than those indicative of case shrinkage) determined by the sensor(s) processor **1210**.

For the example ACC systems **800** and **890** of FIGS. 8A and 8B respectively and FIGS. 9A-11C, the actuator controller **1215** generates and sends a first electrical current via a first electrical signal to the multilayer stack of piezoelectric material **825** located in the actuator **820**. The actuator controller **1215** also generates and sends a second electrical current via a second electrical signal to the multilayer stack of piezoelectric material **830** located in the actuator **820**. In some examples, the actuator controller sends the first electrical current and the second electrical current to the actuator **820** when the sensor(s) processor **1210** determines that the turbine case is expanding. In some examples, the first electrical current causes a first linear displacement in the multilayer stack of piezoelectric material **825** in a first axial direction (e.g., rightward). In some examples, the second electrical current causes a second linear displacement in the multilayer stack of piezoelectric material **830** in the first axial direction (e.g., rightward). In some examples, the second linear displacement is opposite of the first linear displacement. For example, if the first linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **825**, then the second linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **830**. The first linear displacement and the second linear displacement moves the hanger **810** with the slots **840**, **845** along the holding bars **860**, **865**, respectively, in the first axial direction (e.g., rightward) to displace the hanger **810** and the shroud **815** circumferentially in relation to the case **805**. The first linear displacement and the second linear displacement of the multilayer stacks of piezoelectric material **825**, **830** moves the hanger **810** and shroud **815** in the circumferential direction toward the blade **870**, which decreases the distance between the shroud **815** and the blade **870** in the radial direction and decreases the tight clearance **895**. However, the actuator controller **1215** can send the first electrical current and the second electrical current to the actuator **820** for additional and/or alternative flight condi-

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tions (e.g., flight conditions other than those indicative of case shrinkage) determined by the sensor(s) processor **1210**.

In other examples, the actuator controller **1215** generates and sends a third electrical current via a third electrical signal to the multilayer stack of piezoelectric material **825**. The actuator controller **1215** also generates and sends a fourth electrical current via a fourth electrical signal to the multilayer stack of piezoelectric material **830**. In some examples, the actuator controller sends the third electrical current and the fourth electrical current to the actuator **820** when the sensor(s) processor **1210** determines that the case is shrinking. In some examples, the third electrical current causes a third linear displacement in the multilayer stack of piezoelectric material **825** in a second axial direction (e.g., leftward). In some examples, the fourth electrical current causes a fourth linear displacement in the multilayer stack of piezoelectric material **830** in the second axial direction (e.g., leftward). In some examples, the fourth linear displacement is opposite of the third linear displacement. For example, if the third linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **825**, then the fourth linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **830**. The third linear displacement and the fourth linear displacement move the hanger **810** with the slots **840**, **845** along the holding bars **860**, **865**, respectively, in the second axial direction (e.g., leftward) to displace the hanger **810** and the shroud **815** circumferentially in relation to the case **805**. The third linear displacement and the fourth linear displacement of the multilayer stacks of piezoelectric material **825**, **830** moves the hanger **810** and shroud **815** in the circumferential direction away from the blade **870**, which increases the distance between the shroud **815** and the blade **870** in the radial direction and increases the open clearance **875**. However, the actuator controller **1215** can send the third electrical current and the fourth electrical current to the actuator **515** and the actuator **520** for additional and/or alternative flight conditions (e.g., flight conditions other than those indicative of case shrinkage) determined by the sensor(s) processor **1210**.

While an example manner of implementing the controller **1200** is illustrated in FIG. 12, one or more of the elements, processes, and/or devices illustrated in FIG. 12 may be combined, divided, re-arranged, omitted, eliminated, and/or implemented in any other way. Further, the example sensor(s) processor **1210**, the example actuator controller **1215**, and/or, more generally, the example controller **1200** of FIG. 12, may be implemented by hardware alone or by hardware in combination with software and/or firmware. Thus, for example, any of the example sensor(s) processor **1210**, the example actuator controller **1215**, and/or, more generally, the example controller **1200**, could be implemented by processor circuitry, analog circuit(s), digital circuit(s), logic circuit(s), programmable processor(s), programmable microcontroller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)), and/or field programmable logic device(s) (FPLD(s)) such as Field Programmable Gate Arrays (FPGAs). Further still, the example controller **1200** of FIG. 12 may include one or more elements, processes, and/or devices in addition to, or instead of, those illustrated in FIG. 12, and/or may include more than one of any or all of the illustrated elements, processes and devices.

Flowcharts representative of example hardware logic circuitry, machine readable instructions, hardware imple-

mented state machines, and/or any combination thereof for implementing the controller **1200** of FIG. **12** are shown in FIGS. **13**, **14**, **15**, **16**, and **17**. The machine readable instructions may be one or more executable programs or portion(s) of an executable program for execution by processor circuitry, such as the processor circuitry **1812** shown in the example processor platform **1800** discussed below in connection with FIG. **18**. The program may be embodied in software stored on one or more non-transitory computer readable storage media such as a compact disk (CD), a floppy disk, a hard disk drive (HDD), a solid-state drive (SSD), a digital versatile disk (DVD), a Blu-ray disk, a volatile memory (e.g., Random Access Memory (RAM) of any type, etc.), or a non-volatile memory (e.g., electrically erasable programmable read-only memory (EEPROM), FLASH memory, an HDD, an SSD, etc.) associated with processor circuitry located in one or more hardware devices, but the entire program and/or parts thereof could alternatively be executed by one or more hardware devices other than the processor circuitry and/or embodied in firmware or dedicated hardware. The machine readable instructions may be distributed across multiple hardware devices and/or executed by two or more hardware devices (e.g., a server and a client hardware device). For example, the client hardware device may be implemented by an endpoint client hardware device (e.g., a hardware device associated with a user) or an intermediate client hardware device (e.g., a radio access network (RAN)) gateway that may facilitate communication between a server and an endpoint client hardware device). Similarly, the non-transitory computer readable storage media may include one or more mediums located in one or more hardware devices. Further, although the example program is described with reference to the flowcharts illustrated in FIGS. **13**, **14**, **15**, **16**, and **17**, many other methods of implementing the example controller **1200** may alternatively be used. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Additionally or alternatively, any or all of the blocks may be implemented by one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware. The processor circuitry may be distributed in different network locations and/or local to one or more hardware devices (e.g., a single-core processor (e.g., a single core central processor unit (CPU)), a multi-core processor (e.g., a multi-core CPU), etc.) in a single machine, multiple processors distributed across multiple servers of a server rack, multiple processors distributed across one or more server racks, a CPU and/or a FPGA located in the same package (e.g., the same integrated circuit (IC) package or in two or more separate housings, etc.).

The machine readable instructions described herein may be stored in one or more of a compressed format, an encrypted format, a fragmented format, a compiled format, an executable format, a packaged format, etc. Machine readable instructions as described herein may be stored as data or a data structure (e.g., as portions of instructions, code, representations of code, etc.) that may be utilized to create, manufacture, and/or produce machine executable instructions. For example, the machine readable instructions may be fragmented and stored on one or more storage devices and/or computing devices (e.g., servers) located at the same or different locations of a network or collection of networks (e.g., in the cloud, in edge devices, etc.). The

machine readable instructions may require one or more of installation, modification, adaptation, updating, combining, supplementing, configuring, decryption, decompression, unpacking, distribution, reassignment, compilation, etc., in order to make them directly readable, interpretable, and/or executable by a computing device and/or other machine. For example, the machine readable instructions may be stored in multiple parts, which are individually compressed, encrypted, and/or stored on separate computing devices, wherein the parts when decrypted, decompressed, and/or combined form a set of machine executable instructions that implement one or more operations that may together form a program such as that described herein.

In another example, the machine readable instructions may be stored in a state in which they may be read by processor circuitry, but require addition of a library (e.g., a dynamic link library (DLL)), a software development kit (SDK), an application programming interface (API), etc., in order to execute the machine readable instructions on a particular computing device or other device. In another example, the machine readable instructions may need to be configured (e.g., settings stored, data input, network addresses recorded, etc.) before the machine readable instructions and/or the corresponding program(s) can be executed in whole or in part. Thus, machine readable media, as used herein, may include machine readable instructions and/or program(s) regardless of the particular format or state of the machine readable instructions and/or program(s) when stored or otherwise at rest or in transit.

The machine readable instructions described herein can be represented by any past, present, or future instruction language, scripting language, programming language, etc. For example, the machine readable instructions may be represented using any of the following languages: C, C++, Java, C#, Perl, Python, JavaScript, HyperText Markup Language (HTML), Structured Query Language (SQL), Swift, etc.

As mentioned above, the example operations of FIGS. **13**, **14**, **15**, **16**, and **17** may be implemented using executable instructions (e.g., computer and/or machine readable instructions) stored on one or more non-transitory computer and/or machine readable media such as optical storage devices, magnetic storage devices, an HDD, a flash memory, a read-only memory (ROM), a CD, a DVD, a cache, a RAM of any type, a register, and/or any other storage device or storage disk in which information is stored for any duration (e.g., for extended time periods, permanently, for brief instances, for temporarily buffering, and/or for caching of the information). As used herein, the terms non-transitory computer readable medium and non-transitory computer readable storage medium are expressly defined to include any type of computer readable storage device and/or storage disk and to exclude propagating signals and to exclude transmission media.

FIG. **13** is a flowchart representative of example machine readable instructions and/or example operations **1300** that may be executed and/or instantiated by processor circuitry to implement the example controller **1200** of FIG. **12** in conjunction with the first example ACC system **400**, **465** of FIGS. **4A** and **4B**. In examples disclosed herein, the example machine readable instructions and/or example operations **1300** may be executed in a closed loop system and/or an open loop system. In some examples, the example machine readable instructions and/or example operations **1300** may be executed in an open loop system with direct measurement. The machine readable instructions and/or the operations **1300** of FIG. **13** begin at block **1310**, at which the

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example sensor(s) processor **1210** obtains sensor data from the example engine sensor(s) **1205**. In some examples, the sensor data includes the flight condition data obtained by the engine sensor(s) **1205** from an engine (e.g., the gas turbine engine **100** of FIG. 1). In some examples, flight condition data of the sensor data includes values for a plurality of input variables relating to flight conditions (e.g., air density, throttle lever position, engine temperatures, engine pressures, etc.).

At block **1315**, the example sensor(s) processor **1210** monitors engine conditions based on the sensor data from the engine sensor(s) **1205**. For example, the sensor(s) processor **1210** can calculate and monitor the fuel flow, stator vane position, air bleed valve position, etc., using the flight condition data included in the sensor data. At block **1320**, the example sensor(s) processor **1210** determines if the case is expanding. In some examples, the case is a case surrounding a high pressure turbine (e.g., the HP turbine **118** of FIG. 1), a low pressure turbine (e.g., the LP turbine **120** of FIG. 1) or a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. 1). In some example, the sensor(s) processor **1210** determines if the case is expanding based on the engine conditions determined from the obtained flight condition data. When the example sensor(s) processor **1210** determines that the case is expanding, then the example operations **1300** continues to block **1330** at which the example actuator controller **1215** sends a first electrical current to a multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not expanding, then the example operations **1300** continues to block **1325** at which the example sensor(s) processor **1210** determines if the case is shrinking.

At block **1325**, the example sensor(s) processor **1210** determines if the case is shrinking. In some example, the sensor(s) processor **1210** determines if the case is shrinking based on the engine conditions determined from the obtained flight condition data. When the example sensor(s) processor **1210** determines that the case is shrinking, then the example operations **1300** continues to block **1335** at which the example actuator controller **1215** sends a second electrical current to a multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not shrinking, then the example operations **1300** returns to block **1310** at which the example sensor(s) processor **1210** obtains sensor data.

At block **1330**, the example actuator controller **1215** sends a first electrical current to a multilayer stack of piezoelectric material. In some examples, the actuator controller **1215** generates and sends the first electrical current via a first electrical signal to the multilayer stack of piezoelectric material **455** located in the actuator **405** of FIGS. 4A and 4B. In some examples, the first electrical current causes a linear displacement in the multilayer stack of piezoelectric material **455** that moves the shroud **440** in an axial direction towards the blade **450** (similar to the example ACC system **465** of FIG. 4B). After the example actuator controller **1215** sends the first electrical current, the operations **1300** ends.

At block **1335**, the example actuator controller **1215** sends a second electrical current to a multilayer stack of piezoelectric material. In some examples, the multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **455** of FIGS. 4A, 4B. In some examples, the actuator controller **1215** generates and sends the second electrical current via a second electrical signal to the multilayer stack of piezoelectric material **455** located in the actuator **405**. In some examples, the second electrical current causes a linear displacement in

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the multilayer stack of piezoelectric material **455** that moves the shroud **440** in an axial direction away from the blade **450** (similar to the example ACC system **400** of FIG. 4A). After the example actuator controller **1215** sends the second electrical current, the operations **1300** ends.

FIG. 14 is a flowchart representative of example machine readable instructions and/or example operations **1400** that may be executed and/or instantiated by processor circuitry to implement the example controller **1200** of FIG. 12 in conjunction with the second example ACC system **500**, **575** of FIGS. 5A and 5B and/or the fifth example ACC system **800**, **890** of FIGS. 8A-11C. In examples disclosed herein, the example machine readable instructions and/or example operations **1400** may be executed in a closed loop system and/or an open loop system. In some examples, the example machine readable instructions and/or example operations **1400** may be executed in an open loop system with direct measurement. The machine readable instructions and/or the operations **1400** of FIG. 14 begin at block **1410**, at which the example sensor(s) processor **1210** obtains sensor data from the example engine sensor(s) **1205**. In some examples, the sensor data includes the flight condition data obtained by the engine sensor(s) **1205** from an engine (e.g., the gas turbine engine **100** of FIG. 1). In some examples, flight condition data of the sensor data includes values for a plurality of input variables relating to flight conditions (e.g., air density, throttle lever position, engine temperatures, engine pressures, etc.).

At block **1415**, the example sensor(s) processor **1210** monitors engine conditions based on the sensor data from the engine sensor(s) **1205**. For example, the sensor(s) processor **1210** can calculate and monitor the fuel flow, stator vane position, air bleed valve position, etc., using the flight condition data included in the sensor data. At block **1420**, the example sensor(s) processor **1210** determines if the case is expanding. In some examples, the case is a case surrounding a high pressure turbine (e.g., the HP turbine **118** of FIG. 1), a low pressure turbine (e.g., the LP turbine **120** of FIG. 1) or a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. 1). In some examples, the sensor(s) processor **1210** determines if the case is expanding based on the engine conditions determined from the obtained flight condition data. When the example sensor(s) processor **1210** determines that the case is expanding, then the example operations **1400** continues to block **1430** at which the example actuator controller **1215** sends a first electrical current to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not expanding, then the example operations **1400** continues to block **1425** at which the example sensor(s) processor **1210** determines if the case is shrinking.

At block **1425**, the example sensor(s) processor **1210** determines if the case is shrinking. In some examples, the sensor(s) processor **1210** determines if the case is shrinking based on the engine conditions determined from the obtained flight condition data. When the example sensor(s) processor **1210** determines that the case is shrinking, then the example operations **1400** continues to block **1440** at which the example actuator controller **1215** sends a third electrical current to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not shrinking, then the example operations **1400** returns to block **1410** at which the example sensor(s) processor **1210** obtains sensor data.

At block **1430**, the example actuator controller **1215** sends a first electrical current to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material. In some examples, the first multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **550**, and the second multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **560**. In some examples, the actuator controller **1215** generates and sends the first electrical current via a first electrical signal to the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560** located in the actuator **515** and the actuator **520**, respectively. In some examples, the first electrical current causes a first linear displacement in the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560**.

At block **1435**, the example actuator controller **1215** sends a second electrical current to a third multilayer stack of piezoelectric material and a fourth multilayer stack of piezoelectric material. In some examples, the third multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **555**, and the fourth multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **565**. In some examples, the actuator controller **1215** generates and sends the second electrical current via a second electrical signal to the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565** located in the actuator **515** and the actuator **520**, respectively. In some examples, the second electrical current causes a second linear displacement in the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565**. In some examples, the second linear displacement is opposite of the first linear displacement. For example, if the first linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560**, then the second linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565**. While blocks **1430** and **1435** are shown in sequence, they can be executed in parallel. After the example actuator controller **1215** sends the second electrical current to a third multilayer stack of piezoelectric material and a fourth multilayer stack of piezoelectric material, operations **1400** ends.

At block **1440**, the example actuator controller **1215** sends a third electrical current to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material. In some examples, the actuator controller **1215** generates and sends the third electrical current via a third electrical signal to the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560**. In some examples, the third electrical current causes a third linear displacement in the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560**.

At block **1445**, the example actuator controller **1215** sends a fourth electrical current to a third multilayer stack of piezoelectric material and a fourth multilayer stack of piezoelectric material. In some examples, the actuator controller **1215** generates and sends the fourth electrical current via a fourth electrical signal to the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565**. In some examples, the fourth electrical current causes a fourth linear displacement in the multilayer stack of

piezoelectric material **555** and the multilayer stack of piezoelectric material **565**. In some examples, the fourth linear displacement is opposite of the third linear displacement. For example, if the third linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **550** and the multilayer stack of piezoelectric material **560**, then the fourth linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **555** and the multilayer stack of piezoelectric material **565**. While blocks **1440** and **1445** are shown in sequence in the example of FIG. **14**, in certain examples, they can be executed in parallel. After the example actuator controller **1215** sends the fourth electrical current to the third multilayer stack of piezoelectric material and a fourth multilayer stack of piezoelectric material, operations **1400** ends.

FIG. **15** is a flowchart representative of example machine readable instructions and/or example operations **1500** that may be executed and/or instantiated by processor circuitry to implement the example controller **1200** of FIG. **12** in conjunction with the third example ACC system **600**, **675** of FIGS. **6A** and **6B**. In examples disclosed herein, the example machine readable instructions and/or example operations **1500** may be executed in a closed loop system and/or an open loop system. In some examples, the example machine readable instructions and/or example operations **1500** may be executed in an open loop system with direct measurement. The machine readable instructions and/or the operations **1500** of FIG. **15** begin at block **1510**, at which the example sensor(s) processor **1210** obtains sensor data from the example engine sensor(s) **1205**. In some examples, the sensor data includes the flight condition data obtained by the engine sensor(s) **1205** from an engine (e.g., the gas turbine engine **100** of FIG. **1**). In some examples, flight condition data of the sensor data includes values for a plurality of input variables relating to flight conditions (e.g., air density, throttle lever position, engine temperatures, engine pressures, etc.).

At block **1515**, the example sensor(s) processor **1210** monitors engine conditions based on the sensor data from the engine sensor(s) **1205**. For example, the sensor(s) processor **1210** can calculate and monitor the fuel flow, stator vane position, air bleed valve position, etc., using the flight condition data included in the sensor data. At block **1520**, the example sensor(s) processor **1210** determines if the case is expanding. In some examples, the case is a case surrounding a high pressure turbine (e.g., the HP turbine **118** of FIG. **1**), a low pressure turbine (e.g., the LP turbine **120** of FIG. **1**) or a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. **1**). In some examples, the sensor(s) processor **1210** determines if the case is expanding based on the engine conditions determined from the obtained flight condition data. When the example sensor(s) processor **1210** determines that the case is expanding, then the example operations **1500** continues to block **1530** at which the example actuator controller **1215** sends a first electrical current to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not expanding, then the example operations **1500** continues to block **1525** at which the example sensor(s) processor **1210** determines if the case is shrinking.

At block **1525**, the example sensor(s) processor **1210** determines if the case is shrinking. In some examples, the sensor(s) processor **1210** determines if the case is shrinking based on the engine conditions determined from the obtained flight condition data. When the example sensor(s)

processor **1210** determines that the case is shrinking, then the example operations **1500** continues to block **1535** at which the example actuator controller **1215** sends a second electrical current to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not shrinking, then the example operations **1500** returns to block **1510** at which the example sensor(s) processor **1210** obtains sensor data.

At block **1530**, the example actuator controller **1215** sends a first electrical current to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material. In some examples, the first multilayer stack of piezoelectric material is substantially similar to multilayer stack of piezoelectric material **655**, and the second multilayer stack of piezoelectric material is substantially similar to multilayer stack of piezoelectric material **665**. In some examples, the actuator controller **1215** generates and sends the first electrical current via a first electrical signal to the multilayer stack of piezoelectric material **655** and the multilayer stack of piezoelectric material **665** located in the actuator **615** and the actuator **620**, respectively. In some examples, the first electrical current causes a linear displacement in the multilayer stack of piezoelectric material **655** and the multilayer stack of piezoelectric material **665** that moves the shroud **640** in the axial direction (e.g., rightward), which decreases the distance between the shroud **640** and the blade **645** in the radial direction and decreases the tight clearance **680** (similar to the example ACC system **675** of FIG. **6B**). After the example actuator controller **1215** sends the first electrical current, the operations **1500** ends.

At block **1535**, the example actuator controller **1215** sends a second electrical current to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material. In some examples, the actuator controller **1215** generates and sends the second electrical current via a second electrical signal to the multilayer stack of piezoelectric material **655** and the multilayer stack of piezoelectric material **665** located in the actuator **615** and the actuator **620**, respectively. In some examples, the second electrical current causes a linear displacement in the multilayer stack of piezoelectric material **655** and the multilayer stack of piezoelectric material **665** that moves the shroud **640** in the axial direction (e.g., leftward), which increases the distance between the shroud **640** and the blade **645** in the radial direction and increases the open clearance **670** (similar to the example ACC systems **600** of FIG. **6A**). After the example actuator controller **1215** sends the second electrical current, operations **1500** ends.

FIG. **16** is a flowchart representative of example machine readable instructions and/or example operations **1600** that may be executed and/or instantiated by processor circuitry to implement the example controller **1200** of FIG. **12** in conjunction with the fourth example ACC system **700**, **790** of FIGS. **7A** and **7B**. In examples disclosed herein, the example machine readable instructions and/or example operations **1600** may be executed in a closed loop system and/or an open loop system. In some examples, the example machine readable instructions and/or example operations **1600** may be executed in an open loop system with direct measurement. The machine readable instructions and/or the operations **1600** of FIG. **16** begin at block **1610**, at which the example sensor(s) processor **1210** obtains sensor data from the example engine sensor(s) **1205**. In some examples, the sensor data includes the flight condition data obtained by the engine sensor(s) **1205** from an engine (e.g., the gas turbine engine **100** of FIG. **1**). In some examples, flight condition

data of the sensor data includes values for a plurality of input variables relating to flight conditions (e.g., air density, throttle lever position, engine temperatures, engine pressures, etc.).

At block **1615**, the example sensor(s) processor **1210** monitors engine conditions based on the sensor data from the engine sensor(s) **1205**. For example, the sensor(s) processor **1210** can calculate and monitor the fuel flow, stator vane position, air bleed valve position, etc., using the flight condition data included in the sensor data. At block **1620**, the example sensor(s) processor **1210** determines if the case is expanding. In some examples, the case is a case surrounding a high pressure turbine (e.g., the HP turbine **118** of FIG. **1**), a low pressure turbine (e.g., the LP turbine **120** of FIG. **1**) or a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. **1**). In some examples, the sensor(s) processor **1210** determines if the case is expanding based on the engine conditions determined from the obtained flight condition data. When the example sensor(s) processor **1210** determines that the case is expanding, then the example operations **1600** continues to block **1630** at which the example actuator controller **1215** sends a first electrical current to a first multilayer stack of piezoelectric material, a second multilayer stack of piezoelectric material, a third multilayer stack of piezoelectric material, and a fourth multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not expanding, then the example operations **1600** continues to block **1625** at which the example sensor(s) processor **1210** determines if the case is shrinking.

At block **1625**, the example sensor(s) processor **1210** determines if the case is shrinking. In some example, the sensor(s) processor **1210** determines if the case is shrinking based on the engine conditions determined from the obtained flight condition data. When the example sensor(s) processor **1210** determines that the case is shrinking, then the example operations **1600** continues to block **1640** at which the example actuator controller **1215** a third electrical current to a first multilayer stack of piezoelectric material, a second multilayer stack of piezoelectric material, a third multilayer stack of piezoelectric material, and a fourth multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not shrinking, then the example operations **1600** returns to block **1610** at which the example sensor(s) processor **1210** obtains sensor data.

At block **1630**, the example actuator controller **1215** sends a first electrical current to a first multilayer stack of piezoelectric material, a second multilayer stack of piezoelectric material, a third multilayer stack of piezoelectric material, and a fourth multilayer stack of piezoelectric material. In some examples, the first multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **750**, and the second multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **760**, the third multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **770**, and the fourth multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **780**. In some examples, the actuator controller **1215** generates and sends the first electrical current via a first electrical signal to the multilayer stack of piezoelectric material **750** and the multilayer stack of piezoelectric material **760** located in the actuator **715** and the multilayer stack of piezoelectric material **770** and the multilayer stack of piezoelectric material **780** located in the

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actuator **720**. In some examples, the first electrical current causes a first linear displacement in the multilayer stack of piezoelectric material **750**, the multilayer stack of piezoelectric material **760**, the multilayer stack of piezoelectric material **770**, and the multilayer stack of piezoelectric material **780** in a first axial direction (e.g., rightward) and a first radial direction (e.g., downward).

At block **1635**, the example actuator controller **1215** sends a second electrical current to a fifth multilayer stack of piezoelectric material, a sixth multilayer stack of piezoelectric material, a seventh multilayer stack of piezoelectric material, and an eighth multilayer stack of piezoelectric material. In some examples, the fifth multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **755**, the sixth multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **765**, the seventh multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **775**, and the eighth multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **782**. In some examples, the actuator controller **1215** generates and sends the second electrical current via a second electrical signal to the multilayer stack of piezoelectric material **755** and the multilayer stack of piezoelectric material **765** located in the actuator **715** and the multilayer stack of piezoelectric material **775** and the multilayer stack of piezoelectric material **782** located in the actuator **720**. In some examples, the second electrical current causes a second linear displacement in the multilayer stack of piezoelectric material **755**, the multilayer stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782** in the first axial direction (e.g., rightward) and the first radial direction (e.g., downward). In some examples, the second linear displacement is opposite of the first linear displacement. If the first linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **750**, the multilayer stack of piezoelectric material **760**, the multilayer stack of piezoelectric material **770**, and the multilayer stack of piezoelectric material **780**, then the second linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **755**, the multilayer stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782**. While blocks **1630** and **1635** are shown in sequence, they can be executed in parallel. After the example actuator controller **1215** sends the second electrical current to second electrical current to a fifth multilayer stack of piezoelectric material, a sixth multilayer stack of piezoelectric material, a seventh multilayer stack of piezoelectric material, and an eighth multilayer stack of piezoelectric material, operations **1600** ends.

At block **1640**, the example actuator controller **1215** sends a third electrical current to a first multilayer stack of piezoelectric material, a second multilayer stack of piezoelectric material, a third multilayer stack of piezoelectric material, and a fourth multilayer stack of piezoelectric material. In some examples, the actuator controller **1215** generates and sends the third electrical current via a third electrical signal to the multilayer stack of piezoelectric material **750** and the multilayer stack of piezoelectric material **760** located in the actuator **715** and the multilayer stack of piezoelectric material **770** and the multilayer stack of piezoelectric material **780** located in the actuator **720**. In some examples, the third electrical current causes a third

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linear displacement in the multilayer stack of piezoelectric material **750**, the multilayer stack of piezoelectric material **760**, the multilayer stack of piezoelectric material **770**, and the multilayer stack of piezoelectric material **780** in a second axial direction (e.g., leftward) and a second radial direction (e.g., upward).

At block **1645**, the example actuator controller **1215** sends a fourth electrical current to a fifth multilayer stack of piezoelectric material, a sixth multilayer stack of piezoelectric material, a seventh multilayer stack of piezoelectric material, and an eighth multilayer stack of piezoelectric material. In some examples, the actuator controller **1215** generates and sends the fourth electrical current via a fourth electrical signal to the multilayer stack of piezoelectric material **755** and the multilayer stack of piezoelectric material **765** located in the actuator **715** and the multilayer stack of piezoelectric material **775** and the multilayer stack of piezoelectric material **782** located in the actuator **720**. In some examples, the fourth electrical current causes a fourth linear displacement in the multilayer stack of piezoelectric material **755**, the multilayer stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782** in the second axial direction (e.g., leftward) and the second radial direction (e.g., upward). In some examples, the fourth linear displacement is opposite of the third linear displacement. For example, if the third linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **750**, the multilayer stack of piezoelectric material **760**, the multilayer stack of piezoelectric material **770**, and the multilayer stack of piezoelectric material **780**, then the fourth linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **755**, the multilayer stack of piezoelectric material **765**, the multilayer stack of piezoelectric material **775**, and the multilayer stack of piezoelectric material **782**. While blocks **1640** and **1645** are shown in sequence in the example of FIG. **16**, in certain examples, they can be executed in parallel. After the example actuator controller **1215** sends the fourth electrical current to a fifth multilayer stack of piezoelectric material, a sixth multilayer stack of piezoelectric material, a seventh multilayer stack of piezoelectric material, and an eighth multilayer stack of piezoelectric material, operations **1600** ends.

FIG. **17** is a flowchart representative of example machine readable instructions and/or example operations **1700** that may be executed and/or instantiated by processor circuitry to implement the example controller **1200** of FIG. **12** in conjunction with the fifth example ACC system **800**, **890** of FIGS. **8A-11C**. In examples disclosed herein, the example machine readable instructions and/or example operations **1700** may be executed in a closed loop system and/or an open loop system. In some examples, the example machine readable instructions and/or example operations **1700** may be executed in an open loop system with direct measurement. The machine readable instructions and/or the operations **1700** of FIG. **17** begin at block **1710**, at which the example sensor(s) processor **1210** obtains sensor data from the example engine sensor(s) **1205**. In some examples, the sensor data includes the flight condition data obtained by the engine sensor(s) **1205** from an engine (e.g., the gas turbine engine **100** of FIG. **1**). In some examples, flight condition data of the sensor data includes values for a plurality of input variables relating to flight conditions (e.g., air density, throttle lever position, engine temperatures, engine pressures, etc.).

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At block **1715**, the example sensor(s) processor **1210** monitors engine conditions based on the sensor data from the engine sensor(s) **1205**. For example, the sensor(s) processor **1210** can calculate and monitor the fuel flow, stator vane position, air bleed valve position, etc., using the flight condition data included in the sensor data. At block **1720**, the example sensor(s) processor **1210** determines if the case is expanding. In some examples, the case is a case surrounding a high pressure turbine (e.g., the HP turbine **118** of FIG. 1), a low pressure turbine (e.g., the LP turbine **120** of FIG. 1) or a compressor (e.g., the HP compressor **114** and LP compressor **112** of FIG. 1). In some examples, the sensor(s) processor **1210** determines if the case is expanding based on the engine conditions determined from the obtained flight condition data. When the example sensor(s) processor **1210** determines that the case is expanding, then the example operations **1700** continues to block **1730** at which the example actuator controller **1215** sends a first electrical current to a first multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not expanding, then the example operations **1700** continues to block **1725** at which the example sensor(s) processor **1210** determines if the case is shrinking.

At block **1725**, the example sensor(s) processor **1210** determines if the case is shrinking. In some example, the sensor(s) processor **1210** determines if the case is shrinking based on the engine conditions determined from the obtained flight condition data. When the example sensor(s) processor **1210** determines that the case is shrinking, then the example operations **1700** continues to block **1740** at which the example actuator controller **1215** a third electrical current to a first multilayer stack of piezoelectric material. When the example sensor(s) processor **1210** determines that the case is not shrinking, then the example operations **1700** returns to block **1710** at which the example sensor(s) processor **1210** obtains sensor data.

At block **1730**, the example actuator controller **1215** sends a first electrical current to a first multilayer stack of piezoelectric material. In some examples, the first multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **825**. In some examples, the actuator controller **1215** generates and sends the first electrical current via a first electrical signal to the multilayer stack of piezoelectric material **825** located in the actuator **820**. In some examples, the first electrical current causes a first linear displacement in the multilayer stack of piezoelectric material **825**.

At block **1735**, the example actuator controller **1215** sends a second electrical current to a second multilayer stack of piezoelectric material. In some examples, the second multilayer stack of piezoelectric material is substantially similar to the multilayer stack of piezoelectric material **830**. In some examples, the actuator controller **1215** generates and sends the second electrical current via a second electrical signal to the multilayer stack of piezoelectric material **830** located in the actuator **820**. In some examples, the second electrical current causes a second linear displacement in the multilayer stack of piezoelectric material **830**. In some examples, the second linear displacement is opposite of the first linear displacement. For example, if the first linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **825**, then the second linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **830**. While blocks **1730** and **1735** are shown in sequence, they can be executed in parallel. After the example actuator controller **1215** sends the second

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electrical current to a second multilayer stack of piezoelectric material, operations **1700** ends.

At block **1740**, the example actuator controller **1215** sends a third electrical current to a first multilayer stack of piezoelectric material. In some examples, the actuator controller **1215** generates and sends the third electrical current via a third electrical signal to the multilayer stack of piezoelectric material **825**. In some examples, the third electrical current causes a third linear displacement in the multilayer stack of piezoelectric material **825**.

At block **1745**, the example actuator controller **1215** sends a fourth electrical current to a second multilayer stack of piezoelectric material. In some examples, the actuator controller **1215** generates and sends the fourth electrical current via a fourth electrical signal to the multilayer stack of piezoelectric material **830**. In some examples, the fourth electrical current causes a fourth linear displacement in the multilayer stack of piezoelectric material **830**. In some examples, the fourth linear displacement is opposite of the third linear displacement. For example, if the third linear displacement is a decrease in length and an increase in thickness of the multilayer stack of piezoelectric material **825**, then the fourth linear displacement is an increase in length and a decrease in thickness of the multilayer stack of piezoelectric material **830**. While blocks **1740** and **1745** are shown in sequence in the example of FIG. 17, in certain examples, they can be executed in parallel. After the example actuator controller **1215** sends the fourth electrical current to the second multilayer stack of piezoelectric material, operations **1700** ends.

FIG. 18 is a block diagram of an example processor platform **1800** structured to execute the instructions of FIGS. 13, 14, 15, 16 and 17 to implement the example controller **1200** of FIG. 12. The processor platform **1800** can be, for example, a server, a personal computer, a workstation, a self-learning machine (e.g., a neural network), a mobile device (e.g., a tablet such as an iPad™), or any other type of computing device.

The processor platform **1800** of the illustrated example includes a processor **1812**. The processor **1812** of the illustrated example is hardware. For example, the processor **1812** can be implemented by one or more integrated circuits, logic circuits, microprocessors, GPUs, DSPs, or controllers from any desired family or manufacturer. The hardware processor may be a semiconductor based (e.g., silicon based) device. In this example, the processor implements the example sensor(s) processor **1210** and the example actuator controller **1215**.

The processor **1812** of the illustrated example includes a local memory **1813** (e.g., a cache). The processor **1812** of the illustrated example is in communication with a main memory including a volatile memory **1814** and a non-volatile memory **1816** via a bus **1818**. The volatile memory **1814** may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®) and/or any other type of random access memory device. The non-volatile memory **1816** may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory **1814**, **1816** is controlled by a memory controller.

The processor platform **1800** of the illustrated example also includes an interface circuit **1820**. The interface circuit **1820** may be implemented by any type of interface standard, such as an Ethernet interface, a universal serial bus (USB), a Bluetooth® interface, a near field communication (NFC) interface, and/or a PCI express interface.

In the illustrated example, one or more input devices **1822** are connected to the interface circuit **1820**. The input device(s) **1822** permit(s) a user to enter data and/or commands into the processor **1812**. The input device(s) can be implemented by, for example, an audio sensor, a microphone, a camera (still or video), a keyboard, a button, a mouse, a touchscreen, a track-pad, a trackball, isopoint and/or a voice recognition system.

One or more output devices **1824** are also connected to the interface circuit **1820** of the illustrated example. The output devices **1824** can be implemented, for example, by display devices (e.g., a light emitting diode (LED), an organic light emitting diode (OLED), a liquid crystal display (LCD), a cathode ray tube display (CRT), an in-place switching (IPS) display, a touchscreen, etc.), a tactile output device, a printer and/or speaker. The interface circuit **1820** of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip and/or a graphics driver processor.

The interface circuit **1820** of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) via a network **1826**. The communication can be via, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a line-of-site wireless system, a cellular telephone system, etc.

The processor platform **1800** of the illustrated example also includes one or more mass storage devices **1828** for storing software and/or data. Examples of such mass storage devices **1828** include floppy disk drives, hard drive disks, compact disk drives, and redundant array of independent disks (RAID) systems.

The machine executable instructions **1832** of FIGS. **13**, **14**, **15**, **16**, and **17** may be stored in the mass storage device **1828**, in the volatile memory **1814**, in the non-volatile memory **1816**, and/or on a removable non-transitory computer readable storage medium such as a CD or DVD.

From the foregoing, it will be appreciated that example methods, apparatus and articles of manufacture have been disclosed that improve clearance control in a gas turbine engine. The disclosed examples propose improved ACC designs using piezoelectric actuator(s) in the axial, axial/radial combined, and circumferential directions to achieve tighter clearance at any operating conditions with fast mechanical ACC modulation. The disclosed examples use piezoelectric material to generate high mechanical power and provide fast response clearance control in the radial direction with no time delay. The disclosed examples mechanically convert the displacement of the piezoelectric material in the axial, axial/radial combined, and circumferential directions into the clearance control in the radial direction. The disclosed examples use multilayer stacks of the piezoelectric material to manage the range of displacement, which effects the range of the ACC system muscle capability. The disclosed examples propose simpler ACC design with weight reduction and increased space in the undercowl for other components of the gas turbine engine to be installed more freely. The disclosed examples improve engine performance and EGT control capability with additional SFC benefit due to saving airflow because no cooling airflow is needed for the mechanical ACC system.

Further aspects of the present disclosure are provided by the subject matter of the following clauses:

An apparatus to control clearance for a turbine engine, the apparatus comprising an actuator to control clearance

between a blade and at least one of a shroud or a hanger, the actuator including a multilayer stack of material, and wherein the actuator is outside a case, and a rod coupled to the actuator and the at least one of the shroud or the hanger through an opening in the case, the rod to move the at least one of the shroud or the hanger in a radial direction based on movement of the multilayer stack of material.

The apparatus of any preceding clause, wherein the case surrounds at least part of the turbine engine, the at least part of the turbine engine including the at least one of the shroud or the hanger to contain airflow in the at least part of the turbine engine and at least one of a turbine or a compressor.

The apparatus of any preceding clause, wherein the actuator and the multilayer stack of material are oriented in an axial direction, the multilayer stack of material including at least one of a piezoelectric material or a shape memory alloy.

The apparatus of any preceding clause, further including a controller operatively coupled to the actuator, the controller to supply an electrical current to the multilayer stack of material in the actuator, the multilayer stack of material displaced in an axial direction by the electrical current.

The apparatus of any preceding clause, wherein the actuator controls clearance between the blade and the at least one of the shroud or the hanger in the radial direction using the displacement of the multilayer stack of material.

The apparatus of any preceding clause, further including a first seal on an outer surface of the case and a second seal on an inner surface of the case at the opening in the case for the rod, the first seal and the second seal to reduce leakage through the opening in the case.

An apparatus to control clearance for a turbine engine, the apparatus comprising a first actuator to control clearance between a blade and at least one of a shroud or a hanger, the first actuator including a first multilayer stack of material, and wherein the first actuator is coupled to the at least one of the shroud or a first hook of the hanger, and a second actuator to control clearance between the blade and the at least one of the shroud or the hanger, the second actuator including a second multilayer stack of material, the second actuator coupled to the at least one of the shroud or a second hook of the hanger, and wherein the first actuator and the second actuator are to move the at least one of the shroud or the hanger in an axial direction.

The apparatus of any preceding clause, further including a case surrounding at least part of the turbine engine, the at least part of the turbine engine including the at least one of the shroud or the hanger to contain airflow in the turbine engine and at least one of a turbine or a compressor.

The apparatus of any preceding clause, wherein the first multilayer stack of material and the second multilayer stack of material include at least one of piezoelectric material or shape memory alloy.

The apparatus of any preceding clause, the first actuator further including a third multilayer stack of material, and the second actuator further including a fourth multilayer stack of material.

The apparatus of any preceding clause, the apparatus further including a controller operatively coupled to the first actuator and the second actuator, the controller to supply a first electrical current to the first multilayer stack of material and the second multilayer stack of material, the controller to supply a second electrical current to the third multilayer stack of material and the fourth multilayer stack of material, the controller to supply the first electrical current and the second electrical current at a same time.

The apparatus of any preceding clause, wherein the first multilayer stack of material and the third multilayer stack of

material are displaced in the axial direction by the first electrical current, and the third multilayer stack of material and the fourth multilayer stack of material are displaced in the axial direction by the second electrical current, the first multilayer stack of material and the third multilayer stack of material displaced at a same time the third multilayer stack of material and the fourth multilayer stack of material are displaced.

The apparatus of any preceding clause, wherein the first actuator and second actuator control clearance between the at least one of the shroud or the hanger and the blade in a radial direction using the displacement of the first multilayer stack of material, the second multilayer stack of material, the third multilayer stack of material, and the fourth multilayer stack of material.

The apparatus of any preceding clause, the first actuator further including a first spring, and the second actuator further including a second spring.

The apparatus of any preceding clause, the apparatus further including a controller operatively coupled to the first actuator and the second actuator, the controller to supply an electrical current to the first multilayer stack of material and the second multilayer stack of material, the first multilayer stack of material and the second multilayer stack of material displaced in the axial direction by the electrical current.

The apparatus of any preceding clause, wherein the first actuator and second actuator control clearance between the blade and the at least one of the shroud or the hanger in a radial direction using the displacement of the first multilayer stack of material and the second multilayer stack of material, and wherein the first spring supports displacement of the first multilayer stack of material and the second spring supports the displacement the second multilayer stack of material.

The apparatus of any preceding clause, wherein the first actuator further including a third multilayer stack of material, a fourth multilayer stack of material, and a fifth multilayer stack of material, and the second actuator further including a sixth multilayer stack of material, a seventh multilayer stack of material, and an eighth multilayer stack of material.

The apparatus of any preceding clause, the apparatus further including a controller operatively coupled to the first actuator and the second actuator, the controller to supply a first electrical current to the first multilayer stack of material, the second multilayer stack of material, the third multilayer stack of material, and the sixth multilayer stack of material, the controller to supply a second electrical current to the fourth multilayer stack of material, the fifth multilayer stack of material, the seventh multilayer stack of material, and the eighth multilayer stack of material, the controller to supply the first electrical current and the second electrical current at a same time.

The apparatus of any preceding clause, wherein the first multilayer stack of material, the second multilayer stack of material, the third multilayer stack of material, and the sixth multilayer stack of material are displaced by the first electrical current, and the fourth multilayer stack of material, the fifth multilayer stack of material, the seventh multilayer stack of material, and the eighth multilayer stack of material are displaced by the second electrical current.

The apparatus of any preceding clause, wherein the first multilayer stack of material, the second multilayer stack of material, the fourth multilayer stack of material, and the seventh multilayer stack of material are displaced in an axial direction, and the third multilayer stack of material, the fifth multilayer stack of material, the sixth multilayer stack of

material, and the eighth multilayer stack of material are displaced in a radial direction.

The apparatus of any preceding clause, wherein the first actuator and second actuator control clearance between the at least one of the shroud or the hanger and the blade in the axial direction and the radial direction using the displacement of the first multilayer stack of material, the second multilayer stack of material, the third multilayer stack of material, the fourth multilayer stack of material, the fifth multilayer stack of material, the sixth multilayer stack of material, the seventh multilayer stack of material, and the eighth multilayer stack of material.

An apparatus to control clearance for a turbine engine, the apparatus comprising a case surrounding at least part of the turbine engine, the at least part of the turbine engine including a shroud and a hanger to contain airflow in the turbine engine, the case including a first pin and a second pin, the hanger coupled to the shroud, and wherein the hanger includes a first slot and a second slot, the first pin coupled to the first slot and the second pin coupled to the second slot, and an actuator to control clearance between an airfoil and the shroud, the actuator including a first multilayer stack of material and a second multilayer stack of material, and wherein the actuator is inside the hanger between the first slot and the second slot, the actuator to move the hanger and the shroud in a circumferential direction.

The apparatus of any preceding clause, wherein the at least part of the turbine engine includes a turbine or a compressor.

The apparatus of any preceding clause, wherein the first multilayer stack of material and the second multilayer stack of material include at least one of piezoelectric material or shape memory alloy.

The apparatus of any preceding clause, further including a controller operatively coupled to the actuator, the controller to supply a first electrical current to the first multilayer stack of material in the actuator and a second electrical current to the second multilayer stack of material in the actuator, the first multilayer stack of material displaced by the first electrical current and the second multilayer stack of material displaced by the second electrical current.

The apparatus of any preceding clause, wherein the actuator moves the hanger using the displacement of the first multilayer stack of material and the second multilayer stack of material, the actuator to displace the first slot and the second slot in the hanger, the first pin and the second pin in the case to move in the first slot and the second slot.

The apparatus of any preceding clause, wherein the actuator controls clearance between the airfoil and the shroud in the circumferential direction using the displacement of the first slot and the second slot in the hanger.

A clearance control method includes obtaining sensor data and monitoring engine conditions to determine whether an engine case is expanding or shrinking. When the case is expanding, a first electrical current is sent to a multilayer stack of piezoelectric material. When the case is shrinking, a second electrical current is sent to the multilayer stack of piezoelectric material.

A clearance control method includes obtaining sensor data and monitoring engine conditions to determine whether an engine case is expanding or shrinking. When the case is expanding, a first electrical current is sent to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material, and a second electrical current is sent to a third multilayer stack of piezoelectric material and a fourth multilayer stack of piezoelectric material.

rial. When the case is shrinking, a third electrical current is sent to the first multilayer stack of piezoelectric material and the second multilayer stack of piezoelectric material, and a fourth electrical current is sent to the third multilayer stack of piezoelectric material and the fourth multilayer stack of piezoelectric material.

A clearance control method includes obtaining sensor data and monitoring engine conditions to determine whether an engine case is expanding or shrinking. When the case is expanding, a first electrical current is sent to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material. When the case is shrinking, a second electrical current is sent to the first multilayer stack of piezoelectric material and the second multilayer stack of piezoelectric material.

A clearance control method includes obtaining sensor data and monitoring engine conditions to determine whether an engine case is expanding or shrinking. When the case is expanding, a first electrical current is sent to a first multilayer stack of piezoelectric material, and a second electrical current is sent to a second multilayer stack of piezoelectric material. When the case is shrinking, a third electrical current is sent to the first multilayer stack of piezoelectric material, and a fourth electrical current is sent to the second multilayer stack of piezoelectric material.

A clearance control method includes obtaining sensor data and monitoring engine conditions to determine whether an engine case is expanding or shrinking. When the case is expanding, a first electrical current is sent to first, second, third, and fourth multilayer stacks of piezoelectric material, and a second electrical current is sent to fifth, sixth seventh, and eighth multilayer stacks of piezoelectric material. When the case is shrinking, a third electrical current is sent to the first, second, third, and fourth multilayer stacks of piezoelectric material, and a fourth electrical current is sent to the fifth, sixth seventh, and eighth multilayer stacks of piezoelectric material.

The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, methods, apparatus, and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, methods, apparatus, and articles of manufacture fairly falling within the scope of the claims of this patent.

What is claimed is:

1. A clearance control method for a turbine or a compressor of an engine, the method comprising:

obtaining sensor data;

monitoring engine conditions to determine, based on the sensor data, whether an engine case is expanding or shrinking;

when determined that the engine case is expanding, supplying a first electrical current to displace a multilayer stack of material in a first direction, the multilayer stack of material positioned with respect to the engine case; and

when determined that the case is shrinking, supplying a second electrical current to displace the multilayer stack of material in a second direction, the first electrical current different from the second electrical current, the first direction different from the second direction.

2. The clearance control method of claim 1, wherein the multilayer stack of material includes a multilayer stack of piezoelectric material.

3. The clearance control method of claim 1, wherein, when determined that the engine case is expanding, the first electrical current is sent to a first multilayer stack of material and a second multilayer stack of material, and a third electrical current is sent to a third multilayer stack of piezoelectric material and a fourth multilayer stack of piezoelectric material.

4. The clearance control method of claim 3, wherein, when determined that the case is shrinking, the second electrical current is sent to the first multilayer stack of piezoelectric material and the second multilayer stack of piezoelectric material, and a fourth electrical current is sent to the third multilayer stack of piezoelectric material and the fourth multilayer stack of piezoelectric material.

5. The clearance control method of claim 1, wherein, when determined that the engine case is expanding, the first electrical current is sent to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material; and, when determined that the engine case is shrinking, the second electrical current is sent to the first multilayer stack of piezoelectric material and the second multilayer stack of piezoelectric material.

6. The clearance control method of claim 1, wherein, when determined that the engine case is expanding, the first electrical current is sent to a first multilayer stack of piezoelectric material, and a third electrical current is sent to a second multilayer stack of piezoelectric material; and

when the case is shrinking, the second electrical current is sent to the first multilayer stack of piezoelectric material, and a fourth electrical current is sent to the second multilayer stack of piezoelectric material.

7. The clearance control method of claim 1, wherein, when determine that the engine case is expanding, the first electrical current is sent to first, second, third, and fourth multilayer stacks of piezoelectric material, and a third electrical current is sent to fifth, sixth seventh, and eighth multilayer stacks of piezoelectric material; and, when determined that the engine case is shrinking, the second electrical current is sent to the first, second, third, and fourth multilayer stacks of piezoelectric material, and a fourth electrical current is sent to the fifth, sixth seventh, and eighth multilayer stacks of piezoelectric material.

8. The clearance control method of claim 1, wherein the multilayer stack of material is connected to an actuator.

9. The clearance control method of claim 8, wherein the actuator is connected to a rod, the rod connected to the engine case.

10. The clearance control method of claim 9, wherein the rod is to move the engine case in a radial direction based on an axial movement of the multilayer stack of material.

11. At least one non-transitory computer readable storage medium storing instructions that, when executed, cause processor circuitry to at least:

obtain sensor data;

monitor engine conditions to determine, based on the sensor data, whether an engine case is expanding or shrinking;

when determined that the engine case is expanding, supply a first electrical current to displace a multilayer stack of material in a first direction, the multilayer stack of material positioned with respect to the engine case; and

when determined that the case is shrinking, supply a second electrical current to displace the multilayer stack of material in a second direction, the first

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electrical current different from the second electrical current, the first direction different from the second direction.

12. The at least one computer readable storage medium of claim 11, wherein the multilayer stack of material includes a multilayer stack of piezoelectric material.

13. The at least one computer readable storage medium of claim 11, wherein, when determined that the engine case is expanding, the first electrical current is sent to a first multilayer stack of material and a second multilayer stack of material, and a third electrical current is sent to a third multilayer stack of piezoelectric material and a fourth multilayer stack of piezoelectric material.

14. The at least one computer readable storage medium of claim 13, wherein, when determined that the case is shrinking, the second electrical current is sent to the first multilayer stack of piezoelectric material and the second multilayer stack of piezoelectric material, and a fourth electrical current is sent to the third multilayer stack of piezoelectric material and the fourth multilayer stack of piezoelectric material.

15. The at least one computer readable storage medium of claim 11, wherein, when determined that the engine case is expanding, the first electrical current is sent to a first multilayer stack of piezoelectric material and a second multilayer stack of piezoelectric material; and, when determined that the engine case is shrinking, the second electrical current is sent to the first multilayer stack of piezoelectric material and the second multilayer stack of piezoelectric material.

16. The at least one computer readable storage medium of claim 11, wherein, when determined that the engine case is

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expanding, the first electrical current is sent to a first multilayer stack of piezoelectric material, and a third electrical current is sent to a second multilayer stack of piezoelectric material; and

when the case is shrinking, the second electrical current is sent to the first multilayer stack of piezoelectric material, and a fourth electrical current is sent to the second multilayer stack of piezoelectric material.

17. The at least one computer readable storage medium of claim 11, wherein, when determine that the engine case is expanding, the first electrical current is sent to first, second, third, and fourth multilayer stacks of piezoelectric material, and a third electrical current is sent to fifth, sixth seventh, and eighth multilayer stacks of piezoelectric material; and, when determined that the engine case is shrinking, the second electrical current is sent to the first, second, third, and fourth multilayer stacks of piezoelectric material, and a fourth electrical current is sent to the fifth, sixth seventh, and eighth multilayer stacks of piezoelectric material.

18. The at least one computer readable storage medium of claim 11, wherein the multilayer stack of material is connected to an actuator.

19. The at least one computer readable storage medium of claim 18, wherein the actuator is connected to a rod, the rod connected to the engine case.

20. The at least one computer readable storage medium of claim 19, wherein the rod is to move the engine case in a radial direction based on an axial movement of the multilayer stack of material.

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